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(54) **SYSTEMS AND METHODS OF PROVIDING
NEW RADIO POSITIONING**

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(60) Provisional application No. 62/791,469, filed on Jan. 11, 2019.

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H04W 64/00 (2009.01)
H04L 5/00 (2006.01)

(52) **U.S. Cl.**
CPC **H04W 64/003** (2013.01); **H04L 5/0051** (2013.01)

(58) **Field of Classification Search**

CPC H04W 64/003; H04L 5/0051
See application file for complete search history.

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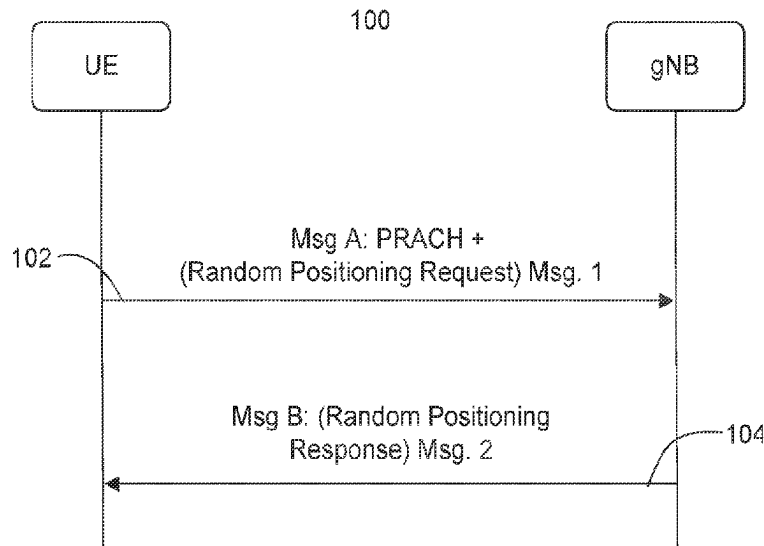
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(57) **ABSTRACT**

Systems and methods are provided to determine an estimated location of a user equipment (UE). For instance, a UE can provide a positioning request to a node while the UE is in a radio resource control idle (RRC_IDLE) or inactive (RRC_INACTIVE) state. The positioning request can be implemented in a random access channel (RACH) message A (MsgA). The UE can then receive a positioning response from the node. The positioning response can be implemented in a RACH message B (MsgB).

20 Claims, 21 Drawing Sheets



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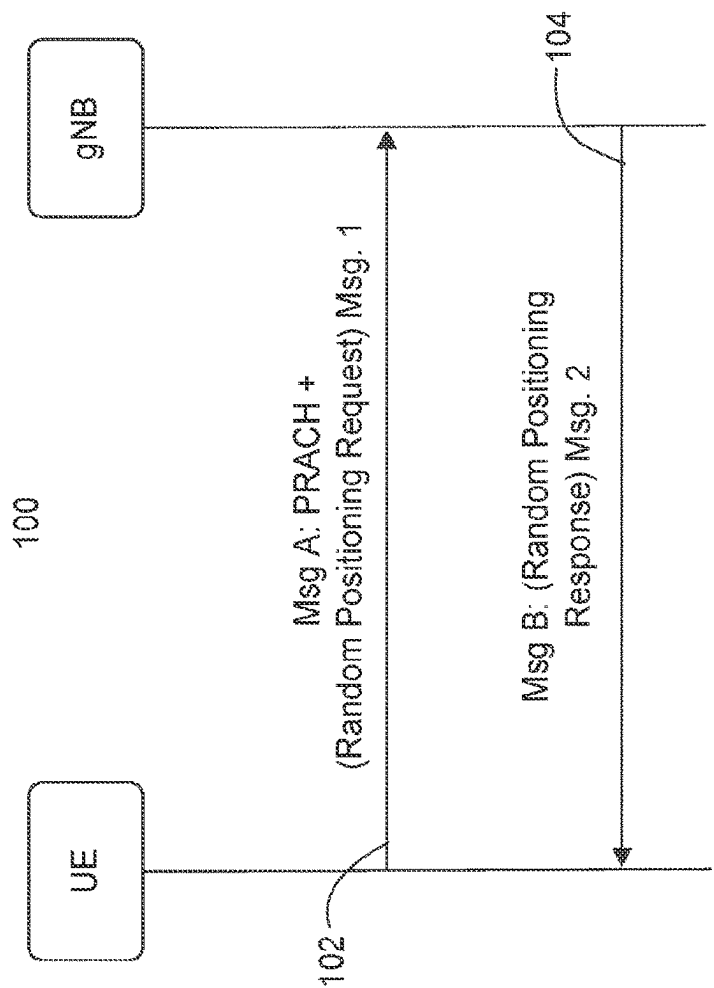


FIG. 1

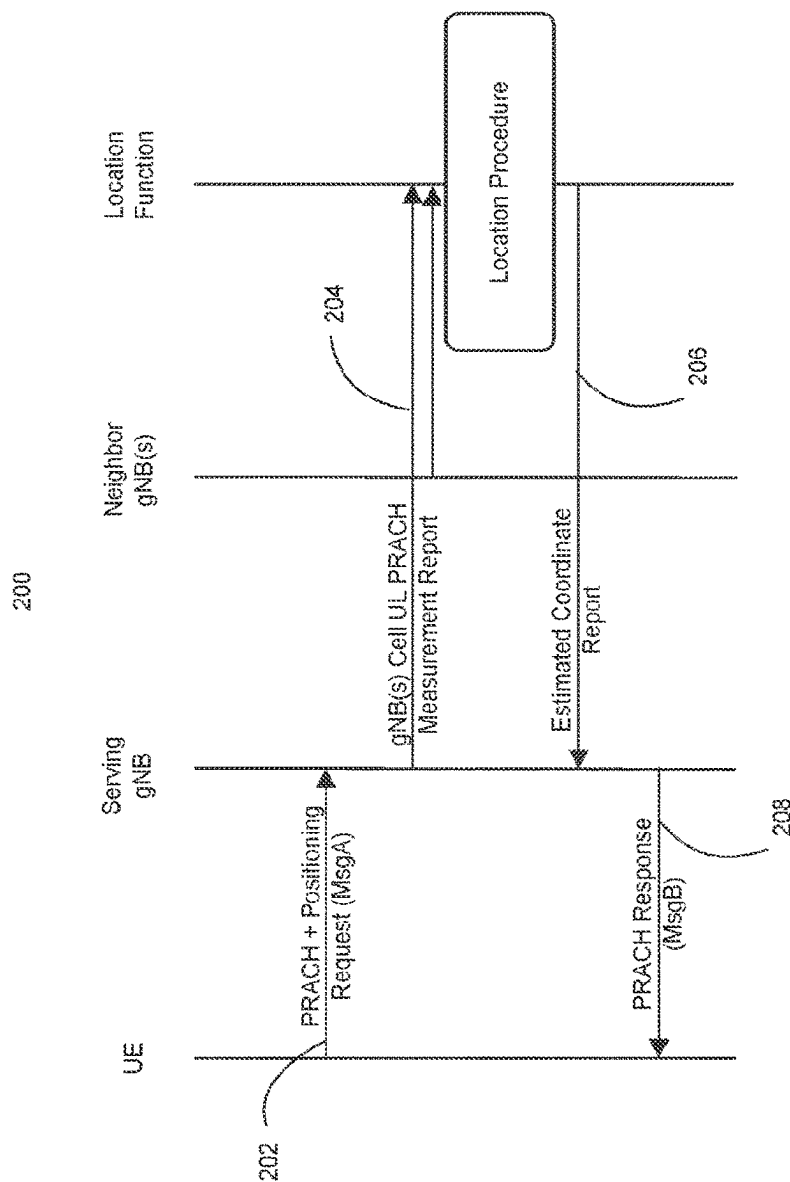


FIG. 2

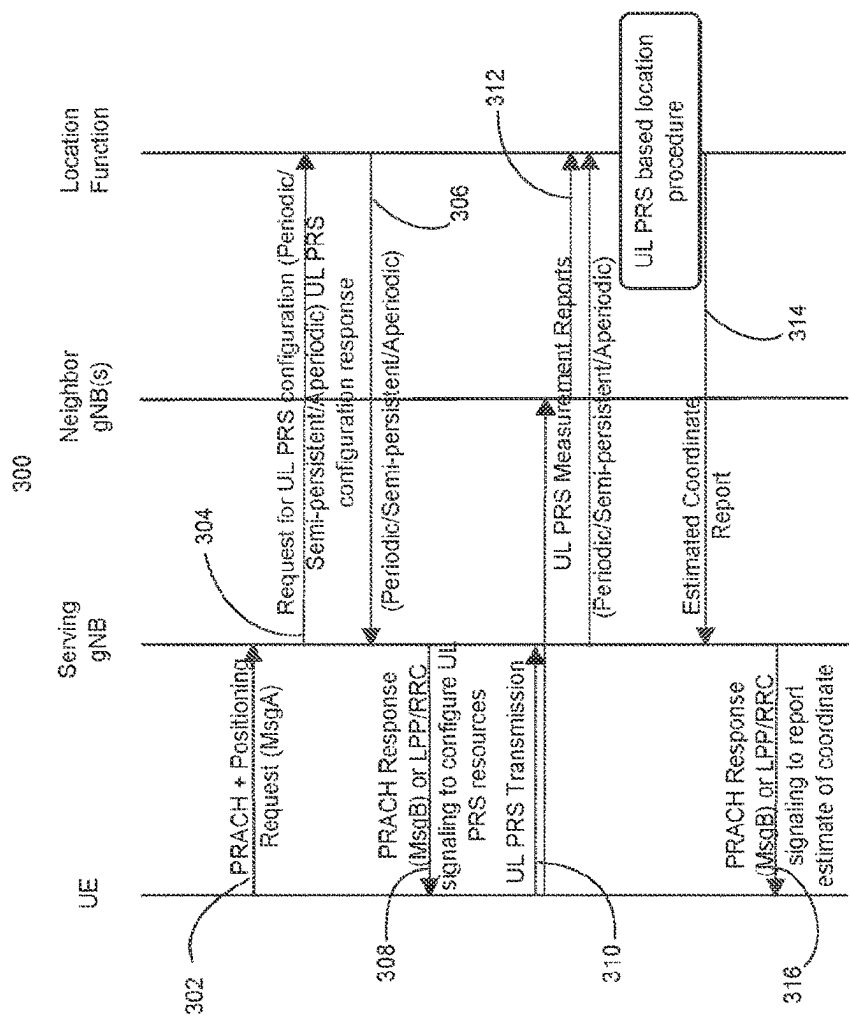


FIG. 3

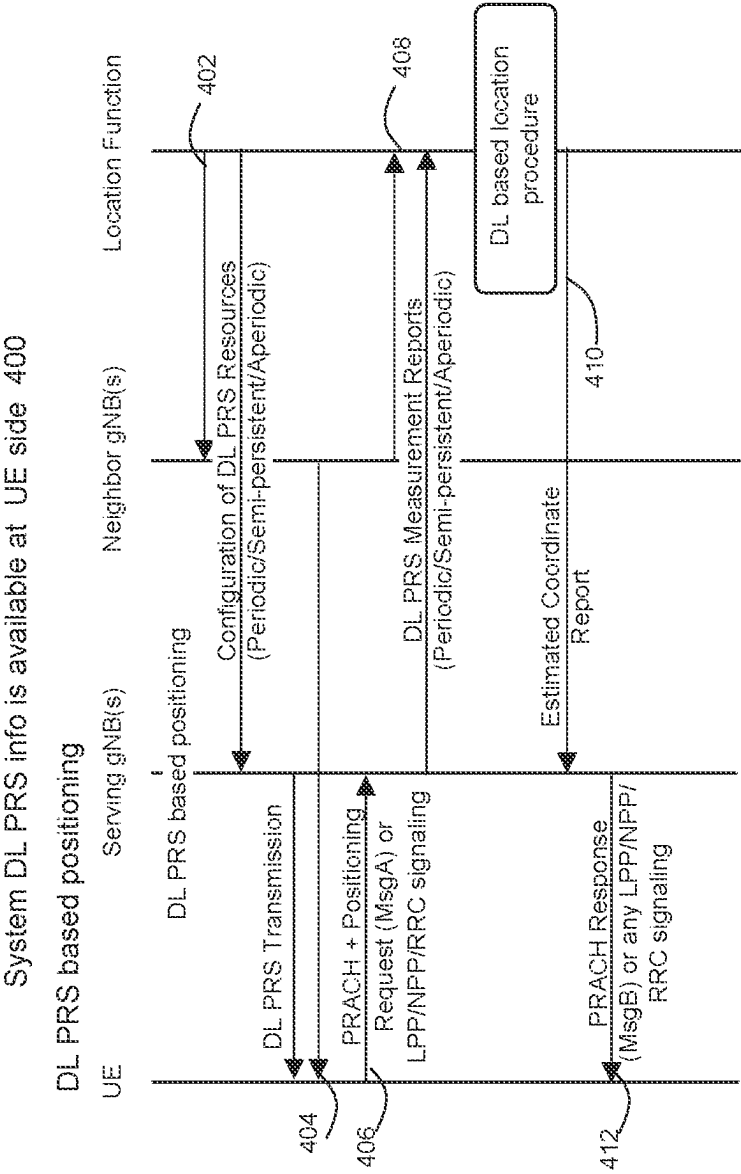


FIG. 4A

System DL PRS info is unavailable at UE side 420

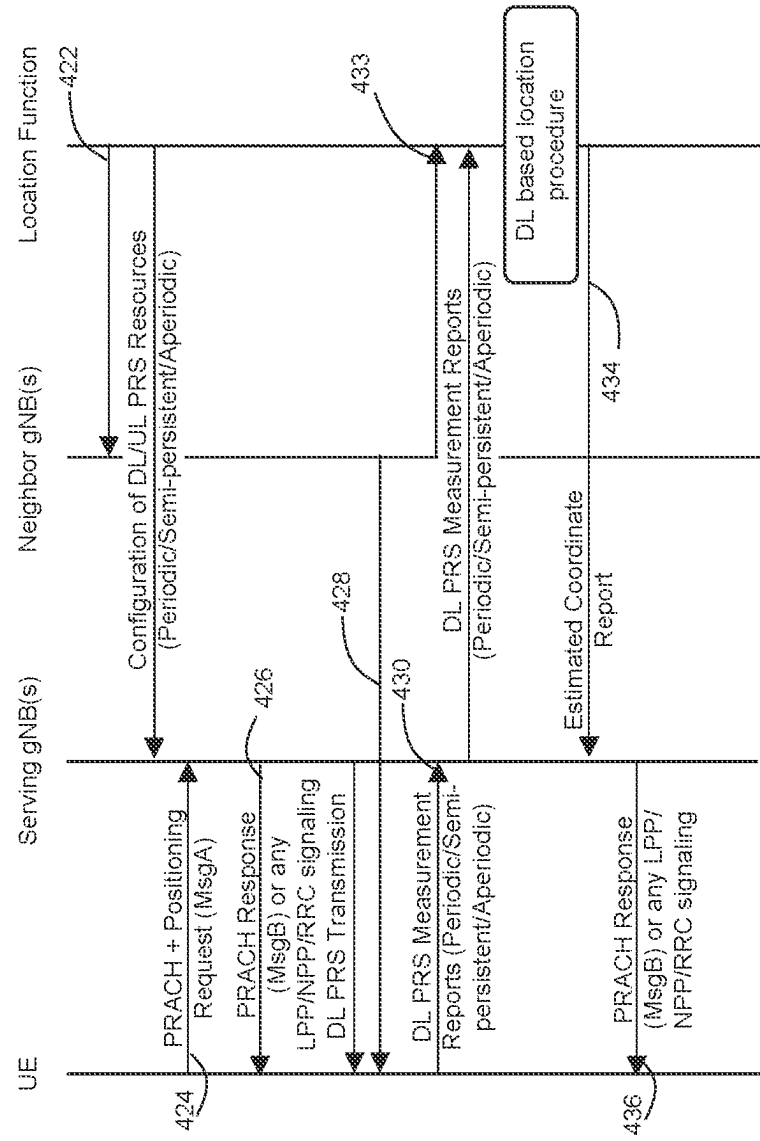


FIG. 4B

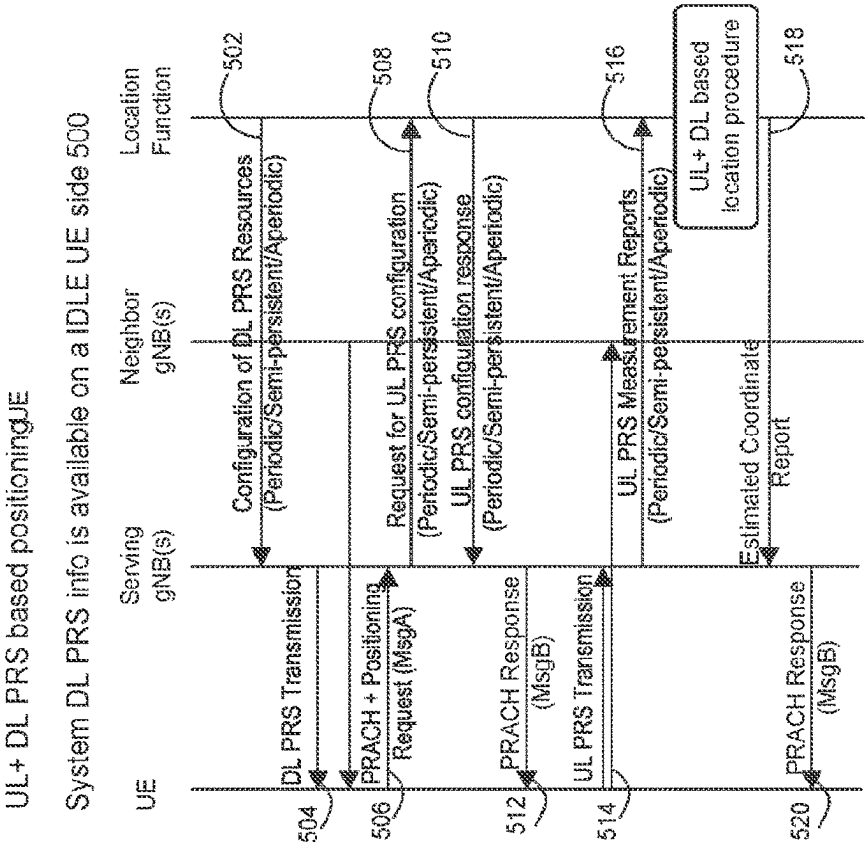


FIG. 5A

System DL PRS info is unavailable on a IDLE UE side 540

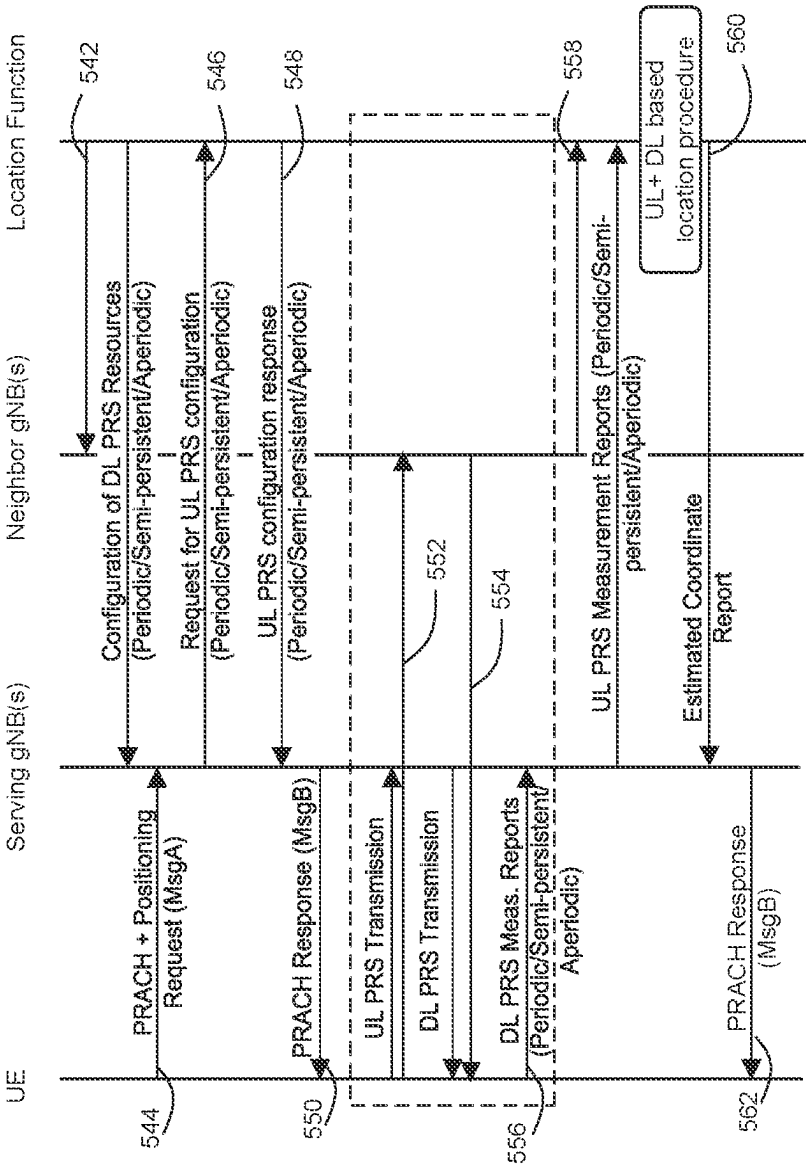


FIG. 5B

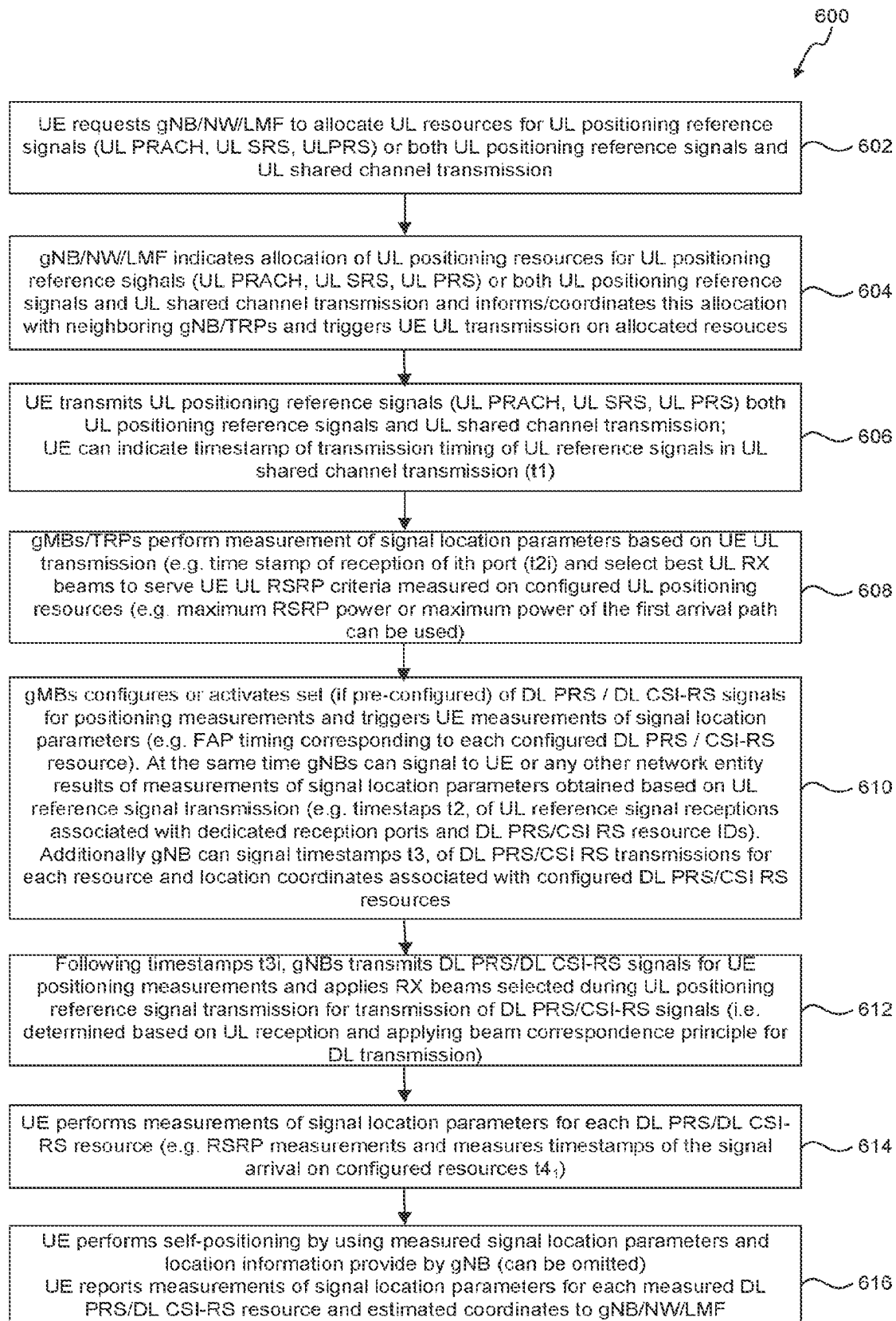


FIG. 6

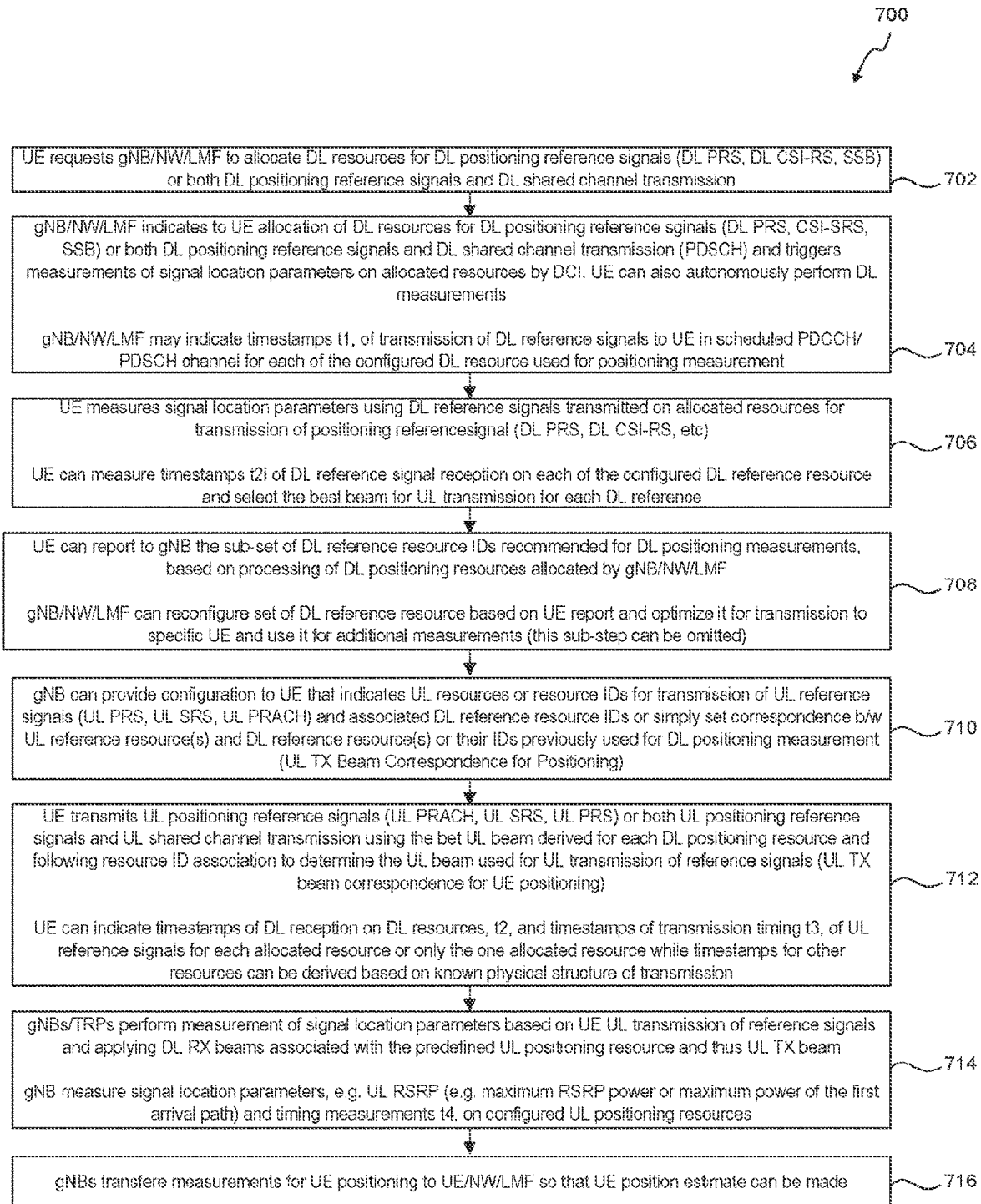


FIG. 7

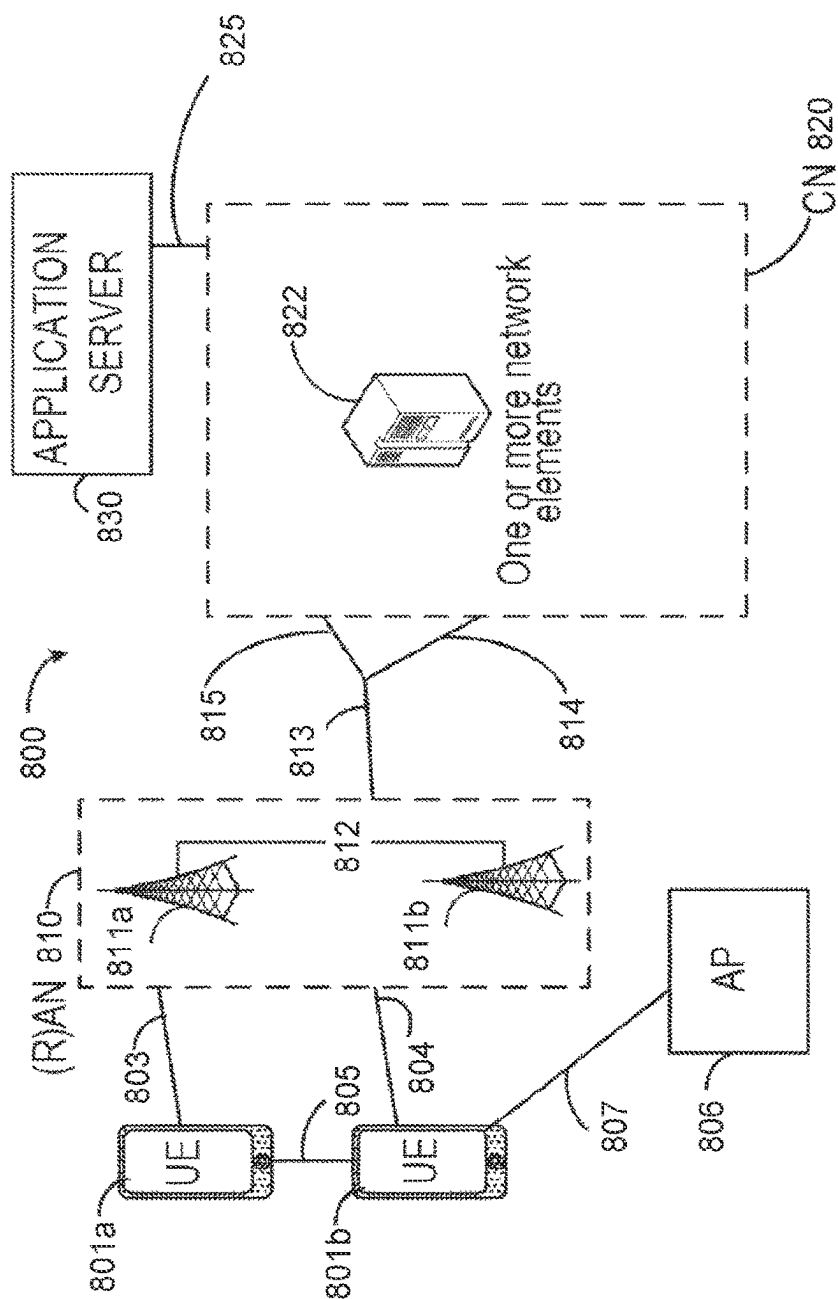


FIG. 8

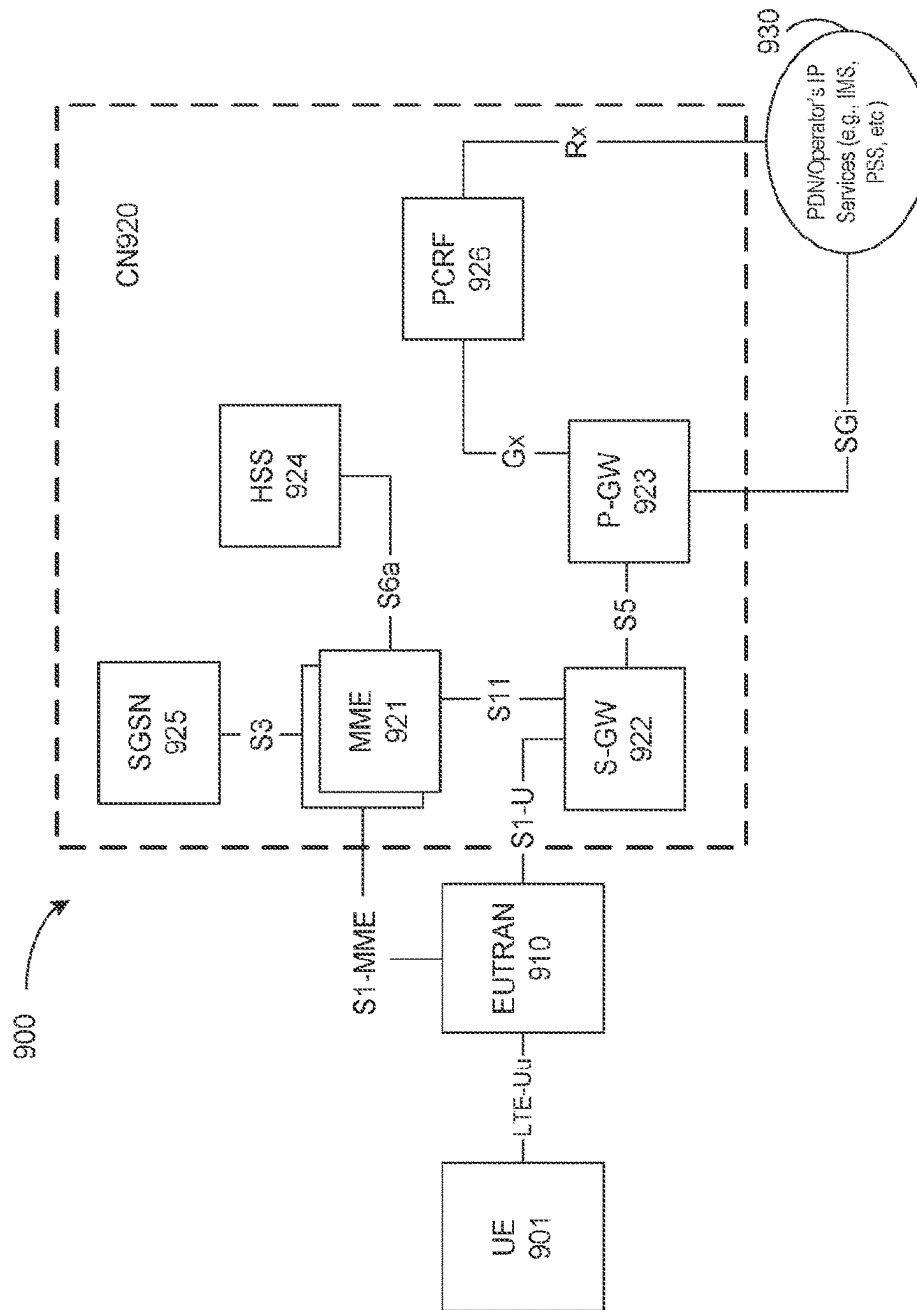


FIG. 9

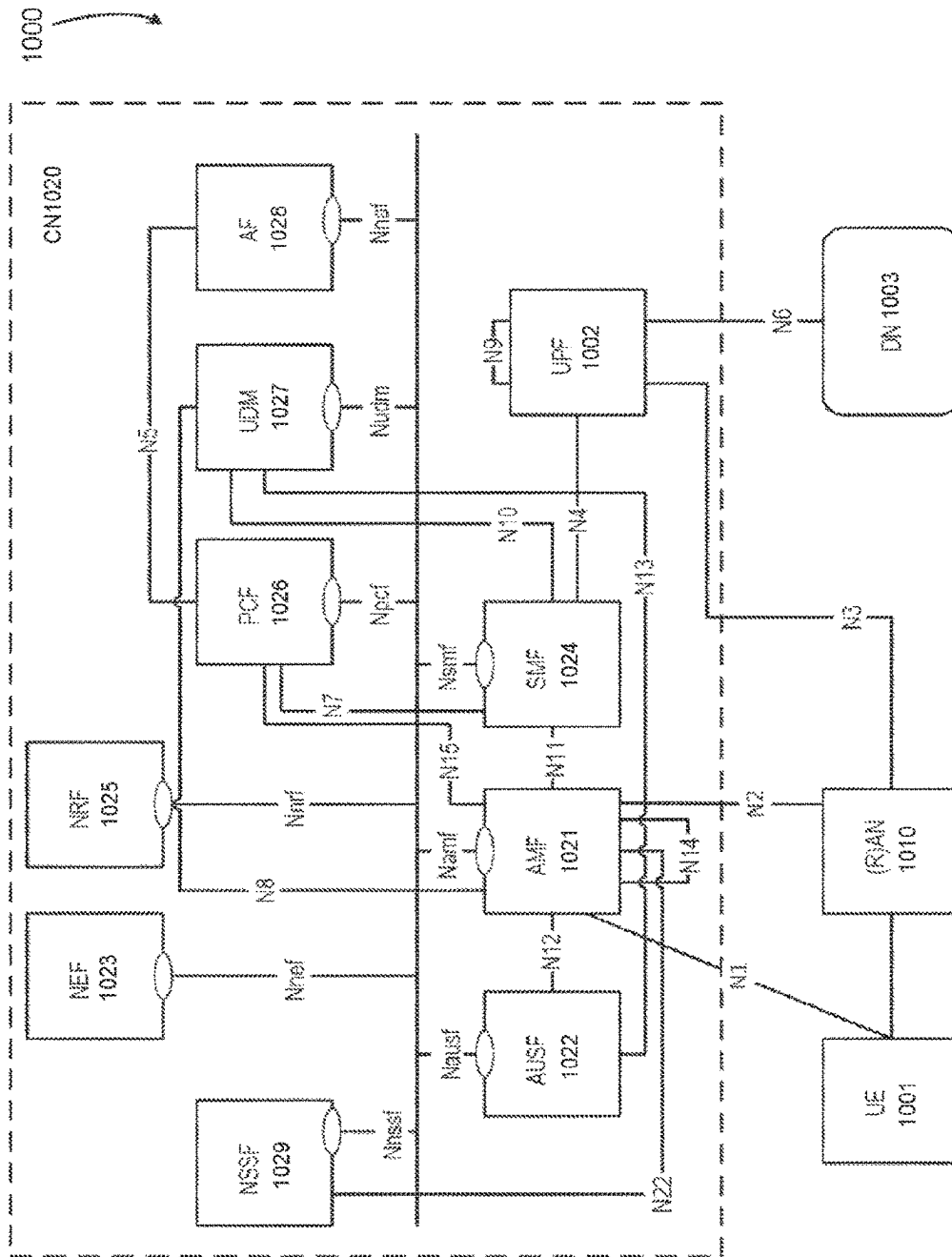


FIG. 10

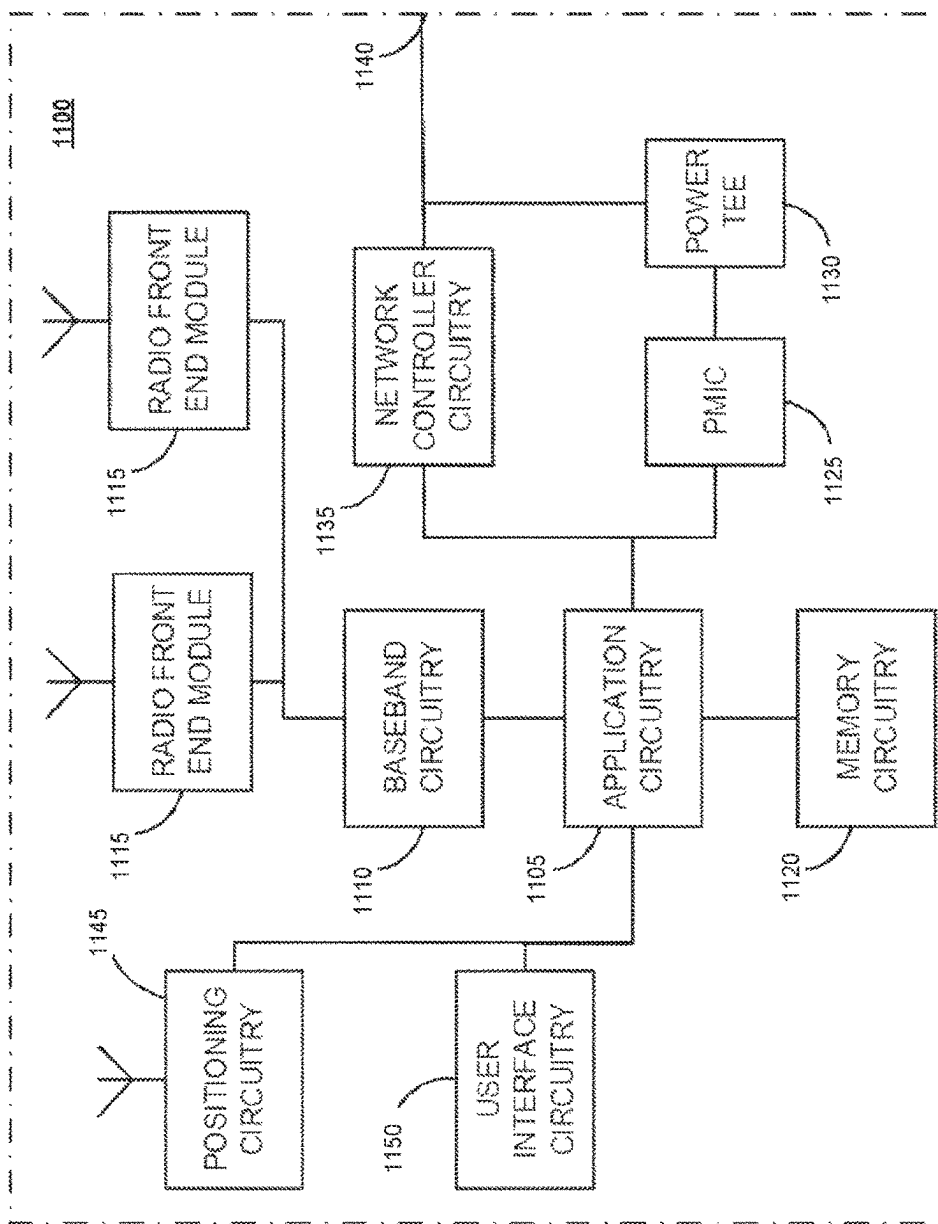


FIG. 11

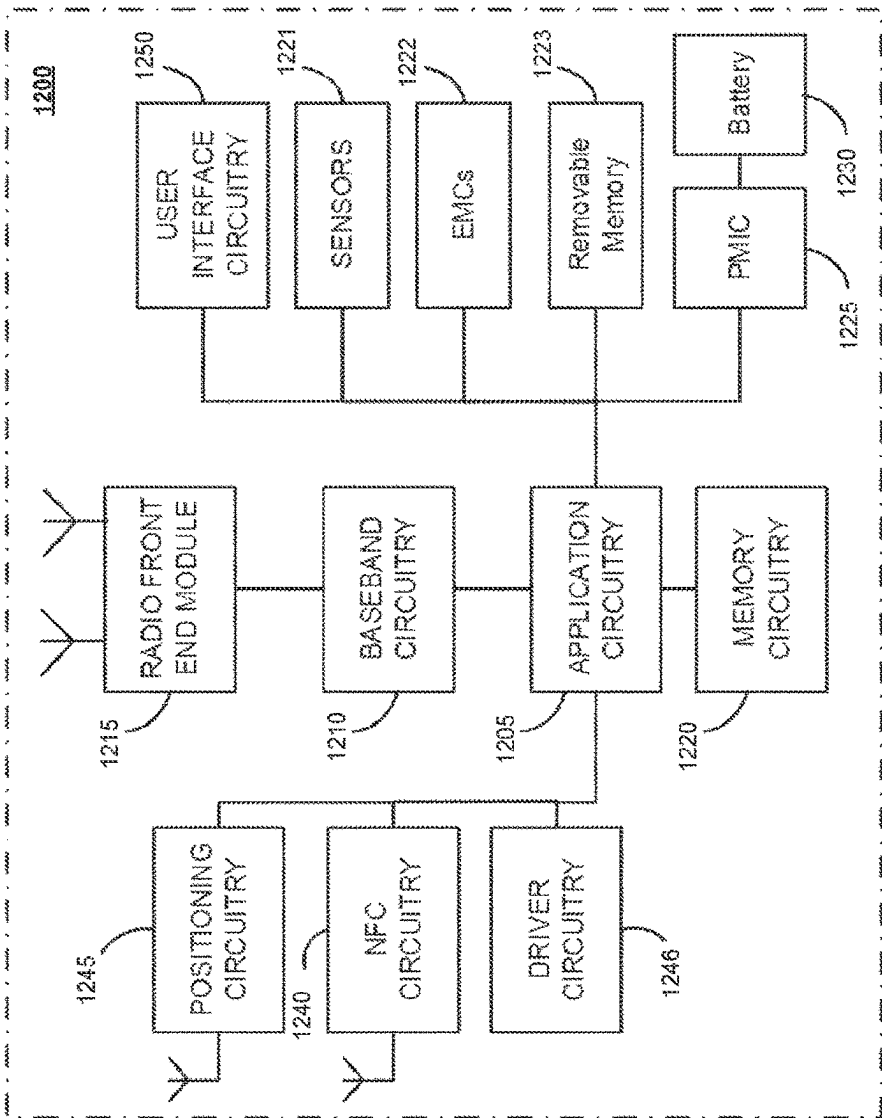


FIG. 12

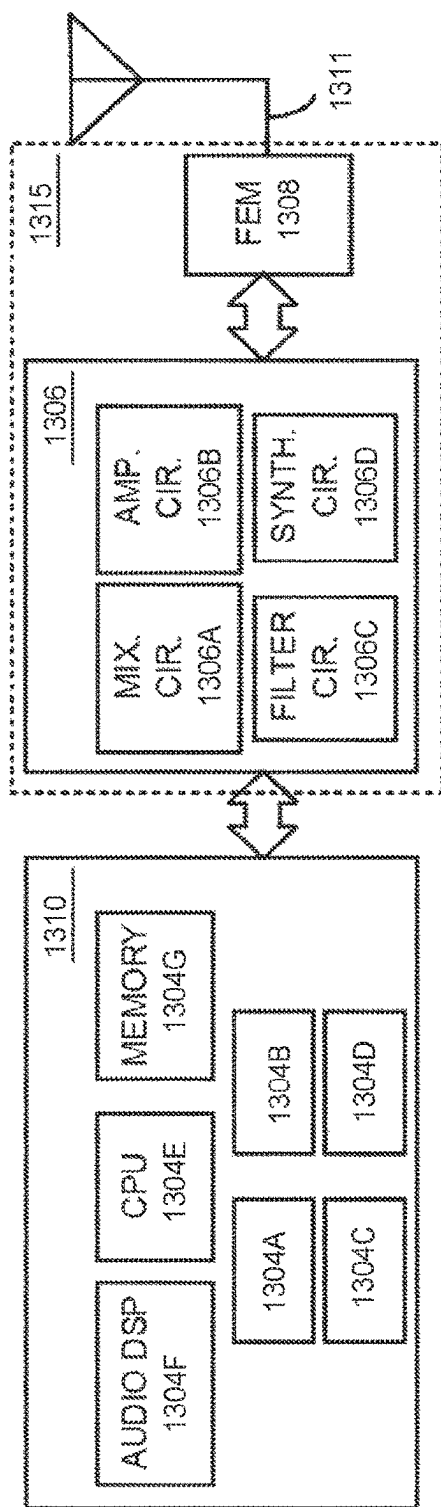


FIG. 13

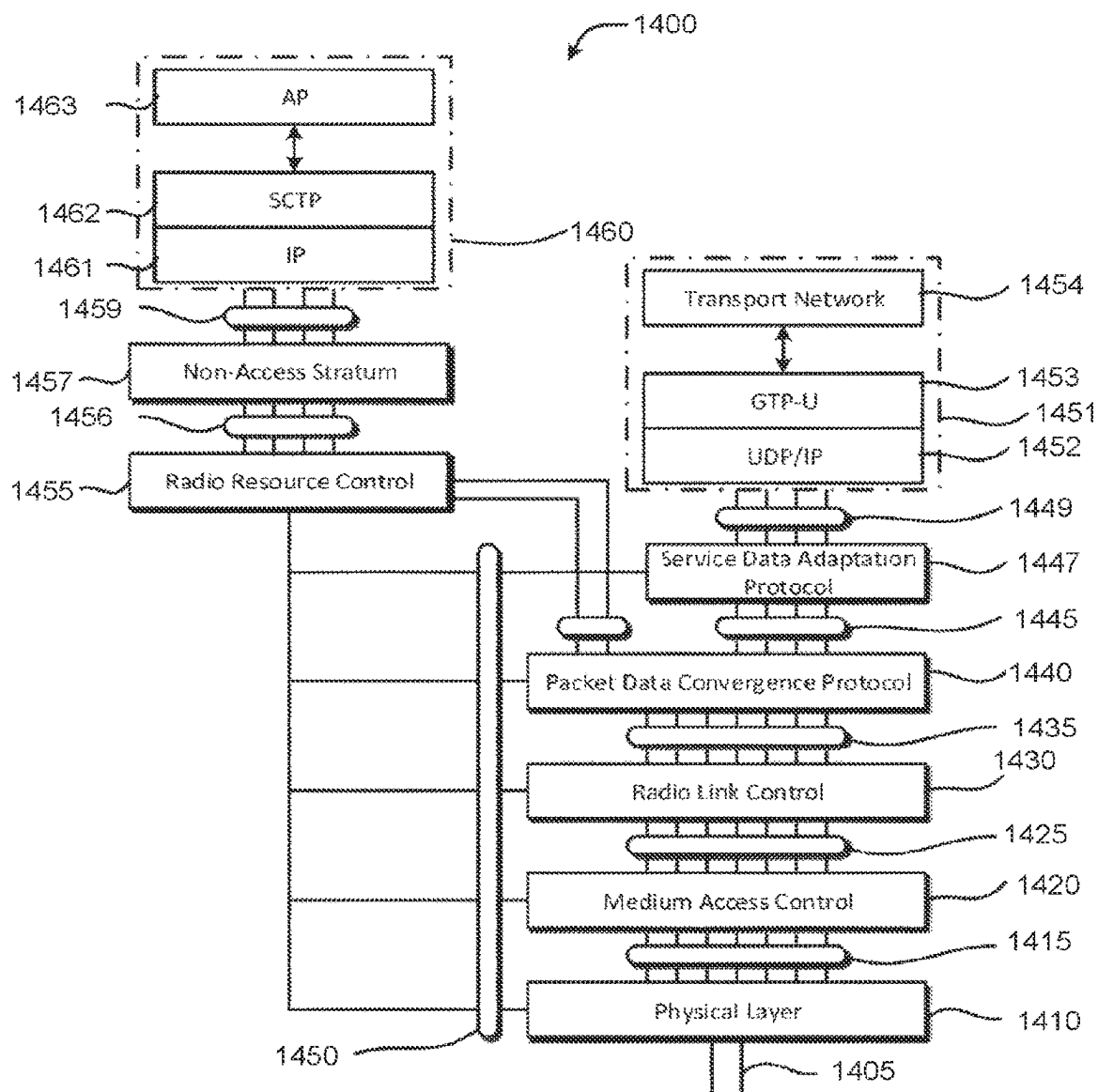


FIG. 14

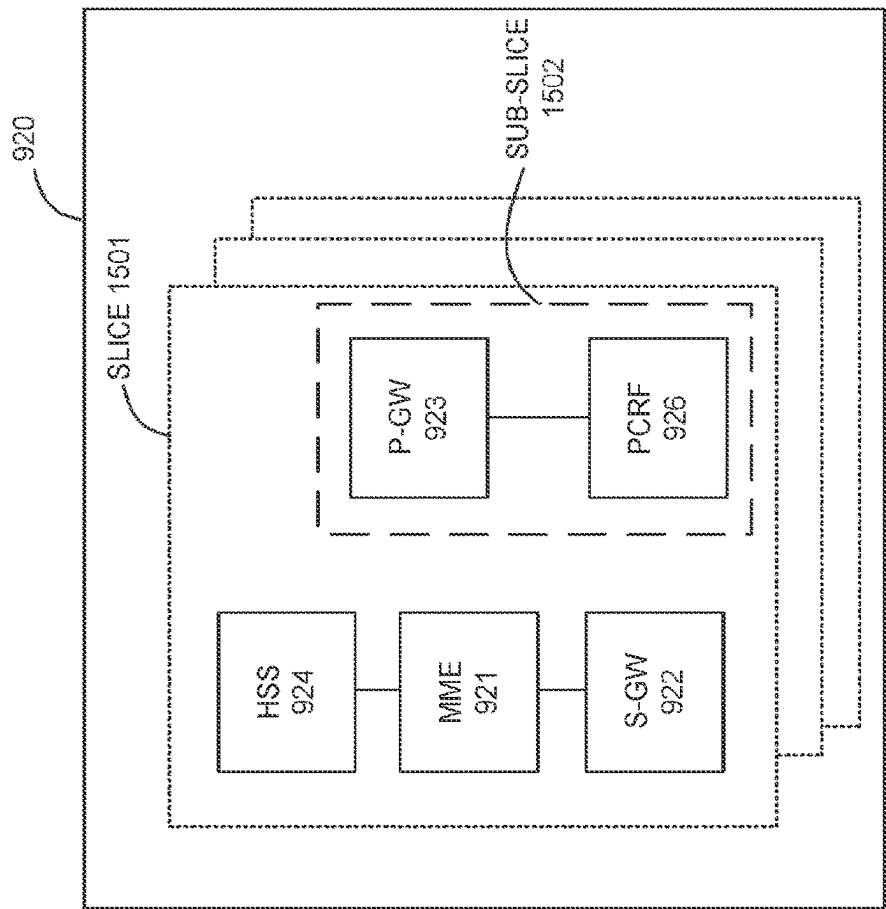


FIG. 15

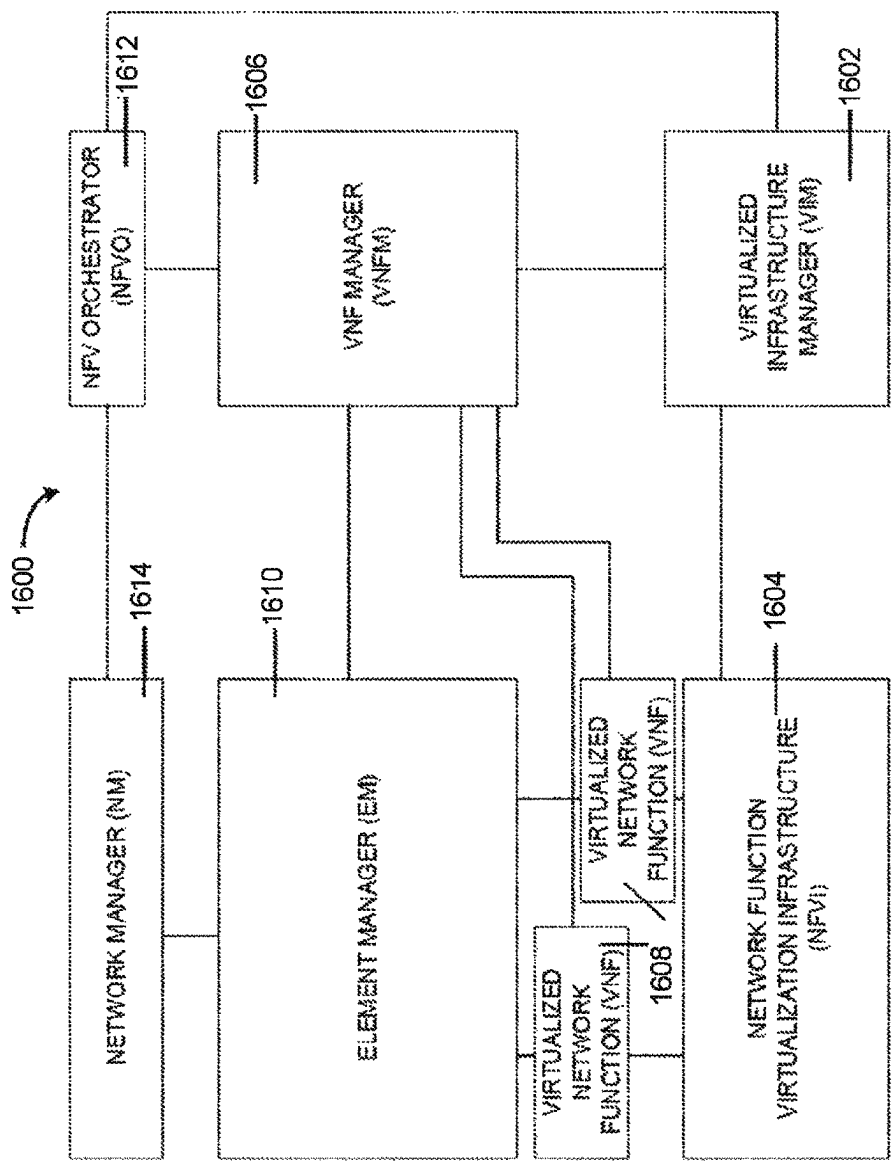


FIG. 16

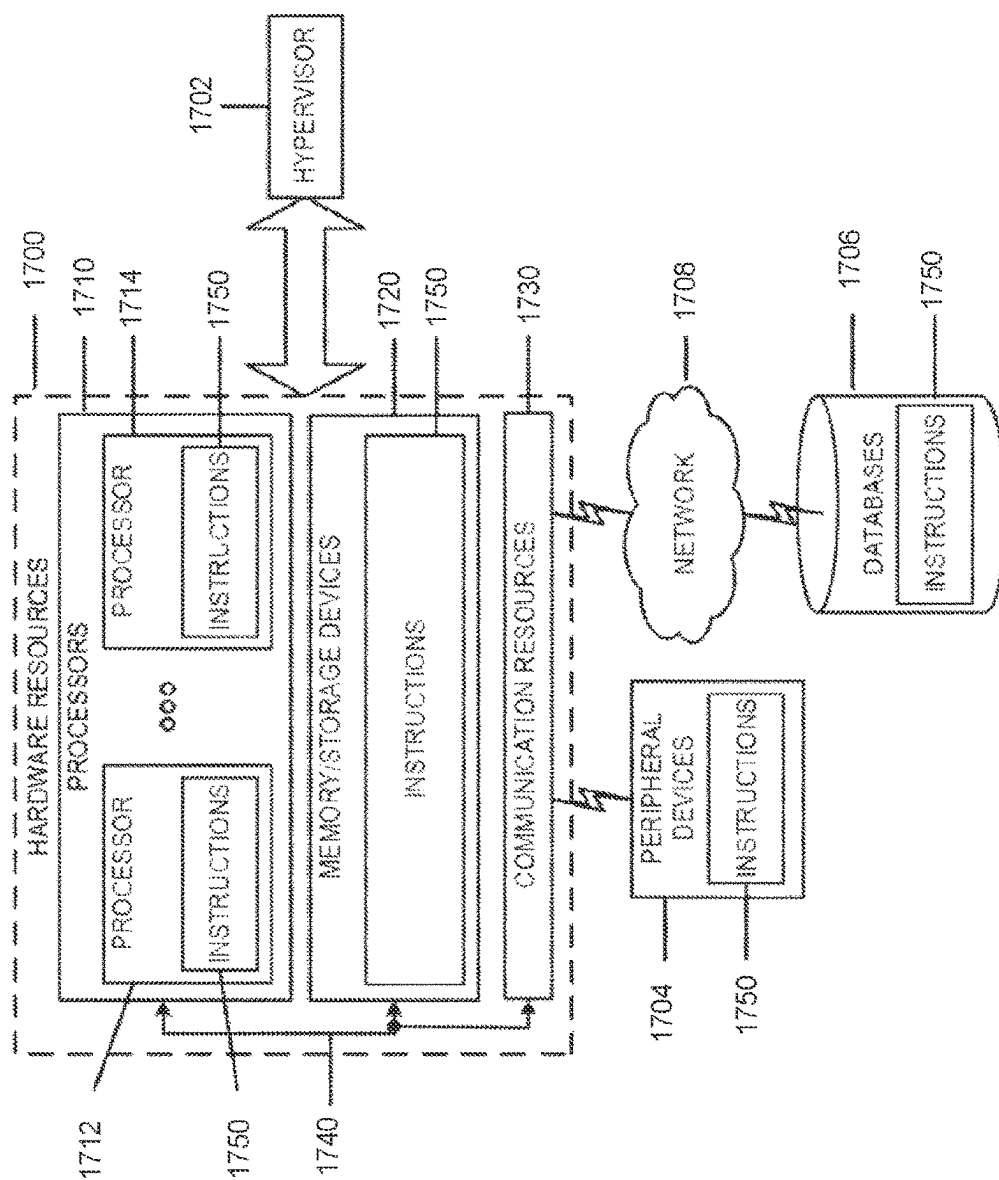


FIG. 17

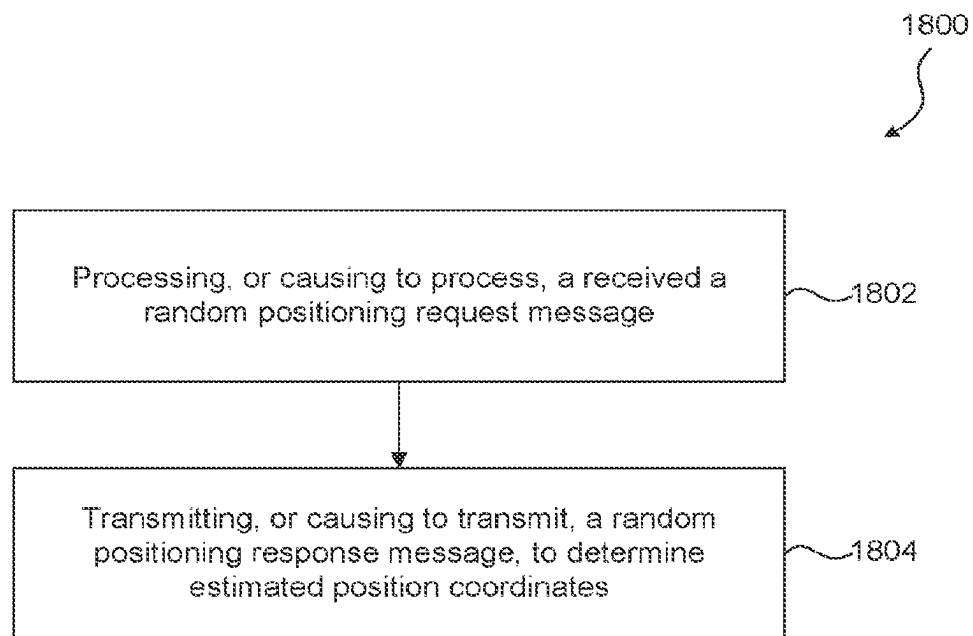


FIG. 18

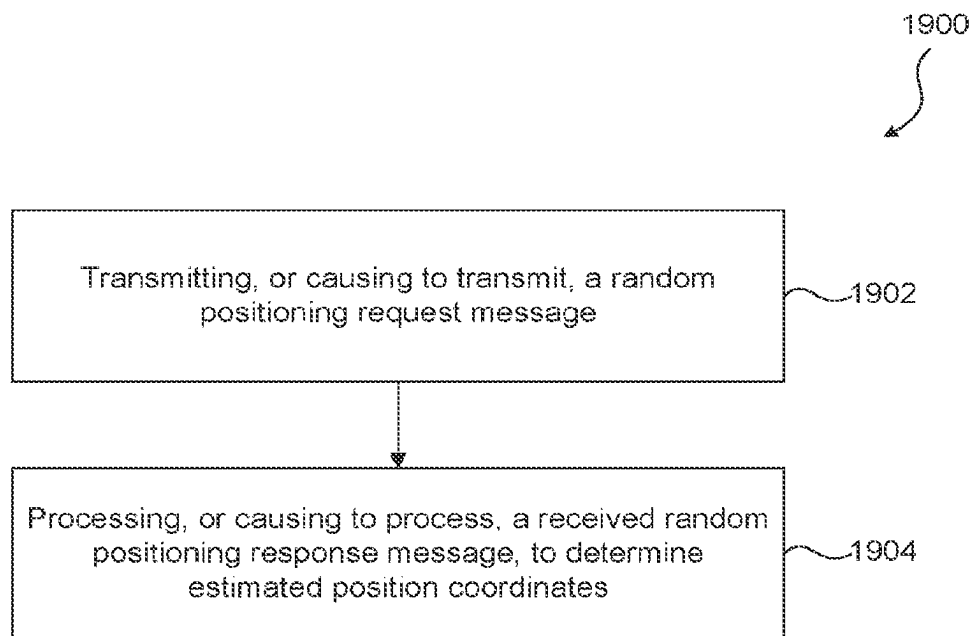


FIG. 19

SYSTEMS AND METHODS OF PROVIDING NEW RADIO POSITIONING

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a Continuation of U.S. application Ser. No. 17/422,101, filed on Jul. 9, 2021, which is a national stage entry of International Application No. PCT/US2020/013093, filed on Jan. 10, 2020, which claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application No. 62/791,469, filed Jan. 11, 2019, all of which are incorporated by reference their entireties.

FIELD

Various embodiments generally may relate to the field of wireless communications.

SUMMARY

Aspects and advantages of embodiments of the present disclosure will be set forth in part in the following description, or may be learned from the description, or may be learned through practice of the embodiments.

One example aspect of the present disclosure is directed to a user equipment (UE) for use in a wireless network. The UE includes one or more processors configured to provide a positioning request to a node while the UE is in a radio resource control idle (RRC_IDLE) or inactive (RRC_INACTIVE) state. The positioning request is implemented in a random access channel (RACH) message A (MsgA). The one or more processors are further configured to receive a positioning response from the node. The positioning response is implemented in a RACH message B (MsgB).

In some embodiments, the MsgA comprises a physical random access channel (PRACH) preamble.

In some embodiments, the MsgA comprises at least one of a requested type of positioning procedure, a request for additional allocation of reference signals, one or more signal location parameter measurements, a UE identification, and a requested type of positioning reference resources.

In some embodiments, the requested type of positioning procedure comprises a downlink (DL) only positioning technique, an uplink (UL) only positioning technique, or a DL and UL positioning technique.

In some embodiments, the positioning response comprises an estimated coordinate of the UE, a metric characterizing coordinate estimation quality, timestamps indicative of a time of reception of one or more reference signals, a request for additional positioning measurements, a configuration or activation of additional resources for transmission of positioning reference signals, and location information associated with downlink (DL) positioning reference resources.

In some embodiments, the MsgB comprises at least one of an estimated location of the UE, a metric characterizing a quality or uncertainty of the estimated location, a timestamp indicating a time of reception of the MsgA by the node, a timestamp indicating a time of transmission of the MsgB by the node, a request for additional positioning measurements, or location information associated with DL positioning reference resources.

In some embodiments, the estimated location of the UE is determined by a location function entity, and provided to the node.

In some embodiments, the one or more processors are further configured to, prior to receipt of the positioning response, receive a UL positioning reference signal (PRS) resource configuration from the node, and to provide UL PRS data to the node, the UL PRS data determined based at least in part on the received UL PRS resource configuration.

In some embodiments, the one or more processors are further configured to, prior to providing the positioning request, receive a DL PRC resource configuration from the node. In such embodiments, the MsgA comprises a DL PRS measurement report determined based at least in part on the received DL PRC resource configuration.

In some embodiments, the positioning response comprises a UL PRS resource configuration. In such embodiments, the one or more processors are further configured to provide UL PRS data to the node, the UL PRS data determined based at least in part on the received UL PRS resource configuration.

Other aspects of the present disclosure are directed to methods, systems, apparatuses, tangible, non-transitory computer-readable media, user interfaces and devices for determining an estimated location of a UE.

These and other features, aspects, and advantages of various embodiments will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in, and constitute part of, this specification, illustrate embodiments of the present disclosure and, together with the description, serve to explain the related principles.

This Summary is provided merely for purposes of reviewing some exemplary embodiments, so as to provide a basic understanding of some aspects of the subject matter described herein. Accordingly, it will be appreciated that the above-described features are merely examples and should not be construed to narrow the scope or spirit of the subject matter described herein in any way. Other features, aspects, and advantages of the subject matter described herein will become apparent from the following Detailed Description, Figures, and Claims.

BRIEF DESCRIPTION OF THE FIGURES

Detailed discussion of embodiments directed to one of ordinary skill in the art is set forth in the specification, which makes reference to the appended figures, in which:

FIG. 1 depicts a diagram of an example method of determining an estimated position of a user equipment according to some embodiments.

FIG. 2 depicts a diagram of another example method of determining an estimated position of a user equipment according to some embodiments.

FIG. 3 depicts a diagram of another example method of determining an estimated position of a user equipment according to some embodiments.

FIG. 4a depicts a diagram of another example method of determining an estimated position of a user equipment according to some embodiments.

FIG. 4b depicts a diagram of another example method of determining an estimated position of a user equipment according to some embodiments.

FIG. 5a depicts a diagram of another example method of determining an estimated position of a user equipment according to some embodiments.

FIG. 5b depicts a diagram of another example method of determining an estimated position of a user equipment according to some embodiments.

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FIG. 6 depicts a diagram of another example method of determining an estimated position of a user equipment according to some embodiments.

FIG. 7 depicts a diagram of another example method of determining an estimated position of a user equipment according to some embodiments.

FIG. 8 depicts an example system architecture according to some embodiments.

FIG. 9 depicts another example system architecture according to some embodiments.

FIG. 10 depicts another example system architecture according to some embodiments.

FIG. 11 depicts a block diagram of an example infrastructure equipment according to some embodiments.

FIG. 12 depicts a block diagram of an example platform according to some embodiments.

FIG. 13 depicts a block diagram of example baseband circuitry and front end modules according to some embodiments.

FIG. 14 depicts a block diagram of example protocol functions that may be implemented in a wireless communication device according to some embodiments.

FIG. 15 depicts a block diagram of example core network components according to some embodiments.

FIG. 16 depicts a block diagram of example components of a system for supporting NFV according to some embodiments.

FIG. 17 depicts a block diagram of an example computer system that can be utilized to implement various some embodiments.

FIG. 18 depicts a flow diagram of an example method of determining an estimated position of a user equipment according to some embodiments.

FIG. 19 depicts a flow diagram of an example method of determining an estimated position of a user equipment according to some embodiments.

The features and advantages of the embodiments will become more apparent from the detailed description set forth below when taken in conjunction with the drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements. The drawing in which an element first appears is indicated by the leftmost digit(s) in the corresponding reference number.

DETAILED DESCRIPTION OF THE INVENTION

Reference will now be made in detail to embodiments, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the embodiments, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modification and variations can be made to the embodiments without departing from the scope or spirit of the present disclosure. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus it is intended that aspects of the present disclosure cover such modifications and variations.

The foregoing description is intended to enable any person skilled in the art to make and use the disclosure, and is provided in the context of a particular application and its requirements. Moreover, the foregoing descriptions of embodiments of the present disclosure have been presented for purposes of illustration and description only. They are

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not intended to be exhaustive or to limit the present disclosure to the forms disclosed. Accordingly, many modifications and variations will be apparent to practitioners skilled in the art, and the general principles defined herein may be applied to other embodiments and applications without departing from the spirit and scope of the present disclosure. Additionally, the discussion of the preceding embodiments is not intended to limit the present disclosure. Thus, the present disclosure is not intended to be limited to the embodiments shown, but is to be accorded the widest scope consistent with the principles and features disclosed herein.

The following detailed description refers to the accompanying drawings. The same reference numbers may be used in different drawings to identify the same or similar elements. In the following description, for purposes of explanation and not limitation, specific details are set forth such as particular structures, architectures, interfaces, techniques, etc. in order to provide a thorough understanding of the various aspects of various embodiments. However, it will be apparent to those skilled in the art having the benefit of the present disclosure that the various aspects of the various embodiments may be practiced in other examples that depart from these specific details. In certain instances, descriptions of well-known devices, circuits, and methods are omitted so as not to obscure the description of the various embodiments with unnecessary detail. For the purposes of the present document, the phrase “A or B” means (A), (B), or (A and B).

Disclosed embodiments include methods and a signaling protocol for enhanced positioning in cellular wireless communication networks of the next generation cellular systems, such as, for example, 3GPP LTE R16+ and 3GPP NR R16+. However, it will be appreciated that various embodiments may be generally applicable to any type of wireless communication system. Various disclosed embodiments aim to decrease the time to first fix (TFF) for a UE by reducing of signaling overhead between UE and a gNBs/LMFs (Location Management Functions) required for UE positioning.

Conventional wireless communication systems do not support positioning procedures for users that are in a ‘stand by’ or IDLE mode. In order to get an estimated coordinate for such type of users, it is required to ‘wake up’ them and apply a positioning procedure. Disclosed embodiments include procedures for an optimized positioning mechanism for UEs in a RRC_IDLE or RRC_INACTIVE state. This mechanism helps to reduce latency between a positioning request and coordinated estimation; additionally, this mechanism helps to spend less power for positioning related signaling as well as spectrum resources.

NR Rel-16 standard will support a 2-step RACH procedure that will help a user more efficiently enter into the ACTIVE/CONNECTED state from the IDLE state. Example aspects of the present disclosure leverage the 2-step RACH procedure for users in an IDLE or INACTIVE state for the coordinate calculation procedure. Skilled artisans will appreciate that this 2-step RACH procedure combines messages 1 and 3 from 4-step RACH into a single message (referred to herein as message A (MsgA)), and combines messages 2 and 4 from 4-step RACH into a single message (referred to herein as message B (MsgB)).

FIG. 1 depicts a flow diagram of an example method for such coordinate calculation according to embodiments. At 102, a UE can provide a MsgA to a gNB node. The MsgA can include a physical random access channel preamble and a random positioning request. The random positioning request can be included within message 1 from 4-step RACH. At 104, the gNB detects PRACH preamble and positioning request, and sends a positioning response as

MsgB. The positing response can be incorporated, for instance, within message 2 of 4-step RACH.

Example embodiments include a novel NR 2-step Positioning RACH mechanism as well as UL SRS/PRS transmission procedures for UE positioning. Various embodiments may provide: (i) improved positioning latency and performance in NR communication systems; (ii) improved energy efficiency for positioning operation NR communication systems.

As used herein, signal location parameters (SLP) are parameters of a signal that can be applied for the purpose of user positioning. Such parameters include phase difference, time of arrival, time of arrival timestamps, time difference of arrival, propagation time/delays, angle of arrivals/departures, received reference signal powers and any other information that can be relevant to facilitate an estimate of UE geographical coordinates. As used herein, positioning reference signals (PRS) are the signals sent by cells/eNB/gNB/TRPs/Network Entities or UEs used to measure SLP. Knowledge of PRS is thus beneficial for UE location. PRS can include specifically designed sequences and signals with good cross and autocorrelation properties, or any data transmission depending on implementation and measurement and reporting type. Examples of such signals in UL are PRACH, SRS or PRS/ranging signals. In downlink, signals may include PRS, CSI-RS, CRS, etc. As used herein, reference resource refers to the resource where PRS is transmitted characterized by stamp/ID, that can be configured by higher layer signaling and may be configured to UE for measurement and reporting. For instance, the following ID may be used: PRS resource ID, CSI-RS resource ID, UL SRS Resource ID, PRACH resource ID, etc.

Generally, NR technology may support three types of RAT dependent positioning: (i) DL only based solutions (gNBs transmit reference signal and UEs perform measurement of SLPs)—Example: D-TDOA DL RSTD measurement; (ii) UL only based solutions (UEs transmit reference signal and gNBs perform measurement of SLPs)—Example: U-TDOA UL RSTD measurement; (iii) DL & UL only based solutions (gNBs and UEs transmit reference signal and gNB and UEs perform measurement of SLPs)—Example—RTT measurement. In some cases, NR technology can support combinations of DL only, UL only, DL and UL based solutions.

Each of the RAT dependent embodiments may be characterized by different sets of DL/UL reference signals used for measurement of signal location parameters and measurement reports and signaling procedures. However, all of the procedures have in common positioning request and positioning response that can be triggered by UE or gNB/NW/LMF.

In various embodiments, a 2-step RACH approach can be used for positioning services, which is characterized by MsgA transmission from UE to gNB and MsgB transmission from gNB to UE. In case of UE-initiated positioning, MsgA can be considered as a positioning request while MsgB as a positioning response. The transmission timing relationship between these two messages (transmission MsgA and reception MsgB) can be predefined, indicated in MsgA or determined by gNB. Such 2-step RACH approach can be used to trigger any type of NR positioning technique and solution: e.g., (DL only (D-TDOA), UL only (U-TDOA), DL&UL E-CID and/or RTT). There may be multiple variations of different NR positioning procedures depending on the requested positioning approach. In some specific embodiments, a 2-step RACH can be used for ranging and RTT type of positioning.

Ranging and RTT Based NR Positioning

In some embodiments, positioning in accordance with aspects of the present disclosure can include ranging and round trip time (RTT) positioning. In case of UE-initiated ranging, a UE requests the network to perform RTT ranging with single or multiple cells. In such an embodiment, the network pre-allocated resources for UL ranging signal transmission (e.g., PRACH/SRS or new UL PRS signals). The UE performs UL PRS transmission at time instance t_1 while gNBs perform measurements of time of arrival and time-stamps them as $t_{2,i}$, where i is the index of gNBs/TRPs performing timing estimation. After timing estimation, gNBs/TRPs perform DL PRS transmissions at predefined estimated time instances $t_{3,i}$. The estimated values of $t_{2,i}$ and $t_{3,i}$ or their differences can be directly reported to the network and/or to the UE through a serving gNB. DL PRS transmissions from gNBs/TRPs are used by UE to estimate time-stamps $t_{4,i}$. The combination of $t_{4,i}$ timestamps jointly with information on t_1 and $t_{2,i}$ and $t_{3,i}$ can be used to estimate $RTT_i = (t_{4,i} - t_1) + (t_{3,i} - t_{2,i})$.

Using 2-step RACH in one embodiment, the UE can transmit RACH signal associated with PUSCH (Message A—msgA) transmission on pre-allocated spectrum resources. The PUSCH transmission payload with MAC IE can indicate that initiated 2-step PRACH procedure is to perform RTT measurements with serving or multiple cells (including serving and neighboring cells). The PUSCH payload may include information on PRACH transmission time-stamp t_1 as well as provide UE identity and indication of specific ranging/positioning request. The UE can also provide additional information on DL CSI-RS Resource IDs or DL PRS resource ID that it has monitored before and conducted measurements of signal location parameters. This information, if available, can be also included into PUSCH content. If there is no information on DL measurements of SLPs available, or corresponding DL reference signals were not configured, the gNB can respond to UE 2-step PRACH request and configure or indicate/activate DL reference signals (e.g., DL PRS/CSI-RS) to be used for estimation of time-stamps of signals arrival on each DL PRS resource and signal this information in PDCCH/PDSCH (Message B—msgB). In addition, the gNB may provide information on estimated timestamps $t_{2,i}$ and $t_{3,i}$ timestamps, including those collected from neighboring cells and associated with DL reference resources used for transmission of DL reference signals applied for measurements of signal location parameters. Alternatively, if no DL PRS is allocated by the serving gNB, the UE can use SSB transmissions to measure $t_{4,i}$ timings.

In alternative embodiments, gNBs/TRPs transmit DL PRS signals at predefined time instances $t_{1,i}$ based on local timers (e.g., loosely synchronized) and trigger UE timing measurements. The UE measures $t_{2,i}$ time-stamps for each neighboring cell/TRPs or configured DL PRS resources using its own local timer and shares this information jointly with common t_3 time-stamp during UL ranging transmission (e.g., PRACH/SRS or new UL PRS signals), that may be represented by UL PRS signals and associated PUSCH payload transmission. For UL transmission, UE may be preconfigured or granted with dedicated UL positioning resource by the serving gNB. gNBs/TRPs measure $t_{4,i}$ time-stamps based on processing of received UL ranging transmission. Given that gNBs/TRPs are aware about $t_{1,i}$ time instances of DL PRS transmissions, the RTT_i can be calculated as $(t_{4,i} - t_{1,i}) + (t_3 - t_{2,i})$ at network side and reported back to UE by serving gNB/TRP.

The NR positioning procedure may use, as an example, a 2-step RACH procedure. Dedicated user plane and control plane signaling can be also defined to support UE positioning in NR systems. Therefore, corresponding signaling and procedures can be generalized and implemented in some embodiments using generic/legacy NR procedures, signaling protocols and functionality such as, for example: Radio-Resource Control (RRC)/LTE Positioning Protocol/NR Positioning Protocol (Cellular Positioning Protocol)/MAC or physical layer signaling.

NR Positioning PRACH Design and Resource Allocation

As indicated, the MsgA from the UE can include a PRACH preamble. In some embodiments, the positioning PRACH preamble can have the same sequence design and physical structure as legacy preamble from Random Access Channel. In such cases, the differentiation is done by set of resources used for Positioning RACH preamble transmission (i.e., dedicated set of allocated resources for NR positioning is assigned by gNB). The dedicated set of allocated resource for positioning PRACH and PRACH for other purpose, e.g., initial access using 2-step or 4-step RACH procedure can be multiplexed in a time division multiplexing (TDM), frequency division multiplexing (FDM) and code division multiplexing (CDM) manner and a combination thereof. In some embodiments, the positioning PRACH preamble shares the same set of resources as the already existing preamble from Random Access Channel, but the sequence design is different. In either case, the UE can randomly select PRACH resource for transmission or use a predefined hashing function that is dependent on UE ID. In another option, the UE may be configured with a dedicated PRACH preamble that can be based on contention free 2-step RACH procedure for positioning.

In some embodiments, the 2 step RACH procedure can be also used for UL-only based positioning solutions (e.g., U-TDOA) or DL-only based positioning (e.g., D-TDOA). From a UE perspective (RRC_IDLE/RRC_INACTIVE/RRC_CONNECTED), two options can be considered for NR positioning procedures: (i) DL/UL PRS resource configuration information is known on the UE; (ii) DL/UL PRS resource configuration information is not known on the UE. Positioning Request (MsgA)

In embodiments, the potential content of MsgA transmitted by UE to gNB at the first step of two step PRACH procedure may include a payload of MsgA. The payload can contain a type of requested positioning procedure (e.g., UL only-UTDOA, DL&UL-RTT with serving cell, RTT with serving+neighboring cells, DL only-D-TDOA, etc.). The payload can also contain a request for additional allocation of reference signals or their activation if preconfigured (e.g., request for UL SRS/PRS resource allocation; request for DL SRS/PRS resource allocation; request for UL SRS/PRS resource allocation followed by DL CSI-RS/PRS; request for DL CSI-RS/PRS resource allocation followed by UL SRS/PRS). The payload can also include a type of requested (either UL or DL) positioning reference resources (periodic/semi-persistent/apperiodic, etc). The payload can also include initial SLP Measurements, such as initial timestamps t_1 of RACH preamble transmission, initial timestamps t_1 of UL SRS/PRS transmission, if RACH transmission is also associated with UL SRS/PRS transmission, a priori measurement on DL PRS/CSI-RS positioning resource and associated resource IDs (e.g., RSRP, time stamps t_2) if available, list of cell IDs detected by UE, list of DL PRS/CSI-RS Resource IDs, and/or list of SSB indexes. The payload can also include a UE ID and/or other positioning related parameters, such as a previous UE coordinate, UE-based location capa-

bility, positioning service related info (e.g., QoS for positioning procedure, e.g., latency, accuracy levels). The payload can further include a PUSCH message type for positioning. Depending on positioning procedure, not all fields may be present. Given that PUSCH payload size can be predetermined the message may need to have a fixed size and zero-padded to fixed size if payload is smaller. In order to properly interpret the message, the message type information can be added. This message type may be included in the MAC CE. The payload can further include Msg B resources. For instance, the UE may indicate time interval or specific resources when it expects from gNB to receive MsgB or monitor PDCCH channel for MsgB reception. This may be useful to reduce UE power consumption and allocate some processing time for network/gNB. The payload can further include information for contention resolution if contention based 2-step RACH positioning procedure is employed.

Positioning Response (MsgB)

Depending on whether the UE requests or uses additional reference signals for NR positioning, four PRS configuration scenarios are possible: Scenario 1—No PRS configuration; Scenario 2—DL only PRS configuration; Scenario 3—UL only PRS configuration; Scenario 4—DL and UL PRS configuration. As indicated, a node can respond to the UE's positioning request by sending a response implemented in a RACH MsgB. Depending on type of PRS configuration, the response message from a gNB to a UE (MsgB) may contain different information fields. To resolve contention, a gNB may respond the same contention resolution information in MsgB that is initiated and carried in MsgA from the UE.

For instance, for scenario 1, if the UE (IDLE/INACTIVE or CONNECTED) does not have any information on DL PRS configuration, the single cell positioning based on PRACH transmission can be used according to the procedure depicted in FIG. 2, which depicts an example flow diagram of a method 200 of single cell positioning for 2-step RACH. At 202, a UE can provide a MsgA to a serving gNB. At 204, the serving gNB can provide a measurement report to a location function, which, at 206, can send an estimated coordinate report (that includes calculated coordinates) back to the serving gNB. At 208, the gNB can provide a positioning response to the UE in MsgB format. The positioning response can contain at least one of the following fields: Estimated UE coordinate based on serving/neighboring cell measurements; Metric characterizing coordinate estimation quality or uncertainty; Time stamps $t_{2,i}$ of PRACH or UL reference signal reception and corresponding Resource IDs and QCL information if any; Time stamps $t_{3,i}$ of DL reference signal transmission (SSB/DL PRS/DL CSI-RS) and corresponding Resource IDs and QCL information if any; Request/trigger for additional positioning measurements and its type; Configuration or activation of additional set of DL/UL resource for transmission of positioning reference signals to conduct additional measurements of SLPs and their type; Location information associated with DL positioning reference resources, so that the UE can perform self-positioning.

In the second scenario, if the UE (IDLE/INACTIVE or CONNECTED) does not have any system information on DL PRS, the positioning procedure based on UL PRS transmission as depicted in FIG. 3 can be applied. Specifically, FIG. 3 depicts a flow diagram of an example method 300 of UL PRS-based positioning for 2-step RACH according to embodiments. At 302, the UE can provide a positioning request (MsgA) to a serving gNB. At 304, the serving gNB can request the gNB/Location Function entity to pro-

vide UL PRS configuration for the UE, and at **306**, the location function entity can provide the configuration to the serving gNB. At **308**, the serving gNB can indicate configuration to the UE via the Positioning Response carrier in MsgB or through any upper layer signaling based on RRC or LTE/NR positioning protocol (LPP/NPP-cellular positioning protocol) or through secure user-plane location (SUPL). The positioning response can contain at least one of the following fields: UL SRS/PRS resource set configuration for a UE requested positioning procedure (e.g., Information on UL SRS/PRS transmission resources (resource ID), their type (periodic/semi-persistent/aperiodic) and/or UL PRS/SRS sequence initialization information (sequence ID)); UL SRS/PRS measurement data (e.g., Information on timestamps of UL SRS/PRS signal reception as well as other SLP measured by each TRP/gNB and/or Information on geolocation coordinates of TRPs/gNBs that measured SLPs); Timestamp of MsgB transmission.

At **310**, the UE can provide the UL PRS to the serving gNB and/or one or more other gNBs. At **312**, all gNB measurements are reported to the Location Function, which provides the calculated coordinate back to the serving gNB at **314**. At **316**, the serving gNB can provide the calculated coordinate to a UE via RACH response with format MsgB or through upper layer signaling based on RRC or LTE/NR positioning protocol (LPP/NPP-cellular positioning protocol or SUPL).

In the third scenario, for an IDLE UE with system DL PRS information, the positioning procedure based on DL PRS can be accomplished as depicted in FIG. **4a**. Specifically, FIG. **4a** depicts a flow diagram of an example method **400** of DL PRS-based positioning for 2-step RACH. At **402**, the location function entity can provide DL PRS resource configuration information to the serving gNB and one or more other gNBs. At **404**, the serving gNB can provide the DL PRS configuration information to the UE. At **406**, the UE transmits to a serving gNB DL PRS measurement report along with a positioning request via MsgA or through any upper layer signaling based on RRC or LTE/NR positioning protocol (LPP/NPP-cellular positioning protocol) or through SUPL. At **408**, the gNB sends this report to the Location Function, which provides the calculated coordinate. At **410**, the calculated coordinate is sent to the serving gNB, and then at **412**, to the UE by the gNB via RACH response with format MsgB, or through any upper layer signaling based on RRC or LTE/NR positioning protocol (LPP/NPP-cellular positioning protocol) or through SUPL.

If the UE does not have any system DL PRS information, the positioning procedure based on DL PRS as shown in FIG. **4b** can be applied. Specifically, FIG. **4b** depicts a flow diagram of an example method **420** for providing DL PRS-based positioning for 2-step RACH. At **422**, the location function entity can provide configuration of DL/UL resources to a serving gNB and/or one or more other gNBs. At **424**, the UE can provide a position request to the serving gNB via MsgA or other signaling format. At **426**, the serving gNB transmits the DL PRS configuration to the UE via the Positioning Response in Msg-B format. The positioning response can contain at least one of the following fields: Configuration of DL PRS/CSI-RS resources for UE requested positioning procedure (e.g., DL PRS transmission resources and the ID (Resource set for DL positioning) and/or DL PRS sequence initialization parameters (sequence IDs)); DL PRS/CSI-RS measurement data (e.g., Information on timestamps of UL SRS/PRS signal reception as well as other SLP measured by each TRP/gNB and/or Information on geolocation coordinates of TRPs/gNBs that measured

SLPs); and Timestamp of MsgB transmission or any associated with MsgB DL PRS or CSI-RS signal.

At **428**, after Transmission of DL PRS, the UE transmits to the serving gNB DL PRS measurement report and, at **430**, the gNB sends this report to the Location Function/Server. At **432**, the gNB(s) provide the measurement report to the location function entity. At **434**, the location function entity provides the calculated coordinate to the serving gNB. At **436**, the calculated coordinate is sent to a UE via RACH response with format MsgB or other format.

In the fourth scenario, if a UE has DL PRS information, the positioning procedure based on UL+DL PRS can be achieved as shown in FIG. **5a**. Specifically, FIG. **5a** depicts a flow diagram of an example method **500** of UL and DL PRS based positioning for 2-step RACH. At **502** and **504**, the DL PRS resource configuration can be provided from the location function entity to the serving gNB, to one or more other gNBs, and then to the UE by the serving gNB and/or the other gNBs. At **506**, the UE can provide a MsgA positioning request to the serving gNB, which, at **508**, can request UL PRS configuration information from the location function entity. At **510**, the location function entity can provide the UL PRS configuration information to the serving gNB, and at **512**, the serving gNB can provide this information to the UE in a MsgB or through any upper layer signaling based on RRC or LTE/NR positioning protocol (LPP/NPP-cellular positioning protocol), or through SUPL. After Transmission of UL PRS by the UE at **514**, the gNB sends measurement reports based on UL PRS to the Location Function at **516**. At **518** the location function provides the calculated coordinate. At **520**, the calculated coordinate is sent to a UE via RACH response with format MsgB or through any upper layer signaling based on RRC or LTE/NR positioning protocol (LPP/NPP-cellular positioning protocol) or through SUPL.

If the UE does not have DL PRS information, the positioning procedure based on UL+DL PRS can be achieved as shown in FIG. **5b**. FIG. **5b** depicts a flow diagram of an example method **540** of providing UL and DL PRS based positioning for 2-step RACH according to embodiments. At **542**, the location function provides DL PRS configuration information to the serving gNB and/or other gNB(s). At **544**, the UE provides a positioning request via MsgA or other format. At **546**, the serving gNB requests the Location Function Node to provide UL PRS configuration for the particular UE, and the location function entity provides this information at **548**. At **550**, the serving gNB transmits the UL and DL PRS configuration to the UE via the Positioning Respond in MsgB format or through any upper layer signaling based on RRC or LTE/NR positioning protocol (LPP/NPP-cellular positioning protocol), or through SUPL. After transmission of UL and DL PRS at **552** and **554**, the UE transmits DL PRS measurement Report to the serving gNB at **556**. The gNB sends measurements reports based on UL and DL PRS to the Location Function at **558**, which provides the calculated coordinate at **560**. The calculated coordinate is sent to a UE at **562** via RACH response with format MsgB or through any upper layer signaling based on RRC or LTE/NR positioning protocol (LPP/NPP-cellular positioning protocol) or through SUPL.

While the 2-step PRACH based approach can serve as a mechanism to run NR positioning procedures, in another more general embodiment, the dedicated user plane and control plane signaling can also be used and defined to support UE positioning in NR systems. Therefore, corre-

sponding signaling and procedures can be generalized and implemented using generic NR procedures, signaling protocols and functionality.

One example implementation can use RRC (radio resource control)/MAC. In such embodiments, the positioning can include one or more of the following features: Configuration or activation of specific NR positioning procedure; Configuration of DL and UL resources and resource types used for positioning measurements as well as reference signals IDs; Association of DL and UL resources and reference; Configuration of UE DL measurement reports for positioning; Signaling of any assistance information to simplify or improve measurements; Activation or deactivation of UE DL measurements for positioning; Activation or deactivation of UE reports for positioning; Activation or deactivation of UE UL reference signals transmission for positioning.

Another example implementation can use LPP/NPP (cellular positioning protocols). In such embodiments, the positioning can include one or more of the following features: Configuration or activation of specific NR positioning procedure; Configuration of DL and UL resources and resource types used for positioning measurements as well as reference signals IDs; Configuration of UE DL measurement reports for positioning; Signaling of any assistance information to simplify or improve measurements; Activation or deactivation of UE DL measurements for positioning; Activation or deactivation of UE reports for positioning; Activation or deactivation of UE UL reference signals transmission for positioning.

Another example implementation can use SUPL (Secure User Plane Positioning Protocol). In such embodiments, the positioning can include one or more of the following features: Configuration or activation of specific NR positioning procedure; Configuration of DL and UL resources and resource types used for positioning measurements as well as reference signals IDs; Configuration of UE DL measurement reports for positioning; Signaling of any assistance information to simplify or improve measurements; Activation or deactivation of UE DL measurements for positioning; Activation or deactivation of UE reports for positioning; Activation or deactivation of UE UL reference signals transmission for positioning.

Another example implementation can use Physical layer signaling such as DCI. In such embodiments, the positioning can include one or more of the following features: Activation or deactivation of UE DL measurements for positioning; Activation or deactivation of UE reports for positioning; Activation or deactivation of UE UL reference signals transmission for positioning.

Independently of the actual signaling being used in the two subsequent charts, NR positioning procedures and signaling may be generalized in terms of UE/gNB behavior for transmission of reference signals and measurements for two practical cases: Case 1: A UE does not have a priori DL measurements based on previous DL PRS/CSI-RS/SSB transmission. The UE transmits UL reference signals for positioning before conducting measurements on DL reference signals. The method of case 1 is described in method 600, depicted in FIG. 6. At (optional) 602, The UE requests gNB/NW/LMF to allocate UL resources for UL positioning reference signals (UL PRACH, UL SRS, UL PRS) or both UL positioning reference signals and UL shared channel transmission. At 604, the gNB/NW/LMF indicates allocation of UL positioning resources for UL positioning reference signals (UL PRACH, UL SRS, UL PRS) or both UL positioning reference signals and UL shared channel trans-

mission and informs/coordinates this allocation with neighboring gNBs/TRPs and triggers UE UL transmission on allocated resources. At 606, the UE transmits UL positioning reference signals (UL PRACH, UL SRS, UL PRS) or both UL positioning reference signals and UL shared channel transmission. The UE can indicate timestamp of transmission timing of UL reference signals in the UL shared channel transmission.

At 608, the gNBs/TRPs perform measurement of signal location parameters based on UE UL transmission, and select UL RX beams to serve UE. The UL RSRP criteria is measured on configured UL positioning resources (e.g., maximum RSRP power or maximum power of the first arrival path can be used). At 610, the gNB configures or activates the set of DL PRS/DL CSI-RS signals for positioning measurements and triggers UE measurements of signal location parameters (e.g., FAP timing corresponding to each configured DL PRS/CSI-RS resource). At the same time, the gNBs can signal to UE or any other network entity result of measurements of signal location parameters obtained based on UL reference signal transmission. Also, the gNB can signal timestamps of DL PRS/CSI-RS transmissions for each resource and location coordinates associated with configured DL PRS/CSI-RS resources.

At 612, the gNB transmits DL PRS/CSI-RS signals for UE positioning measurements and applies the selected RX beams for transmission of DL PRS/CSI-RS signals. At 614, the UE performs measurements of signal location parameters for each DL-PRS/DL CSI-RS resource. At 616, the UE performs self-positioning by using measured signal location parameters and location information provided by the gNB. The UE can report measurements of signal location parameters and estimated coordinates to the gNB, etc.

In Case 2, a UE has a priori DL measurements based on previous DL PRS/CSI-RS/SSB transmission. The UE processes DL reference signals for positioning before transmission of UL reference signals for positioning. Case 2 is described in FIG. 7, which depicts an example method 700 according to embodiments. At 702, the UE requests the gNB, etc. to allocate DL resources for DL positioning reference signals, or both DL positioning reference signals and DL shared channel transmission. At 704, the gNB etc. indicates to UE allocation of DL resources for DL positioning reference signals and/or DL shared channel transmission and triggers measurements of signal location parameters on allocated resources by DCI. The UE can also autonomously perform the DL measurements. At 706, the UE measures signal location parameters using DL reference signals. At 708, the UE can report to gNB the subset of DL reference resource IDs recommended for DL positioning measurements. At 710, the gNB can provide configuration to UE that indicate UL resources or resource IDs for transmission of UL reference signals and associated DL reference resource IDs, or to set correspondence between the UL reference resources and DL reference resources.

At 712, the UE transmits UL positioning reference signals and/or UL shared channel transmission using the best UL beam for each DL positioning resource. At 714, gNB etc. performs measurement of signal location parameters based on UE UL transmission of reference signals and applying DL RX beams associated with the predefined UL positioning resource. At 716, the gNB transfers measurements for UE positioning to UE so that UE position estimate can be made.

The processes and functions in FIGS. 6-7 can be performed by one or more of application circuitry 1105 or 1205, baseband circuitry 1110 or 1210, or processors 1710.

Systems and Implementations

FIG. 8 illustrates an example architecture of a system **800** of a network, in accordance with various embodiments. The following description is provided for an example system **800** that operates in conjunction with the LTE system standards and 5G or NR system standards as provided by 3GPP technical specifications. However, the example embodiments are not limited in this regard and the described embodiments may apply to other networks that benefit from the principles described herein, such as future 3GPP systems (e.g., Sixth Generation (6G)) systems, IEEE 802.16 protocols (e.g., WMAN, WiMAX, etc.), or the like.

As shown by FIG. 8, the system **800** includes UE **801a** and UE **801b** (collectively referred to as “UEs **801**” or “UE **801**”). In this example, UEs **801** are illustrated as smartphones (e.g., handheld touchscreen mobile computing devices connectable to one or more cellular networks), but may also comprise any mobile or non-mobile computing device, such as consumer electronics devices, cellular phones, smartphones, feature phones, tablet computers, wearable computer devices, personal digital assistants (PDAs), pagers, wireless handsets, desktop computers, laptop computers, in-vehicle infotainment (IVI), in-car entertainment (ICE) devices, an Instrument Cluster (IC), head-up display (HUD) devices, onboard diagnostic (OBD) devices, dashtop mobile equipment (DME), mobile data terminals (MDTs), Electronic Engine Management System (EEMS), electronic/engine control units (ECUs), electronic/engine control modules (ECMs), embedded systems, microcontrollers, control modules, engine management systems (EMS), networked or “smart” appliances, MTC devices, M2M, IoT devices, and/or the like.

In some embodiments, any of the UEs **801** may be IoT UEs, which may comprise a network access layer designed for low-power IoT applications utilizing short-lived UE connections. An IoT UE can utilize technologies such as M2M or MTC for exchanging data with an MTC server or device via a PLMN, ProSe or D2D communication, sensor networks, or IoT networks. The M2M or MTC exchange of data may be a machine-initiated exchange of data. An IoT network describes interconnecting IoT UEs, which may include uniquely identifiable embedded computing devices (within the Internet infrastructure), with short-lived connections. The IoT UEs may execute background applications (e.g., keep-alive messages, status updates, etc.) to facilitate the connections of the IoT network.

The UEs **801** may be configured to connect, for example, communicatively couple, with an or RAN **810**. In embodiments, the RAN **810** may be an NG RAN or a 5G RAN, an E-UTRAN, or a legacy RAN, such as a UTRAN or GERAN. As used herein, the term “NG RAN” or the like may refer to a RAN **810** that operates in an NR or 5G system **800**, and the term “E-UTRAN” or the like may refer to a RAN **810** that operates in an LTE or 4G system **800**. The UEs **801** utilize connections (or channels) **803** and **804**, respectively, each of which comprises a physical communications interface or layer (discussed in further detail below).

In this example, the connections **803** and **804** are illustrated as an air interface to enable communicative coupling, and can be consistent with cellular communications protocols, such as a GSM protocol, a CDMA network protocol, a PTT protocol, a POC protocol, a UMTS protocol, a 3GPP LTE protocol, a 5G protocol, a NR protocol, and/or any of the other communications protocols discussed herein. In embodiments, the UEs **801** may directly exchange communication data via a ProSe interface **805**. The ProSe interface **805** may alternatively be referred to as a SL interface **805**

and may comprise one or more logical channels, including but not limited to a PSCCH, a PSSCH, a PSDCH, and a PSBCH.

The UE **801b** is shown to be configured to access an AP **806** (also referred to as “WLAN node **806**,” “WLAN **806**,” “WLAN Termination **806**,” “WT **806**” or the like) via connection **807**. The connection **807** can comprise a local wireless connection, such as a connection consistent with any IEEE 802.11 protocol, wherein the AP **806** would comprise a wireless fidelity (Wi-Fi®) router. In this example, the AP **806** is shown to be connected to the Internet without connecting to the core network of the wireless system (described in further detail below). In various embodiments, the UE **801b**, RAN **810**, and AP **806** may be configured to utilize LWA operation and/or LWIP operation. The LWA operation may involve the UE **801b** in RRC_CONNECTED being configured by a RAN node **811a-b** to utilize radio resources of LTE and WLAN. LWIP operation may involve the UE **801b** using WLAN radio resources (e.g., connection **807**) via IPsec protocol tunneling to authenticate and encrypt packets (e.g., IP packets) sent over the connection **807**. IPsec tunneling may include encapsulating the entirety of original IP packets and adding a new packet header, thereby protecting the original header of the IP packets.

The RAN **810** can include one or more AN nodes or RAN nodes **811a** and **811b** (collectively referred to as “RAN nodes **811**” or “RAN node **811**”) that enable the connections **803** and **804**. As used herein, the terms “access node,” “access point,” or the like may describe equipment that provides the radio baseband functions for data and/or voice connectivity between a network and one or more users. These access nodes can be referred to as BS, gNBs, RAN nodes, eNBs, NodeBs, RSUs, TRxPs or TRPs, and so forth, and can comprise ground stations (e.g., terrestrial access points) or satellite stations providing coverage within a geographic area (e.g., a cell). As used herein, the term “NG RAN node” or the like may refer to a RAN node **811** that operates in an NR or 5G system **800** (for example, a gNB), and the term “E-UTRAN node” or the like may refer to a RAN node **811** that operates in an LTE or 4G system **800** (e.g., an eNB). According to various embodiments, the RAN nodes **811** may be implemented as one or more of a dedicated physical device such as a macrocell base station, and/or a low power (LP) base station for providing femtocells, picocells or other like cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells.

In some embodiments, all or parts of the RAN nodes **811** may be implemented as one or more software entities running on server computers as part of a virtual network, which may be referred to as a CRAN and/or a virtual baseband unit pool (vBBUP). In these embodiments, the CRAN or vBBUP may implement a RAN function split, such as a PDCP split wherein RRC and PDCP layers are operated by the CRAN/vBBUP and other L2 protocol entities are operated by individual RAN nodes **811**; a MAC/PHY split wherein RRC, PDCP, RLC, and MAC layers are operated by the CRAN/vBBUP and the PHY layer is operated by individual RAN nodes **811**; or a “lower PHY” split wherein RRC, PDCP, RLC, MAC layers and upper portions of the PHY layer are operated by the CRAN/vBBUP and lower portions of the PHY layer are operated by individual RAN nodes **811**. This virtualized framework allows the freed-up processor cores of the RAN nodes **811** to perform other virtualized applications. In some embodiments, an individual RAN node **811** may represent individual gNB-

DUs that are connected to a gNB-CU via individual F1 interfaces (not shown by FIG. 8). In these embodiments, the gNB-DUs may include one or more remote radio heads or RFEMs (see, e.g., FIG. 11), and the gNB-CU may be operated by a server that is located in the RAN 810 (not shown) or by a server pool in a similar manner as the CRAN/vBBUP. Additionally or alternatively, one or more of the RAN nodes 811 may be next generation eNBs (ng-eNBs), which are RAN nodes that provide E-UTRA user plane and control plane protocol terminations toward the UEs 801, and are connected to a 5GC (e.g., CN 1020 of FIG. 10) via an NG interface (discussed infra).

In V2X scenarios one or more of the RAN nodes 811 may be or act as RSUs. The term “Road Side Unit” or “RSU” may refer to any transportation infrastructure entity used for V2X communications. An RSU may be implemented in or by a suitable RAN node or a stationary (or relatively stationary) UE, where an RSU implemented in or by a UE may be referred to as a “UE-type RSU,” an RSU implemented in or by an eNB may be referred to as an “eNB-type RSU,” an RSU implemented in or by a gNB may be referred to as a “gNB-type RSU,” and the like. In one example, an RSU is a computing device coupled with radio frequency circuitry located on a roadside that provides connectivity support to passing vehicle UEs 801 (vUEs 801). The RSU may also include internal data storage circuitry to store intersection map geometry, traffic statistics, media, as well as applications/software to sense and control ongoing vehicular and pedestrian traffic. The RSU may operate on the 5.9 GHz Direct Short Range Communications (DSRC) band to provide very low latency communications required for high speed events, such as crash avoidance, traffic warnings, and the like. Additionally or alternatively, the RSU may operate on the cellular V2X band to provide the aforementioned low latency communications, as well as other cellular communications services. Additionally or alternatively, the RSU may operate as a Wi-Fi hotspot (2.4 GHz band) and/or provide connectivity to one or more cellular networks to provide uplink and downlink communications. The computing device(s) and some or all of the radiofrequency circuitry of the RSU may be packaged in a weatherproof enclosure suitable for outdoor installation, and may include a network interface controller to provide a wired connection (e.g., Ethernet) to a traffic signal controller and/or a backhaul network.

Any of the RAN nodes 811 can terminate the air interface protocol and can be the first point of contact for the UEs 801. In some embodiments, any of the RAN nodes 811 can fulfill various logical functions for the RAN 810 including, but not limited to, radio network controller (RNC) functions such as radio bearer management, uplink and downlink dynamic radio resource management and data packet scheduling, and mobility management.

In embodiments, the UEs 801 can be configured to communicate using OFDM communication signals with each other or with any of the RAN nodes 811 over a multicarrier communication channel in accordance with various communication techniques, such as, but not limited to, an OFDMA communication technique (e.g., for downlink communications) or a SC-FDMA communication technique (e.g., for uplink and ProSe or sidelink communications), although the scope of the embodiments is not limited in this respect. The OFDM signals can comprise a plurality of orthogonal subcarriers.

In some embodiments, a downlink resource grid can be used for downlink transmissions from any of the RAN nodes 811 to the UEs 801, while uplink transmissions can utilize

similar techniques. The grid can be a time-frequency grid, called a resource grid or time-frequency resource grid, which is the physical resource in the downlink in each slot. Such a time-frequency plane representation is a common practice for OFDM systems, which makes it intuitive for radio resource allocation. Each column and each row of the resource grid corresponds to one OFDM symbol and one OFDM subcarrier, respectively. The duration of the resource grid in the time domain corresponds to one slot in a radio frame. The smallest time-frequency unit in a resource grid is denoted as a resource element. Each resource grid comprises a number of resource blocks, which describe the mapping of certain physical channels to resource elements. Each resource block comprises a collection of resource elements; in the frequency domain, this may represent the smallest quantity of resources that currently can be allocated. There are several different physical downlink channels that are conveyed using such resource blocks.

According to various embodiments, the UEs 801 and the RAN nodes 811 communicate data (for example, transmit and receive) data over a licensed medium (also referred to as the “licensed spectrum” and/or the “licensed band”) and an unlicensed shared medium (also referred to as the “unlicensed spectrum” and/or the “unlicensed band”). The licensed spectrum may include channels that operate in the frequency range of approximately 400 MHz to approximately 3.8 GHz, whereas the unlicensed spectrum may include the 5 GHz band.

To operate in the unlicensed spectrum, the UEs 801 and the RAN nodes 811 may operate using LAA, eLAA, and/or feLAA mechanisms. In these embodiments, the UEs 801 and the RAN nodes 811 may perform one or more known medium-sensing operations and/or carrier-sensing operations in order to determine whether one or more channels in the unlicensed spectrum is unavailable or otherwise occupied prior to transmitting in the unlicensed spectrum. The medium/carrier sensing operations may be performed according to a listen-before-talk (LBT) protocol.

LBT is a mechanism whereby equipment (for example, UEs 801 RAN nodes 811 etc.) senses a medium (for example, a channel or carrier frequency) and transmits when the medium is sensed to be idle (or when a specific channel in the medium is sensed to be unoccupied). The medium sensing operation may include CCA, which utilizes at least ED to determine the presence or absence of other signals on a channel in order to determine if a channel is occupied or clear. This LBT mechanism allows cellular/LAA networks to coexist with incumbent systems in the unlicensed spectrum and with other LAA networks. ED may include sensing RF energy across an intended transmission band for a period of time and comparing the sensed RF energy to a predefined or configured threshold.

Typically, the incumbent systems in the 5 GHz band are WLANs based on IEEE 802.11 technologies. WLAN employs a contention-based channel access mechanism, called CSMA/CA. Here, when a WLAN node (e.g., a mobile station (MS) such as UE 801, AP 806, or the like) intends to transmit, the WLAN node may first perform CCA before transmission. Additionally, a backoff mechanism is used to avoid collisions in situations where more than one WLAN node senses the channel as idle and transmits at the same time. The backoff mechanism may be a counter that is drawn randomly within the CWS, which is increased exponentially upon the occurrence of collision and reset to a minimum value when the transmission succeeds. The LBT mechanism designed for LAA is somewhat similar to the CSMA/CA of WLAN. In some embodiments, the LBT procedure for DL

or UL transmission bursts including PDSCH or PUSCH transmissions, respectively, may have an LAA contention window that is variable in length between X and Y ECCA slots, where X and Y are minimum and maximum values for the CWSs for LAA. In one example, the minimum CWS for an LAA transmission may be 9 microseconds (μ s); however, the size of the CWS and a MCOT (for example, a transmission burst) may be based on governmental regulatory requirements.

The LAA mechanisms are built upon CA technologies of LTE-Advanced systems. In CA, each aggregated carrier is referred to as a CC. A CC may have a bandwidth of 1.4, 3, 5, 10, 15 or 20 MHz and a maximum of five CCs can be aggregated, and therefore, a maximum aggregated bandwidth is 100 MHz. In FDD systems, the number of aggregated carriers can be different for DL and UL, where the number of UL CCs is equal to or lower than the number of DL component carriers. In some cases, individual CCs can have a different bandwidth than other CCs. In TDD systems, the number of CCs as well as the bandwidths of each CC is usually the same for DL and UL.

CA also comprises individual serving cells to provide individual CCs. The coverage of the serving cells may differ, for example, because CCs on different frequency bands will experience different path loss. A primary service cell or PCell may provide a PCC for both UL and DL, and may handle RRC and NAS related activities. The other serving cells are referred to as SCells, and each SCell may provide an individual SCC for both UL and DL. The SCCs may be added and removed as required, while changing the PCC may require the UE 801 to undergo a handover. In LAA, eLAA, and fLAA, some or all of the SCells may operate in the unlicensed spectrum (referred to as "LAA SCells"), and the LAA SCells are assisted by a PCell operating in the licensed spectrum. When a UE is configured with more than one LAA SCell, the UE may receive UL grants on the configured LAA SCells indicating different PUSCH starting positions within a same subframe.

The PDSCH carries user data and higher-layer signaling to the UEs 801. The PDCCH carries information about the transport format and resource allocations related to the PDSCH channel, among other things. It may also inform the UEs 801 about the transport format, resource allocation, and HARQ information related to the uplink shared channel. Typically, downlink scheduling (assigning control and shared channel resource blocks to the UE 801b within a cell) may be performed at any of the RAN nodes 811 based on channel quality information fed back from any of the UEs 801. The downlink resource assignment information may be sent on the PDCCH used for (e.g., assigned to) each of the UEs 801.

The PDCCH uses CCEs to convey the control information. Before being mapped to resource elements, the PDCCH complex-valued symbols may first be organized into quadruplets, which may then be permuted using a sub-block interleaver for rate matching. Each PDCCH may be transmitted using one or more of these CCEs, where each CCE may correspond to nine sets of four physical resource elements known as REGs. Four Quadrature Phase Shift Keying (QPSK) symbols may be mapped to each REG. The PDCCH can be transmitted using one or more CCEs, depending on the size of the DCI and the channel condition. There can be four or more different PDCCH formats defined in LTE with different numbers of CCEs (e.g., aggregation level, $L=1, 2, 4$, or 8).

Some embodiments may use concepts for resource allocation for control channel information that are an extension

of the above-described concepts. For example, some embodiments may utilize an EPDCCH that uses PDSCH resources for control information transmission. The EPDCCH may be transmitted using one or more ECCEs. Similar to above, each ECCE may correspond to nine sets of four physical resource elements known as an EREGs. An ECCE may have other numbers of EREGs in some situations.

The RAN nodes 811 may be configured to communicate with one another via interface 812. In embodiments where the system 800 is an LTE system (e.g., when CN 820 is an EPC 920 as in FIG. 9), the interface 812 may be an X2 interface 812. The X2 interface may be defined between two or more RAN nodes 811 (e.g., two or more eNBs and the like) that connect to EPC 820, and/or between two eNBs connecting to EPC 820. In some embodiments, the X2 interface may include an X2 user plane interface (X2-U) and an X2 control plane interface (X2-C). The X2-U may provide flow control mechanisms for user data packets transferred over the X2 interface, and may be used to communicate information about the delivery of user data between eNBs. For example, the X2-U may provide specific sequence number information for user data transferred from a MeNB to an SeNB; information about successful in sequence delivery of PDCP PDUs to a UE 801 from an SeNB for user data; information of PDCP PDUs that were not delivered to a UE 801; information about a current minimum desired buffer size at the SeNB for transmitting to the UE user data; and the like. The X2-C may provide intra-LTE access mobility functionality, including context transfers from source to target eNBs, user plane transport control, etc.; load management functionality; as well as inter-cell interference coordination functionality.

In embodiments where the system 800 is a 5G or NR system (e.g., when CN 820 is a 5GC 1020 as in FIG. 10), the interface 812 may be an Xn interface 812. The Xn interface is defined between two or more RAN nodes 811 (e.g., two or more gNBs and the like) that connect to 5GC 820, between a RAN node 811 (e.g., a gNB) connecting to 5GC 820 and an eNB, and/or between two eNBs connecting to 5GC 820. In some embodiments, the Xn interface may include an Xn user plane (Xn-U) interface and an Xn control plane (Xn-C) interface. The Xn-U may provide non-guaranteed delivery of user plane PDUs and support/provide data forwarding and flow control functionality. The Xn-C may provide management and error handling functionality, functionality to manage the Xn-C interface; mobility support for UE 801 in a connected mode (e.g., CM-CONNECTED) including functionality to manage the UE mobility for connected mode between one or more RAN nodes 811. The mobility support may include context transfer from an old (source) serving RAN node 811 to new (target) serving RAN node 811; and control of user plane tunnels between old (source) serving RAN node 811 to new (target) serving RAN node 811. A protocol stack of the Xn-U may include a transport network layer built on Internet Protocol (IP) transport layer, and a GTP-U layer on top of a UDP and/or IP layer(s) to carry user plane PDUs. The Xn-C protocol stack may include an application layer signaling protocol (referred to as Xn Application Protocol (Xn-AP)) and a transport network layer that is built on SCTP. The SCTP may be on top of an IP layer, and may provide the guaranteed delivery of application layer messages. In the transport IP layer, point-to-point transmission is used to deliver the signaling PDUs. In other embodiments, the Xn-U protocol stack

and/or the Xn-C protocol stack may be same or similar to the user plane and/or control plane protocol stack(s) shown and described herein.

The RAN **810** is shown to be communicatively coupled to a core network—in this embodiment, core network (CN) **820**. The CN **820** may comprise a plurality of network elements **822**, which are configured to offer various data and telecommunications services to customers/subscribers (e.g., users of UEs **801**) who are connected to the CN **820** via the RAN **810**. The components of the CN **820** may be implemented in one physical node or separate physical nodes including components to read and execute instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium). In some embodiments, NFV may be utilized to virtualize any or all of the above-described network node functions via executable instructions stored in one or more computer-readable storage mediums (described in further detail below). A logical instantiation of the CN **820** may be referred to as a network slice, and a logical instantiation of a portion of the CN **820** may be referred to as a network sub-slice. NFV architectures and infrastructures may be used to virtualize one or more network functions, alternatively performed by proprietary hardware, onto physical resources comprising a combination of industry-standard server hardware, storage hardware, or switches. In other words, NFV systems can be used to execute virtual or reconfigurable embodiments of one or more EPC components/functions.

Generally, the application server **830** may be an element offering applications that use IP bearer resources with the core network (e.g., UMTS PS domain, LTE PS data services, etc.). The application server **830** can also be configured to support one or more communication services (e.g., VoIP sessions, PTT sessions, group communication sessions, social networking services, etc.) for the UEs **801** via the EPC **820**.

In embodiments, the CN **820** may be a 5GC (referred to as “5GC **820**” or the like), and the RAN **810** may be connected with the CN **820** via an NG interface **813**. In embodiments, the NG interface **813** may be split into two parts, an NG user plane (NG-U) interface **814**, which carries traffic data between the RAN nodes **811** and a UPF, and the S1 control plane (NG-C) interface **815**, which is a signaling interface between the RAN nodes **811** and AMFs. Embodiments where the CN **820** is a 5GC **820** are discussed in more detail with regard to FIG. 10.

In embodiments, the CN **820** may be a 5G CN (referred to as “5G CN **820**” or the like), while in other embodiments, the CN **820** may be an EPC. Where CN **820** is an EPC (referred to as “EPC **820**” or the like), the RAN **810** may be connected with the CN **820** via an S1 interface **813**. In embodiments, the S1 interface **813** may be split into two parts, an S1 user plane (S1-U) interface **814**, which carries traffic data between the RAN nodes **811** and the S-GW, and the S1-MME interface **815**, which is a signaling interface between the RAN nodes **811** and MMEs. An example architecture wherein the CN **820** is an EPC **820** is shown by FIG. 9.

FIG. 9 illustrates an example architecture of a system **900** including a first CN **920**, in accordance with various embodiments. In this example, system **900** may implement the LTE standard wherein the CN **920** is an EPC **920** that corresponds with CN **820** of FIG. 8. Additionally, the UE **901** may be the same or similar as the UEs **801** of FIG. 8, and the E-UTRAN **910** may be a RAN that is the same or similar to the RAN **810** of FIG. 8, and which may include

RAN nodes **811** discussed previously. The CN **920** may comprise MMEs **921**, an S-GW **922**, a P-GW **923**, a HSS **924**, and a SGSN **925**.

The MMEs **921** may be similar in function to the control plane of legacy SGSN, and may implement MM functions to keep track of the current location of a UE **901**. The MMEs **921** may perform various MM procedures to manage mobility aspects in access such as gateway selection and tracking area list management. MM (also referred to as “EPS MM” or “EMM” in E-UTRAN systems) may refer to all applicable procedures, methods, data storage, etc. that are used to maintain knowledge about a present location of the UE **901**, provide user identity confidentiality, and/or perform other like services to users/subscribers. Each UE **901** and the MME **921** may include an MM or EMM sublayer, and an MM context may be established in the UE **901** and the MME **921** when an attach procedure is successfully completed. The MM context may be a data structure or database object that stores MM-related information of the UE **901**. The MMEs **921** may be coupled with the HSS **924** via an S6a reference point, coupled with the SGSN **925** via an S3 reference point, and coupled with the S-GW **922** via an S11 reference point.

The SGSN **925** may be a node that serves the UE **901** by tracking the location of an individual UE **901** and performing security functions. In addition, the SGSN **925** may perform Inter-EPC node signaling for mobility between 2G/3G and E-UTRAN 3GPP access networks; PDN and S-GW selection as specified by the MMEs **921**; handling of UE **901** time zone functions as specified by the MMEs **921**; and MME selection for handovers to E-UTRAN 3GPP access network. The S3 reference point between the MMEs **921** and the SGSN **925** may enable user and bearer information exchange for inter-3GPP access network mobility in idle and/or active states.

The HSS **924** may comprise a database for network users, including subscription-related information to support the network entities’ handling of communication sessions. The EPC **920** may comprise one or several HSSs **924**, depending on the number of mobile subscribers, on the capacity of the equipment, on the organization of the network, etc. For example, the HSS **924** can provide support for routing/roaming, authentication, authorization, naming/addressing resolution, location dependencies, etc. An S6a reference point between the HSS **924** and the MMEs **921** may enable transfer of subscription and authentication data for authenticating/authorizing user access to the EPC **920** between HSS **924** and the MMEs **921**.

The S-GW **922** may terminate the S1 interface **813** (“S1-U” in FIG. 9) toward the RAN **910**, and routes data packets between the RAN **910** and the EPC **920**. In addition, the S-GW **922** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other responsibilities may include lawful intercept, charging, and some policy enforcement. The S11 reference point between the S-GW **922** and the MMEs **921** may provide a control plane between the MMEs **921** and the S-GW **922**. The S-GW **922** may be coupled with the P-GW **923** via an S5 reference point.

The P-GW **923** may terminate an Sgi interface toward a PDN **930**. The P-GW **923** may route data packets between the EPC **920** and external networks such as a network including the application server **830** (alternatively referred to as an “AF”) via an IP interface **825** (see e.g., FIG. 8). In embodiments, the P-GW **923** may be communicatively coupled to an application server (application server **830** of FIG. 8 or PDN **930** in FIG. 9) via an IP communications

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interface **825** (see, e.g., FIG. **8**). The S5 reference point between the P-GW **923** and the S-GW **922** may provide user plane tunneling and tunnel management between the P-GW **923** and the S-GW **922**. The S5 reference point may also be used for S-GW **922** relocation due to UE **901** mobility and if the S-GW **922** needs to connect to a non-collocated P-GW **923** for the required PDN connectivity. The P-GW **923** may further include a node for policy enforcement and charging data collection (e.g., PCEF (not shown)). Additionally, the SGi reference point between the P-GW **923** and the packet data network (PDN) **930** may be an operator external public, a private PDN, or an intra operator packet data network, for example, for provision of IMS services. The P-GW **923** may be coupled with a PCRF **926** via a Gx reference point.

PCRF **926** is the policy and charging control element of the EPC **920**. In a non-roaming scenario, there may be a single PCRF **926** in the Home Public Land Mobile Network (HPLMN) associated with a UE **901**'s Internet Protocol Connectivity Access Network (IP-CAN) session. In a roaming scenario with local breakout of traffic, there may be two PCRFs associated with a UE **901**'s IP-CAN session, a Home PCRF (H-PCRF) within an HPLMN and a Visited PCRF (V-PCRF) within a Visited Public Land Mobile Network (VPLMN). The PCRF **926** may be communicatively coupled to the application server **930** via the P-GW **923**. The application server **930** may signal the PCRF **926** to indicate a new service flow and select the appropriate QoS and charging parameters. The PCRF **926** may provision this rule into a PCEF (not shown) with the appropriate TFT and QCI, which commences the QoS and charging as specified by the application server **930**. The Gx reference point between the PCRF **926** and the P-GW **923** may allow for the transfer of QoS policy and charging rules from the PCRF **926** to PCEF in the P-GW **923**. An Rx reference point may reside between the PDN **930** (or "AF **930**") and the PCRF **926**.

FIG. **10** illustrates an architecture of a system **1000** including a second CN **1020** in accordance with various embodiments. The system **1000** is shown to include a UE **1001**, which may be the same or similar to the UEs **801** and UE **901** discussed previously; a (R)AN **1010**, which may be the same or similar to the RAN **810** and RAN **910** discussed previously, and which may include RAN nodes **811** discussed previously; and a DN **1003**, which may be, for example, operator services, Internet access or 3rd party services; and a 5GC **1020**. The 5GC **1020** may include an AUSF **1022**; an AMF **1021**; a SMF **1024**; a NEF **1023**; a PCF **1026**; a NRF **1025**; a UDM **1027**; an AF **1028**; a UPF **1002**; and a NSSF **1029**.

The UPF **1002** may act as an anchor point for intra-RAT and inter-RAT mobility, an external PDU session point of interconnect to DN **1003**, and a branching point to support multi-homed PDU session. The UPF **1002** may also perform packet routing and forwarding, perform packet inspection, enforce the user plane part of policy rules, lawfully intercept packets (UP collection), perform traffic usage reporting, perform QoS handling for a user plane (e.g., packet filtering, gating, UL/DL rate enforcement), perform Uplink Traffic verification (e.g., SDF to QoS flow mapping), transport level packet marking in the uplink and downlink, and perform downlink packet buffering and downlink data notification triggering. UPF **1002** may include an uplink classifier to support routing traffic flows to a data network. The DN **1003** may represent various network operator services, Internet access, or third party services. DN **1003** may include, or be similar to, application server **830** discussed previously. The UPF **1002** may interact with the SMF **1024** via an N4 reference point between the SMF **1024** and the UPF **1002**.

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The AUSF **1022** may store data for authentication of UE **1001** and handle authentication-related functionality. The AUSF **1022** may facilitate a common authentication framework for various access types. The AUSF **1022** may communicate with the AMF **1021** via an N12 reference point between the AMF **1021** and the AUSF **1022**; and may communicate with the UDM **1027** via an N13 reference point between the UDM **1027** and the AUSF **1022**. Additionally, the AUSF **1022** may exhibit an Nausf service-based interface.

The AMF **1021** may be responsible for registration management (e.g., for registering UE **1001**, etc.), connection management, reachability management, mobility management, and lawful interception of AMF-related events, and access authentication and authorization. The AMF **1021** may be a termination point for the an N11 reference point between the AMF **1021** and the SMF **1024**. The AMF **1021** may provide transport for SM messages between the UE **1001** and the SMF **1024**, and act as a transparent proxy for routing SM messages. AMF **1021** may also provide transport for SMS messages between UE **1001** and an SMSF (not shown by FIG. **10**). AMF **1021** may act as SEAF, which may include interaction with the AUSF **1022** and the UE **1001**, receipt of an intermediate key that was established as a result of the UE **1001** authentication process. Where USIM based authentication is used, the AMF **1021** may retrieve the security material from the AUSF **1022**. AMF **1021** may also include a SCM function, which receives a key from the SEA that it uses to derive access-network specific keys. Furthermore, AMF **1021** may be a termination point of a RAN CP interface, which may include or be an N2 reference point between the (R)AN **1010** and the AMF **1021**; and the AMF **1021** may be a termination point of NAS (N1) signalling, and perform NAS ciphering and integrity protection.

AMF **1021** may also support NAS signalling with a UE **1001** over an N3 IWF interface. The N3IWF may be used to provide access to untrusted entities. N3IWF may be a termination point for the N2 interface between the (R)AN **1010** and the AMF **1021** for the control plane, and may be a termination point for the N3 reference point between the (R)AN **1010** and the UPF **1002** for the user plane. As such, the AMF **1021** may handle N2 signalling from the SMF **1024** and the AMF **1021** for PDU sessions and QoS, encapsulate/de-encapsulate packets for IPsec and N3 tunnelling, mark N3 user-plane packets in the uplink, and enforce QoS corresponding to N3 packet marking taking into account QoS requirements associated with such marking received over N2. N3IWF may also relay uplink and downlink control-plane NAS signalling between the UE **1001** and AMF **1021** via an N1 reference point between the UE **1001** and the AMF **1021**, and relay uplink and downlink user-plane packets between the UE **1001** and UPF **1002**. The N3IWF also provides mechanisms for IPsec tunnel establishment with the UE **1001**. The AMF **1021** may exhibit an Namf service-based interface, and may be a termination point for an N14 reference point between two AMFs **1021** and an N17 reference point between the AMF **1021** and a 5G-EIR (not shown by FIG. **10**).

The UE **1001** may need to register with the AMF **1021** in order to receive network services. RM is used to register or deregister the UE **1001** with the network (e.g., AMF **1021**), and establish a UE context in the network (e.g., AMF **1021**). The UE **1001** may operate in an RM-REGISTERED state or an RM-DEREGISTERED state. In the RM-DEREGISTERED state, the UE **1001** is not registered with the network, and the UE context in AMF **1021** holds no valid location or routing information for the UE **1001** so the UE

1001 is not reachable by the AMF **1021**. In the RM-REGISTERED state, the UE **1001** is registered with the network, and the UE context in AMF **1021** may hold a valid location or routing information for the UE **1001** so the UE **1001** is reachable by the AMF **1021**. In the RM-REGISTERED state, the UE **1001** may perform mobility Registration Update procedures, perform periodic Registration Update procedures triggered by expiration of the periodic update timer (e.g., to notify the network that the UE **1001** is still active), and perform a Registration Update procedure to update UE capability information or to re-negotiate protocol parameters with the network, among others.

The AMF **1021** may store one or more RM contexts for the UE **1001**, where each RM context is associated with a specific access to the network. The RM context may be a data structure, database object, etc. that indicates or stores, inter alia, a registration state per access type and the periodic update timer. The AMF **1021** may also store a 5GC MM context that may be the same or similar to the (E)MM context discussed previously. In various embodiments, the AMF **1021** may store a CE mode B Restriction parameter of the UE **1001** in an associated MM context or RM context. The AMF **1021** may also derive the value, when needed, from the UE's usage setting parameter already stored in the UE context (and/or MM/RM context).

CM may be used to establish and release a signaling connection between the UE **1001** and the AMF **1021** over the N1 interface. The signaling connection is used to enable NAS signaling exchange between the UE **1001** and the CN **1020**, and comprises both the signaling connection between the UE and the AN (e.g., RRC connection or UE-N3IWF connection for non-3GPP access) and the N2 connection for the UE **1001** between the AN (e.g., RAN **1010**) and the AMF **1021**. The UE **1001** may operate in one of two CM states, CM-IDLE mode or CM-CONNECTED mode. When the UE **1001** is operating in the CM-IDLE state/mode, the UE **1001** may have no NAS signaling connection established with the AMF **1021** over the N1 interface, and there may be (R)AN **1010** signaling connection (e.g., N2 and/or N3 connections) for the UE **1001**. When the UE **1001** is operating in the CM-CONNECTED state/mode, the UE **1001** may have an established NAS signaling connection with the AMF **1021** over the N1 interface, and there may be a (R)AN **1010** signaling connection (e.g., N2 and/or N3 connections) for the UE **1001**. Establishment of an N2 connection between the (R)AN **1010** and the AMF **1021** may cause the UE **1001** to transition from CM-IDLE mode to CM-CONNECTED mode, and the UE **1001** may transition from the CM-CONNECTED mode to the CM-IDLE mode when N2 signaling between the (R)AN **1010** and the AMF **1021** is released.

The SMF **1024** may be responsible for SM (e.g., session establishment, modify and release, including tunnel maintain between UPF and AN node); UE IP address allocation and management (including optional authorization); selection and control of UP function; configuring traffic steering at UPF to route traffic to proper destination; termination of interfaces toward policy control functions; controlling part of policy enforcement and QoS; lawful intercept (for SM events and interface to LI system); termination of SM parts of NAS messages; downlink data notification; initiating AN specific SM information, sent via AMF over N2 to AN; and determining SSC mode of a session. SM may refer to management of a PDU session, and a PDU session or "session" may refer to a PDU connectivity service that provides or enables the exchange of PDUs between a UE **1001** and a data network (DN) **1003** identified by a Data

Network Name (DNN). PDU sessions may be established upon UE **1001** request, modified upon UE **1001** and 5GC **1020** request, and released upon UE **1001** and 5GC **1020** request using NAS SM signaling exchanged over the N1 reference point between the UE **1001** and the SMF **1024**. Upon request from an application server, the 5GC **1020** may trigger a specific application in the UE **1001**. In response to receipt of the trigger message, the UE **1001** may pass the trigger message (or relevant parts/information of the trigger message) to one or more identified applications in the UE **1001**. The identified application(s) in the UE **1001** may establish a PDU session to a specific DNN. The SMF **1024** may check whether the UE **1001** requests are compliant with user subscription information associated with the UE **1001**. In this regard, the SMF **1024** may retrieve and/or request to receive update notifications on SMF **1024** level subscription data from the UDM **1027**.

The SMF **1024** may include the following roaming functionality: handling local enforcement to apply QoS SLAs (VPLMN); charging data collection and charging interface (VPLMN); lawful intercept (in VPLMN for SM events and interface to LI system); and support for interaction with external DN for transport of signalling for PDU session authorization/authentication by external DN. An N16 reference point between two SMFs **1024** may be included in the system **1000**, which may be between another SMF **1024** in a visited network and the SMF **1024** in the home network in roaming scenarios. Additionally, the SMF **1024** may exhibit the Nsmf service-based interface.

The NEF **1023** may provide means for securely exposing the services and capabilities provided by 3GPP network functions for third party, internal exposure/re-exposure, Application Functions (e.g., AF **1028**), edge computing or fog computing systems, etc. In such embodiments, the NEF **1023** may authenticate, authorize, and/or throttle the AFs. NEF **1023** may also translate information exchanged with the AF **1028** and information exchanged with internal network functions. For example, the NEF **1023** may translate between an AF-Service-Identifier and an internal 5GC information. NEF **1023** may also receive information from other network functions (NFs) based on exposed capabilities of other network functions. This information may be stored at the NEF **1023** as structured data, or at a data storage NF using standardized interfaces. The stored information can then be re-exposed by the NEF **1023** to other NFs and AFs, and/or used for other purposes such as analytics. Additionally, the NEF **1023** may exhibit an Nnef service-based interface.

The NRF **1025** may support service discovery functions, receive NF discovery requests from NF instances, and provide the information of the discovered NF instances to the NF instances. NRF **1025** also maintains information of available NF instances and their supported services. As used herein, the terms "instantiate," "instantiation," and the like may refer to the creation of an instance, and an "instance" may refer to a concrete occurrence of an object, which may occur, for example, during execution of program code. Additionally, the NRF **1025** may exhibit the Nnrf service-based interface.

The PCF **1026** may provide policy rules to control plane function(s) to enforce them, and may also support unified policy framework to govern network behaviour. The PCF **1026** may also implement an FE to access subscription information relevant for policy decisions in a UDR of the UDM **1027**. The PCF **1026** may communicate with the AMF **1021** via an N15 reference point between the PCF **1026** and the AMF **1021**, which may include a PCF **1026** in a visited

network and the AMF 1021 in case of roaming scenarios. The PCF 1026 may communicate with the AF 1028 via an N5 reference point between the PCF 1026 and the AF 1028; and with the SMF 1024 via an N7 reference point between the PCF 1026 and the SMF 1024. The system 1000 and/or CN 1020 may also include an N24 reference point between the PCF 1026 (in the home network) and a PCF 1026 in a visited network. Additionally, the PCF 1026 may exhibit an Npcf service-based interface.

The UDM 1027 may handle subscription-related information to support the network entities' handling of communication sessions, and may store subscription data of UE 1001. For example, subscription data may be communicated between the UDM 1027 and the AMF 1021 via an N8 reference point between the UDM 1027 and the AMF. The UDM 1027 may include two parts, an application FE and a UDR (the FE and UDR are not shown by FIG. 10). The UDR may store subscription data and policy data for the UDM 1027 and the PCF 1026, and/or structured data for exposure and application data (including PFDs for application detection, application request information for multiple UEs 1001) for the NEF 1023. The Nudr service-based interface may be exhibited by the UDR 221 to allow the UDM 1027, PCF 1026, and NEF 1023 to access a particular set of the stored data, as well as to read, update (e.g., add, modify), delete, and subscribe to notification of relevant data changes in the UDR. The UDM may include a UDM-FE, which is in charge of processing credentials, location management, subscription management and so on. Several different front ends may serve the same user in different transactions. The UDM-FE accesses subscription information stored in the UDR and performs authentication credential processing, user identification handling, access authorization, registration/mobility management, and subscription management. The UDR may interact with the SMF 1024 via an N10 reference point between the UDM 1027 and the SMF 1024. UDM 1027 may also support SMS management, wherein an SMS-FE implements the similar application logic as discussed previously. Additionally, the UDM 1027 may exhibit the Nudm service-based interface.

The AF 1028 may provide application influence on traffic routing, provide access to the NCE, and interact with the policy framework for policy control. The NCE may be a mechanism that allows the 5GC 1020 and AF 1028 to provide information to each other via NEF 1023, which may be used for edge computing embodiments. In such embodiments, the network operator and third party services may be hosted close to the UE 1001 access point of attachment to achieve an efficient service delivery through the reduced end-to-end latency and load on the transport network. For edge computing embodiments, the 5GC may select a UPF 1002 close to the UE 1001 and execute traffic steering from the UPF 1002 to DN 1003 via the N6 interface. This may be based on the UE subscription data, UE location, and information provided by the AF 1028. In this way, the AF 1028 may influence UPF (re)selection and traffic routing. Based on operator deployment, when AF 1028 is considered to be a trusted entity, the network operator may permit AF 1028 to interact directly with relevant NFs. Additionally, the AF 1028 may exhibit an Naf service-based interface.

The NSSF 1029 may select a set of network slice instances serving the UE 1001. The NSSF 1029 may also determine allowed NSSAI and the mapping to the subscribed S-NSSAIs, if needed. The NSSF 1029 may also determine the AMF set to be used to serve the UE 1001, or a list of candidate AMF(s) 1021 based on a suitable configuration and possibly by querying the NRF 1025. The

selection of a set of network slice instances for the UE 1001 may be triggered by the AMF 1021 with which the UE 1001 is registered by interacting with the NSSF 1029, which may lead to a change of AMF 1021. The NSSF 1029 may interact with the AMF 1021 via an N22 reference point between AMF 1021 and NSSF 1029; and may communicate with another NSSF 1029 in a visited network via an N31 reference point (not shown by FIG. 10). Additionally, the NSSF 1029 may exhibit an Nnssf service-based interface.

As discussed previously, the CN 1020 may include an SMSF, which may be responsible for SMS subscription checking and verification, and relaying SM messages to/from the UE 1001 to/from other entities, such as an SMS-GMSC/IW MSC/SMS-router. The SMS may also interact with AMF 1021 and UDM 1027 for a notification procedure that the UE 1001 is available for SMS transfer (e.g., set a UE not reachable flag, and notifying UDM 1027 when UE 1001 is available for SMS).

The CN 120 may also include other elements that are not shown by FIG. 10, such as a Data Storage system/architecture, a 5G-EIR, a SEPP, and the like. The Data Storage system may include a SDSF, an UDSF, and/or the like. Any NF may store and retrieve unstructured data into/from the UDSF (e.g., UE contexts), via N18 reference point between any NF and the UDSF (not shown by FIG. 10). Individual NFs may share a UDSF for storing their respective unstructured data or individual NFs may each have their own UDSF located at or near the individual NFs. Additionally, the UDSF may exhibit an Nudsf service-based interface (not shown by FIG. 10). The 5G-EIR may be an NF that checks the status of PEI for determining whether particular equipment/entities are blacklisted from the network; and the SEPP may be a non-transparent proxy that performs topology hiding, message filtering, and policing on inter-PLMN control plane interfaces.

Additionally, there may be many more reference points and/or service-based interfaces between the NF services in the NFs; however, these interfaces and reference points have been omitted from FIG. 10 for clarity. In one example, the CN 1020 may include an Nx interface, which is an inter-CN interface between the MME (e.g., MME 921) and the AMF 1021 in order to enable interworking between CN 1020 and CN 920. Other example interfaces/reference points may include an N5g-EIR service-based interface exhibited by a 5G-EIR, an N27 reference point between the NRF in the visited network and the NRF in the home network; and an N31 reference point between the NSSF in the visited network and the NSSF in the home network.

FIG. 11 illustrates an example of infrastructure equipment 1100 in accordance with various embodiments. The infrastructure equipment 1100 (or "system 1100") may be implemented as a base station, radio head, RAN node such as the RAN nodes 811 and/or AP 806 shown and described previously, application server(s) 830, and/or any other element/device discussed herein. In other examples, the system 1100 could be implemented in or by a UE.

The system 1100 includes application circuitry 1105, baseband circuitry 1110, one or more radio front end modules (RFEMs) 1115, memory circuitry 1120, power management integrated circuitry (PMIC) 1125, power tee circuitry 1130, network controller circuitry 1135, network interface connector 1140, satellite positioning circuitry 1145, and user interface 1150. In some embodiments, the device 1100 may include additional elements such as, for example, memory/storage, display, camera, sensor, or input/output (I/O) interface. In other embodiments, the components described below may be included in more than one

device. For example, said circuitries may be separately included in more than one device for CRAN, vBBU, or other like embodiments.

Application circuitry **1105** includes circuitry such as, but not limited to one or more processors (or processor cores), cache memory, and one or more of low drop-out voltage regulators (LDOs), interrupt controllers, serial interfaces such as SPI, I²C or universal programmable serial interface module, real time clock (RTC), timer-counters including interval and watchdog timers, general purpose input/output (I/O or IO), memory card controllers such as Secure Digital (SD) Multi Media Card (MMC) or similar, Universal Serial Bus (USB) interfaces, Mobile Industry Processor Interface (MIPI) interfaces and Joint Test Access Group (JTAG) test access ports. The processors (or cores) of the application circuitry **1105** may be coupled with or may include memory/storage elements and may be configured to execute instructions stored in the memory/storage to enable various applications or operating systems to run on the system **1100**. In some embodiments, the memory/storage elements may be on-chip memory circuitry, which may include any suitable volatile and/or non-volatile memory, such as DRAM, SRAM, EPROM, EEPROM, Flash memory, solid-state memory, and/or any other type of memory device technology, such as those discussed herein.

The processor(s) of application circuitry **1105** may include, for example, one or more processor cores (CPUs), one or more application processors, one or more graphics processing units (GPUs), one or more reduced instruction set computing (RISC) processors, one or more Acorn RISC Machine (ARM) processors, one or more complex instruction set computing (CISC) processors, one or more digital signal processors (DSP), one or more FPGAs, one or more PLDs, one or more ASICs, one or more microprocessors or controllers, or any suitable combination thereof. In some embodiments, the application circuitry **1105** may comprise, or may be, a special-purpose processor/controller to operate according to the various embodiments herein. As examples, the processor(s) of application circuitry **1105** may include one or more Intel Pentium®, Core®, or Xeon® processor(s); Advanced Micro Devices (AMD) Ryzen® processor(s), Accelerated Processing Units (APUs), or Epyc® processors; ARM-based processor(s) licensed from ARM Holdings, Ltd. such as the ARM Cortex-A family of processors and the ThunderX2® provided by Cavium™, Inc.; a MIPS-based design from MIPS Technologies, Inc. such as MIPS Warrior P-class processors; and/or the like. In some embodiments, the system **1100** may not utilize application circuitry **1105**, and instead may include a special-purpose processor/controller to process IP data received from an EPC or 5GC, for example.

In some embodiments, the application circuitry **1105** may include one or more hardware accelerators, which may be microprocessors, programmable processing devices, or the like. The one or more hardware accelerators may include, for example, computer vision (CV) and/or deep learning (DL) accelerators. As examples, the programmable processing devices may be one or more a field-programmable devices (FPDs) such as field-programmable gate arrays (FPGAs) and the like; programmable logic devices (PLDs) such as complex PLDs (CPLDs), high-capacity PLDs (HCPLDs), and the like; ASICs such as structured ASICs and the like; programmable SoCs (PSoCs); and the like. In such embodiments, the circuitry of application circuitry **1105** may comprise logic blocks or logic fabric, and other interconnected resources that may be programmed to perform various functions, such as the procedures, methods, functions, etc. of

the various embodiments discussed herein. In such embodiments, the circuitry of application circuitry **1105** may include memory cells (e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), flash memory, static memory (e.g., static random access memory (SRAM), antifuses, etc.)) used to store logic blocks, logic fabric, data, etc. in look-up-tables (LUTs) and the like.

The baseband circuitry **1110** may be implemented, for example, as a solder-down substrate including one or more integrated circuits, a single packaged integrated circuit soldered to a main circuit board or a multi-chip module containing two or more integrated circuits. The various hardware electronic elements of baseband circuitry **1110** are discussed infra with regard to FIG. **13**.

User interface circuitry **1150** may include one or more user interfaces designed to enable user interaction with the system **1100** or peripheral component interfaces designed to enable peripheral component interaction with the system **1100**. User interfaces may include, but are not limited to, one or more physical or virtual buttons (e.g., a reset button), one or more indicators (e.g., light emitting diodes (LEDs)), a physical keyboard or keypad, a mouse, a touchpad, a touchscreen, speakers or other audio emitting devices, microphones, a printer, a scanner, a headset, a display screen or display device, etc. Peripheral component interfaces may include, but are not limited to, a nonvolatile memory port, a universal serial bus (USB) port, an audio jack, a power supply interface, etc.

The radio front end modules (RFEMs) **1115** may comprise a millimeter wave (mmWave) RFEM and one or more sub-mmWave radio frequency integrated circuits (RFICs). In some embodiments, the one or more sub-mmWave RFICs may be physically separated from the mmWave RFEM. The RFICs may include connections to one or more antennas or antenna arrays (see e.g., antenna array **1311** of FIG. **13** infra), and the RFEM may be connected to multiple antennas. In alternative embodiments, both mmWave and sub-mmWave radio functions may be implemented in the same physical RFEM **1115**, which incorporates both mmWave antennas and sub-mmWave.

The memory circuitry **1120** may include one or more of volatile memory including dynamic random access memory (DRAM) and/or synchronous dynamic random access memory (SDRAM), and nonvolatile memory (NVM) including high-speed electrically erasable memory (commonly referred to as Flash memory), phase change random access memory (PRAM), magnetoresistive random access memory (MRAM), etc., and may incorporate the three-dimensional (3D) cross-point (XPOINT) memories from Intel® and Micron®. Memory circuitry **1120** may be implemented as one or more of solder down packaged integrated circuits, socketed memory modules and plug-in memory cards.

The PMIC **1125** may include voltage regulators, surge protectors, power alarm detection circuitry, and one or more backup power sources such as a battery or capacitor. The power alarm detection circuitry may detect one or more of brown out (under-voltage) and surge (over-voltage) conditions. The power tee circuitry **1130** may provide for electrical power drawn from a network cable to provide both power supply and data connectivity to the infrastructure equipment **1100** using a single cable.

The network controller circuitry **1135** may provide connectivity to a network using a standard network interface protocol such as Ethernet, Ethernet over GRE Tunnels, Ethernet over Multiprotocol Label Switching (MPLS), or

some other suitable protocol. Network connectivity may be provided to/from the infrastructure equipment **1100** via network interface connector **1140** using a physical connection, which may be electrical (commonly referred to as a “copper interconnect”), optical, or wireless. The network controller circuitry **1135** may include one or more dedicated processors and/or FPGAs to communicate using one or more of the aforementioned protocols. In some embodiments, the network controller circuitry **1135** may include multiple controllers to provide connectivity to other networks using the same or different protocols.

The positioning circuitry **1145** includes circuitry to receive and decode signals transmitted/broadcasted by a positioning network of a global navigation satellite system (GNSS). Examples of navigation satellite constellations (or GNSS) include United States’ Global Positioning System (GPS), Russia’s Global Navigation System (GLONASS), the European Union’s Galileo system, China’s BeiDou Navigation Satellite System, a regional navigation system or GNSS augmentation system (e.g., Navigation with Indian Constellation (NAVIC), Japan’s Quasi-Zenith Satellite System (QZSS), France’s Doppler Orbitography and Radio-positioning Integrated by Satellite (DORIS), etc.), or the like. The positioning circuitry **1145** comprises various hardware elements (e.g., including hardware devices such as switches, filters, amplifiers, antenna elements, and the like to facilitate OTA communications) to communicate with components of a positioning network, such as navigation satellite constellation nodes. In some embodiments, the positioning circuitry **1145** may include a Micro-Technology for Positioning, Navigation, and Timing (Micro-PNT) IC that uses a master timing clock to perform position tracking/estimation without GNSS assistance. The positioning circuitry **1145** may also be part of, or interact with, the baseband circuitry **1110** and/or RFEMs **1115** to communicate with the nodes and components of the positioning network. The positioning circuitry **1145** may also provide position data and/or time data to the application circuitry **1105**, which may use the data to synchronize operations with various infrastructure (e.g., RAN nodes **811**, etc.), or the like.

The components shown by FIG. **11** may communicate with one another using interface circuitry, which may include any number of bus and/or interconnect (IX) technologies such as industry standard architecture (ISA), extended ISA (EISA), peripheral component interconnect (PCI), peripheral component interconnect extended (PCIx), PCI express (PCIe), or any number of other technologies. The bus/IX may be a proprietary bus, for example, used in a SoC based system. Other bus/IX systems may be included, such as an I²C interface, an SPI interface, point to point interfaces, and a power bus, among others.

FIG. **12** illustrates an example of a platform **1200** (or “device **1200**”) in accordance with various embodiments. In embodiments, the computer platform **1200** may be suitable for use as UEs **801**, **901**, application servers **830**, and/or any other element/device discussed herein. The platform **1200** may include any combinations of the components shown in the example. The components of platform **1200** may be implemented as integrated circuits (ICs), portions thereof, discrete electronic devices, or other modules, logic, hardware, software, firmware, or a combination thereof adapted in the computer platform **1200**, or as components otherwise incorporated within a chassis of a larger system. The block diagram of FIG. **12** is intended to show a high level view of components of the computer platform **1200**. However, some of the components shown may be omitted, additional com-

ponents may be present, and different arrangement of the components shown may occur in other embodiments.

Application circuitry **1205** includes circuitry such as, but not limited to one or more processors (or processor cores), cache memory, and one or more of LDOs, interrupt controllers, serial interfaces such as SPI, I²C or universal programmable serial interface module, RTC, timer-counters including interval and watchdog timers, general purpose I/O, memory card controllers such as SD MMC or similar, USB interfaces, MIPI interfaces, and JTAG test access ports. The processors (or cores) of the application circuitry **1205** may be coupled with or may include memory/storage elements and may be configured to execute instructions stored in the memory/storage to enable various applications or operating systems to run on the system **1200**. In some embodiments, the memory/storage elements may be on-chip memory circuitry, which may include any suitable volatile and/or non-volatile memory, such as DRAM, SRAM, EPROM, EEPROM, Flash memory, solid-state memory, and/or any other type of memory device technology, such as those discussed herein.

The processor(s) of application circuitry **1105** may include, for example, one or more processor cores, one or more application processors, one or more GPUs, one or more RISC processors, one or more ARM processors, one or more CISC processors, one or more DSP, one or more FPGAs, one or more PLDs, one or more ASICs, one or more microprocessors or controllers, a multithreaded processor, an ultra-low voltage processor, an embedded processor, some other known processing element, or any suitable combination thereof. In some embodiments, the application circuitry **1105** may comprise, or may be, a special-purpose processor/controller to operate according to the various embodiments herein.

As examples, the processor(s) of application circuitry **1205** may include an Intel® Architecture Core™ based processor, such as a Quark™, an Atom™, an i3, an i5, an i7, or an MCU-class processor, or another such processor available from Intel® Corporation, Santa Clara, CA. The processors of the application circuitry **1205** may also be one or more of Advanced Micro Devices (AMD) Ryzen® processor(s) or Accelerated Processing Units (APUs); A5-A9 processor(s) from Apple® Inc., Snapdragon™ processor(s) from Qualcomm® Technologies, Inc., Texas Instruments, Inc.® Open Multimedia Applications Platform (OMAP)™ processor(s); a MIPS-based design from MIPS Technologies, Inc. such as MIPS Warrior M-class, Warrior I-class, and Warrior P-class processors; an ARM-based design licensed from ARM Holdings, Ltd., such as the ARM Cortex-A, Cortex-R, and Cortex-M family of processors; or the like. In some embodiments, the application circuitry **1205** may be a part of a system on a chip (SoC) in which the application circuitry **1205** and other components are formed into a single integrated circuit, or a single package, such as the Edison™ or Galileo™ SoC boards from Intel® Corporation.

Additionally or alternatively, application circuitry **1205** may include circuitry such as, but not limited to, one or more a field-programmable devices (FPDs) such as FPGAs and the like; programmable logic devices (PLDs) such as complex PLDs (CPLDs), high-capacity PLDs (HCPLDs), and the like; ASICs such as structured ASICs and the like; programmable SoCs (PSoCs); and the like. In such embodiments, the circuitry of application circuitry **1205** may comprise logic blocks or logic fabric, and other interconnected resources that may be programmed to perform various functions, such as the procedures, methods, functions, etc. of

the various embodiments discussed herein. In such embodiments, the circuitry of application circuitry **1205** may include memory cells (e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), flash memory, static memory (e.g., static random access memory (SRAM), antifuses, etc.)) used to store logic blocks, logic fabric, data, etc. in look-up tables (LUTs) and the like.

The baseband circuitry **1210** may be implemented, for example, as a solder-down substrate including one or more integrated circuits, a single packaged integrated circuit soldered to a main circuit board or a multi-chip module containing two or more integrated circuits. The various hardware electronic elements of baseband circuitry **1210** are discussed infra with regard to FIG. **13**.

The RFEMs **1215** may comprise a millimeter wave (mm-Wave) RFEM and one or more sub-mmWave radio frequency integrated circuits (RFICs). In some embodiments, the one or more sub-mmWave RFICs may be physically separated from the mmWave RFEM. The RFICs may include connections to one or more antennas or antenna arrays (see e.g., antenna array **1311** of FIG. **13** infra), and the RFEM may be connected to multiple antennas. In alternative embodiments, both mmWave and sub-mmWave radio functions may be implemented in the same physical RFEM **1215**, which incorporates both mmWave antennas and sub-mm-Wave.

The memory circuitry **1220** may include any number and type of memory devices used to provide for a given amount of system memory. As examples, the memory circuitry **1220** may include one or more of volatile memory including random access memory (RAM), dynamic RAM (DRAM) and/or synchronous dynamic RAM (SDRAM), and nonvolatile memory (NVM) including high-speed electrically erasable memory (commonly referred to as Flash memory), phase change random access memory (PRAM), magnetoresistive random access memory (MRAM), etc. The memory circuitry **1220** may be developed in accordance with a Joint Electron Devices Engineering Council (JEDEC) low power double data rate (LPDDR)-based design, such as LPDDR2, LPDDR3, LPDDR4, or the like. Memory circuitry **1220** may be implemented as one or more of solder down packaged integrated circuits, single die package (SDP), dual die package (DDP) or quad die package (Q17P), socketed memory modules, dual inline memory modules (DIMMs) including microDIMMs or MiniDIMMs, and/or soldered onto a motherboard via a ball grid array (BGA). In low power embodiments, the memory circuitry **1220** may be on-die memory or registers associated with the application circuitry **1205**. To provide for persistent storage of information such as data, applications, operating systems and so forth, memory circuitry **1220** may include one or more mass storage devices, which may include, inter alia, a solid state disk drive (SSDD), hard disk drive (HDD), a micro HDD, resistance change memories, phase change memories, holographic memories, or chemical memories, among others. For example, the computer platform **1200** may incorporate the three-dimensional (3D) cross-point (XPOINT) memories from Intel® and Micron®.

Removable memory circuitry **1223** may include devices, circuitry, enclosures/housings, ports or receptacles, etc. used to couple portable data storage devices with the platform **1200**. These portable data storage devices may be used for mass storage purposes, and may include, for example, flash memory cards (e.g., Secure Digital (SD) cards, microSD cards, xD picture cards, and the like), and USB flash drives, optical discs, external HDDs, and the like.

The platform **1200** may also include interface circuitry (not shown) that is used to connect external devices with the platform **1200**. The external devices connected to the platform **1200** via the interface circuitry include sensor circuitry **1221** and electro-mechanical components (EMCs) **1222**, as well as removable memory devices coupled to removable memory circuitry **1223**.

The sensor circuitry **1221** include devices, modules, or subsystems whose purpose is to detect events or changes in its environment and send the information (sensor data) about the detected events to some other a device, module, subsystem, etc. Examples of such sensors include, inter alia, inertia measurement units (IMUs) comprising accelerometers, gyroscopes, and/or magnetometers; microelectromechanical systems (MEMS) or nanoelectromechanical systems (NEMS) comprising 3-axis accelerometers, 3-axis gyroscopes, and/or magnetometers; level sensors; flow sensors; temperature sensors (e.g., thermistors); pressure sensors; barometric pressure sensors; gravimeters; altimeters; image capture devices (e.g., cameras or lensless apertures); light detection and ranging (LiDAR) sensors; proximity sensors (e.g., infrared radiation detector and the like), depth sensors, ambient light sensors, ultrasonic transceivers; microphones or other like audio capture devices; etc.

EMCs **1222** include devices, modules, or subsystems whose purpose is to enable platform **1200** to change its state, position, and/or orientation, or move or control a mechanism or (sub)system. Additionally, EMCs **1222** may be configured to generate and send messages/signalling to other components of the platform **1200** to indicate a current state of the EMCs **1222**. Examples of the EMCs **1222** include one or more power switches, relays including electromechanical relays (EMRs) and/or solid state relays (SSRs), actuators (e.g., valve actuators, etc.), an audible sound generator, a visual warning device, motors (e.g., DC motors, stepper motors, etc.), wheels, thrusters, propellers, claws, clamps, hooks, and/or other like electro-mechanical components. In embodiments, platform **1200** is configured to operate one or more EMCs **1222** based on one or more captured events and/or instructions or control signals received from a service provider and/or various clients.

In some embodiments, the interface circuitry may connect the platform **1200** with positioning circuitry **1245**. The positioning circuitry **1245** includes circuitry to receive and decode signals transmitted/broadcasted by a positioning network of a GNSS. Examples of navigation satellite constellations (or GNSS) include United States' GPS, Russia's GLONASS, the European Union's Galileo system, China's BeiDou Navigation Satellite System, a regional navigation system or GNSS augmentation system (e.g., NAVIC), Japan's QZSS, France's DORIS, etc.), or the like. The positioning circuitry **1245** comprises various hardware elements (e.g., including hardware devices such as switches, filters, amplifiers, antenna elements, and the like to facilitate OTA communications) to communicate with components of a positioning network, such as navigation satellite constellation nodes. In some embodiments, the positioning circuitry **1245** may include a Micro-PNT IC that uses a master timing clock to perform position tracking/estimation without GNSS assistance. The positioning circuitry **1245** may also be part of, or interact with, the baseband circuitry **1110** and/or RFEMs **1215** to communicate with the nodes and components of the positioning network. The positioning circuitry **1245** may also provide position data and/or time data to the application circuitry **1205**, which may use the data to

synchronize operations with various infrastructure (e.g., radio base stations), for turn-by-turn navigation applications, or the like

In some embodiments, the interface circuitry may connect the platform 1200 with Near-Field Communication (NFC) circuitry 1240. NFC circuitry 1240 is configured to provide contactless, short-range communications based on radio frequency identification (RFID) standards, wherein magnetic field induction is used to enable communication between NFC circuitry 1240 and NFC-enabled devices external to the platform 1200 (e.g., an “NFC touchpoint”). NFC circuitry 1240 comprises an NFC controller coupled with an antenna element and a processor coupled with the NFC controller. The NFC controller may be a chip/IC providing NFC functionalities to the NFC circuitry 1240 by executing NFC controller firmware and an NFC stack. The NFC stack may be executed by the processor to control the NFC controller, and the NFC controller firmware may be executed by the NFC controller to control the antenna element to emit short-range RF signals. The RF signals may power a passive NFC tag (e.g., a microchip embedded in a sticker or wristband) to transmit stored data to the NFC circuitry 1240, or initiate data transfer between the NFC circuitry 1240 and another active NFC device (e.g., a smartphone or an NFC-enabled POS terminal) that is proximate to the platform 1200.

The driver circuitry 1246 may include software and hardware elements that operate to control particular devices that are embedded in the platform 1200, attached to the platform 1200, or otherwise communicatively coupled with the platform 1200. The driver circuitry 1246 may include individual drivers allowing other components of the platform 1200 to interact with or control various input/output (I/O) devices that may be present within, or connected to, the platform 1200. For example, driver circuitry 1246 may include a display driver to control and allow access to a display device, a touchscreen driver to control and allow access to a touchscreen interface of the platform 1200, sensor drivers to obtain sensor readings of sensor circuitry 1221 and control and allow access to sensor circuitry 1221, EMC drivers to obtain actuator positions of the EMCs 1222 and/or control and allow access to the EMCs 1222, a camera driver to control and allow access to an embedded image capture device, audio drivers to control and allow access to one or more audio devices.

The power management integrated circuitry (PMIC) 1225 (also referred to as “power management circuitry 1225”) may manage power provided to various components of the platform 1200. In particular, with respect to the baseband circuitry 1210, the PMIC 1225 may control power-source selection, voltage scaling, battery charging, or DC-to-DC conversion. The PMIC 1225 may often be included when the platform 1200 is capable of being powered by a battery 1230, for example, when the device is included in a UE 801, 901.

In some embodiments, the PMIC 1225 may control, or otherwise be part of, various power saving mechanisms of the platform 1200. For example, if the platform 1200 is in an RRC_Connected state, where it is still connected to the RAN node as it expects to receive traffic shortly, then it may enter a state known as Discontinuous Reception Mode (DRX) after a period of inactivity. During this state, the platform 1200 may power down for brief intervals of time and thus save power. If there is no data traffic activity for an extended period of time, then the platform 1200 may transition off to an RRC_Idle state, where it disconnects from the network and does not perform operations such as channel quality

feedback, handover, etc. The platform 1200 goes into a very low power state and it performs paging where again it periodically wakes up to listen to the network and then powers down again. The platform 1200 may not receive data in this state; in order to receive data, it can then transition back to RRC_Connected state. An additional power saving mode may allow a device to be unavailable to the network for periods longer than a paging interval (ranging from seconds to a few hours). During this time, the device is unreachable to the network and may power down completely. Any data sent during this time incurs a large delay and it is assumed the delay is acceptable.

A battery 1230 may power the platform 1200, although in some examples the platform 1200 may be mounted deployed in a fixed location, and may have a power supply coupled to an electrical grid. The battery 1230 may be a lithium ion battery, a metal-air battery, such as a zinc-air battery, an aluminum-air battery, a lithium-air battery, and the like. In some embodiments, such as in V2X applications, the battery 1230 may be a typical lead-acid automotive battery.

In some embodiments, the battery 1230 may be a “smart battery,” which includes or is coupled with a Battery Management System (BMS) or battery monitoring integrated circuitry. The BMS may be included in the platform 1200 to track the state of charge (SoCh) of the battery 1230. The BMS may be used to monitor other parameters of the battery 1230 to provide failure predictions, such as the state of health (SoH) and the state of function (SoF) of the battery 1230. The BMS may communicate the information of the battery 1230 to the application circuitry 1205 or other components of the platform 1200. The BMS may also include an analog-to-digital (ADC) convertor that allows the application circuitry 1205 to directly monitor the voltage of the battery 1230 or the current flow from the battery 1230. The battery parameters may be used to determine actions that the platform 1200 may perform, such as transmission frequency, network operation, sensing frequency, and the like.

A power block, or other power supply coupled to an electrical grid may be coupled with the BMS to charge the battery 1230. In some examples, the power block XS30 may be replaced with a wireless power receiver to obtain the power wirelessly, for example, through a loop antenna in the computer platform 1200. In these examples, a wireless battery charging circuit may be included in the BMS. The specific charging circuits chosen may depend on the size of the battery 1230, and thus, the current required. The charging may be performed using the Airfuel standard promulgated by the Airfuel Alliance, the Qi wireless charging standard promulgated by the Wireless Power Consortium, or the Rezence charging standard promulgated by the Alliance for Wireless Power, among others.

User interface circuitry 1250 includes various input/output (I/O) devices present within, or connected to, the platform 1200, and includes one or more user interfaces designed to enable user interaction with the platform 1200 and/or peripheral component interfaces designed to enable peripheral component interaction with the platform 1200. The user interface circuitry 1250 includes input device circuitry and output device circuitry. Input device circuitry includes any physical or virtual means for accepting an input including, inter alia, one or more physical or virtual buttons (e.g., a reset button), a physical keyboard, keypad, mouse, touchpad, touchscreen, microphones, scanner, headset, and/or the like. The output device circuitry includes any physical or virtual means for showing information or otherwise conveying information, such as sensor readings, actuator

position(s), or other like information. Output device circuitry may include any number and/or combinations of audio or visual display, including, inter alia, one or more simple visual outputs/indicators (e.g., binary status indicators (e.g., light emitting diodes (LEDs)) and multi-character visual outputs, or more complex outputs such as display devices or touchscreens (e.g., Liquid Chrystal Displays (LCD), LED displays, quantum dot displays, projectors, etc.), with the output of characters, graphics, multimedia objects, and the like being generated or produced from the operation of the platform **1200**. The output device circuitry may also include speakers or other audio emitting devices, printer(s), and/or the like. In some embodiments, the sensor circuitry **1221** may be used as the input device circuitry (e.g., an image capture device, motion capture device, or the like) and one or more EMCs may be used as the output device circuitry (e.g., an actuator to provide haptic feedback or the like). In another example, NFC circuitry comprising an NFC controller coupled with an antenna element and a processing device may be included to read electronic tags and/or connect with another NFC-enabled device. Peripheral component interfaces may include, but are not limited to, a non-volatile memory port, a USB port, an audio jack, a power supply interface, etc.

Although not shown, the components of platform **1200** may communicate with one another using a suitable bus or interconnect (IX) technology, which may include any number of technologies, including ISA, EISA, PCI, PCIX, PCIe, a Time-Trigger Protocol (TTP) system, a FlexRay system, or any number of other technologies. The bus/IX may be a proprietary bus/IX, for example, used in a SoC based system. Other bus/IX systems may be included, such as an I²C interface, an SPI interface, point-to-point interfaces, and a power bus, among others.

FIG. **13** illustrates example components of baseband circuitry **1310** and radio front end modules (RFEM) **1315** in accordance with various embodiments. The baseband circuitry **1310** corresponds to the baseband circuitry **1110** and **1210** of FIGS. **11** and **12**, respectively. The RFEM **1315** corresponds to the RFEM **1115** and **1215** of FIGS. **11** and **12**, respectively. As shown, the RFEMs **1315** may include Radio Frequency (RF) circuitry **1306**, front-end module (FEM) circuitry **1308**, antenna array **1311** coupled together at least as shown.

The baseband circuitry **1310** includes circuitry and/or control logic configured to carry out various radio/network protocol and radio control functions that enable communication with one or more radio networks via the RF circuitry **1306**. The radio control functions may include, but are not limited to, signal modulation/demodulation, encoding/decoding, radio frequency shifting, etc. In some embodiments, modulation/demodulation circuitry of the baseband circuitry **1310** may include Fast-Fourier Transform (FFT), precoding, or constellation mapping/demapping functionality. In some embodiments, encoding/decoding circuitry of the baseband circuitry **1310** may include convolution, tail-biting convolution, turbo, Viterbi, or Low Density Parity Check (LDPC) encoder/decoder functionality. Embodiments of modulation/demodulation and encoder/decoder functionality are not limited to these examples and may include other suitable functionality in other embodiments. The baseband circuitry **1310** is configured to process baseband signals received from a receive signal path of the RF circuitry **1306** and to generate baseband signals for a transmit signal path of the RF circuitry **1306**. The baseband circuitry **1310** is configured to interface with application circuitry **1105/1205** (see FIGS. **11** and **12**) for generation and processing of the

baseband signals and for controlling operations of the RF circuitry **1306**. The baseband circuitry **1310** may handle various radio control functions.

The aforementioned circuitry and/or control logic of the baseband circuitry **1310** may include one or more single or multi-core processors. For example, the one or more processors may include a 3G baseband processor **1304A**, a 4G/LTE baseband processor **1304B**, a 5G/NR baseband processor **1304C**, or some other baseband processor(s) **1304D** for other existing generations, generations in development or to be developed in the future (e.g., sixth generation (6G), etc.). In other embodiments, some or all of the functionality of baseband processors **1304A-D** may be included in modules stored in the memory **1304G** and executed via a Central Processing Unit (CPU) **1304E**. In other embodiments, some or all of the functionality of baseband processors **1304A-D** may be provided as hardware accelerators (e.g., FPGAs, ASICs, etc.) loaded with the appropriate bit streams or logic blocks stored in respective memory cells. In various embodiments, the memory **1304G** may store program code of a real-time OS (RTOS), which when executed by the CPU **1304E** (or other baseband processor), is to cause the CPU **1304E** (or other baseband processor) to manage resources of the baseband circuitry **1310**, schedule tasks, etc. Examples of the RTOS may include Operating System Embedded (OSE)[™] provided by Enea®, Nucleus RTOS[™] provided by Mentor Graphics®, Versatile Real-Time Executive (VRTX) provided by Mentor Graphics®, ThreadX[™] provided by Express Logic®, FreeRTOS, REX OS provided by Qualcomm®, OKL4 provided by Open Kernel (OK) Labs®, or any other suitable RTOS, such as those discussed herein. In addition, the baseband circuitry **1310** includes one or more audio digital signal processor(s) (DSP) **1304F**. The audio DSP(s) **1304F** include elements for compression/decompression and echo cancellation and may include other suitable processing elements in other embodiments.

In some embodiments, each of the processors **1304A-1304E** include respective memory interfaces to send/receive data to/from the memory **1304G**. The baseband circuitry **1310** may further include one or more interfaces to communicatively couple to other circuitries/devices, such as an interface to send/receive data to/from memory external to the baseband circuitry **1310**; an application circuitry interface to send/receive data to/from the application circuitry **1105/1205** of FIGS. **11-XT**); an RF circuitry interface to send/receive data to/from RF circuitry **1306** of FIG. **13**; a wireless hardware connectivity interface to send/receive data to/from one or more wireless hardware elements (e.g., Near Field Communication (NFC) components, Bluetooth®/Bluetooth® Low Energy components, Wi-Fi® components, and/or the like); and a power management interface to send/receive power or control signals to/from the PMIC **1225**.

In alternate embodiments (which may be combined with the above described embodiments), baseband circuitry **1310** comprises one or more digital baseband systems, which are coupled with one another via an interconnect subsystem and to a CPU subsystem, an audio subsystem, and an interface subsystem. The digital baseband subsystems may also be coupled to a digital baseband interface and a mixed-signal baseband subsystem via another interconnect subsystem. Each of the interconnect subsystems may include a bus system, point-to-point connections, network-on-chip (NOC) structures, and/or some other suitable bus or interconnect technology, such as those discussed herein. The audio subsystem may include DSP circuitry, buffer memory, program

memory, speech processing accelerator circuitry, data converter circuitry such as analog-to-digital and digital-to-analog converter circuitry, analog circuitry including one or more of amplifiers and filters, and/or other like components. In an aspect of the present disclosure, baseband circuitry **1310** may include protocol processing circuitry with one or more instances of control circuitry (not shown) to provide control functions for the digital baseband circuitry and/or radio frequency circuitry (e.g., the radio front end modules **1315**).

Although not shown by FIG. **13**, in some embodiments, the baseband circuitry **1310** includes individual processing device(s) to operate one or more wireless communication protocols (e.g., a “multi-protocol baseband processor” or “protocol processing circuitry”) and individual processing device(s) to implement PHY layer functions. In these embodiments, the PHY layer functions include the aforementioned radio control functions. In these embodiments, the protocol processing circuitry operates or implements various protocol layers/entities of one or more wireless communication protocols. In a first example, the protocol processing circuitry may operate LTE protocol entities and/or 5G/NR protocol entities when the baseband circuitry **1310** and/or RF circuitry **1306** are part of mmWave communication circuitry or some other suitable cellular communication circuitry. In the first example, the protocol processing circuitry would operate MAC, RLC, PDCP, SDAP, RRC, and NAS functions. In a second example, the protocol processing circuitry may operate one or more IEEE-based protocols when the baseband circuitry **1310** and/or RF circuitry **1306** are part of a Wi-Fi communication system. In the second example, the protocol processing circuitry would operate Wi-Fi MAC and logical link control (LLC) functions. The protocol processing circuitry may include one or more memory structures (e.g., **1304G**) to store program code and data for operating the protocol functions, as well as one or more processing cores to execute the program code and perform various operations using the data. The baseband circuitry **1310** may also support radio communications for more than one wireless protocol.

The various hardware elements of the baseband circuitry **1310** discussed herein may be implemented, for example, as a solder-down substrate including one or more integrated circuits (ICs), a single packaged IC soldered to a main circuit board or a multi-chip module containing two or more ICs. In one example, the components of the baseband circuitry **1310** may be suitably combined in a single chip or chipset, or disposed on a same circuit board. In another example, some or all of the constituent components of the baseband circuitry **1310** and RF circuitry **1306** may be implemented together such as, for example, a system on a chip (SoC) or System-in-Package (SiP). In another example, some or all of the constituent components of the baseband circuitry **1310** may be implemented as a separate SoC that is communicatively coupled with and RF circuitry **1306** (or multiple instances of RF circuitry **1306**). In yet another example, some or all of the constituent components of the baseband circuitry **1310** and the application circuitry **1105/1205** may be implemented together as individual SoCs mounted to a same circuit board (e.g., a “multi-chip package”).

In some embodiments, the baseband circuitry **1310** may provide for communication compatible with one or more radio technologies. For example, in some embodiments, the baseband circuitry **1310** may support communication with an E-UTRAN or other WMAN, a WLAN, a WPAN. Embodiments in which the baseband circuitry **1310** is con-

figured to support radio communications of more than one wireless protocol may be referred to as multi-mode baseband circuitry.

RF circuitry **1306** may enable communication with wireless networks using modulated electromagnetic radiation through a non-solid medium. In various embodiments, the RF circuitry **1306** may include switches, filters, amplifiers, etc. to facilitate the communication with the wireless network. RF circuitry **1306** may include a receive signal path, which may include circuitry to down-convert RF signals received from the FEM circuitry **1308** and provide baseband signals to the baseband circuitry **1310**. RF circuitry **1306** may also include a transmit signal path, which may include circuitry to up-convert baseband signals provided by the baseband circuitry **1310** and provide RF output signals to the FEM circuitry **1308** for transmission.

In some embodiments, the receive signal path of the RF circuitry **1306** may include mixer circuitry **1306a**, amplifier circuitry **1306b** and filter circuitry **1306c**. In some embodiments, the transmit signal path of the RF circuitry **1306** may include filter circuitry **1306c** and mixer circuitry **1306a**. RF circuitry **1306** may also include synthesizer circuitry **1306d** for synthesizing a frequency for use by the mixer circuitry **1306a** of the receive signal path and the transmit signal path. In some embodiments, the mixer circuitry **1306a** of the receive signal path may be configured to down-convert RF signals received from the FEM circuitry **1308** based on the synthesized frequency provided by synthesizer circuitry **1306d**. The amplifier circuitry **1306b** may be configured to amplify the down-converted signals and the filter circuitry **1306c** may be a low-pass filter (LPF) or band-pass filter (BPF) configured to remove unwanted signals from the down-converted signals to generate output baseband signals. Output baseband signals may be provided to the baseband circuitry **1310** for further processing. In some embodiments, the output baseband signals may be zero-frequency baseband signals, although this is not a requirement. In some embodiments, mixer circuitry **1306a** of the receive signal path may comprise passive mixers, although the scope of the embodiments is not limited in this respect.

In some embodiments, the mixer circuitry **1306a** of the transmit signal path may be configured to up-convert input baseband signals based on the synthesized frequency provided by the synthesizer circuitry **1306d** to generate RF output signals for the FEM circuitry **1308**. The baseband signals may be provided by the baseband circuitry **1310** and may be filtered by filter circuitry **1306c**.

In some embodiments, the mixer circuitry **1306a** of the receive signal path and the mixer circuitry **1306a** of the transmit signal path may include two or more mixers and may be arranged for quadrature downconversion and upconversion, respectively. In some embodiments, the mixer circuitry **1306a** of the receive signal path and the mixer circuitry **1306a** of the transmit signal path may include two or more mixers and may be arranged for image rejection (e.g., Hartley image rejection). In some embodiments, the mixer circuitry **1306a** of the receive signal path and the mixer circuitry **1306a** of the transmit signal path may be arranged for direct down-conversion and direct up-conversion, respectively. In some embodiments, the mixer circuitry **1306a** of the receive signal path and the mixer circuitry **1306a** of the transmit signal path may be configured for super-heterodyne operation.

In some embodiments, the output baseband signals and the input baseband signals may be analog baseband signals, although the scope of the embodiments is not limited in this respect. In some alternate embodiments, the output baseband

signals and the input baseband signals may be digital baseband signals. In these alternate embodiments, the RF circuitry **1306** may include analog-to-digital converter (ADC) and digital-to-analog converter (DAC) circuitry and the baseband circuitry **1310** may include a digital baseband interface to communicate with the RF circuitry **1306**.

In some dual-mode embodiments, a separate radio IC circuitry may be provided for processing signals for each spectrum, although the scope of the embodiments is not limited in this respect.

In some embodiments, the synthesizer circuitry **1306d** may be a fractional-N synthesizer or a fractional N/N+1 synthesizer, although the scope of the embodiments is not limited in this respect as other types of frequency synthesizers may be suitable. For example, synthesizer circuitry **1306d** may be a delta-sigma synthesizer, a frequency multiplier, or a synthesizer comprising a phase-locked loop with a frequency divider.

The synthesizer circuitry **1306d** may be configured to synthesize an output frequency for use by the mixer circuitry **1306a** of the RF circuitry **1306** based on a frequency input and a divider control input. In some embodiments, the synthesizer circuitry **1306d** may be a fractional N/N+1 synthesizer.

In some embodiments, frequency input may be provided by a voltage controlled oscillator (VCO), although that is not a requirement. Divider control input may be provided by either the baseband circuitry **1310** or the application circuitry **1105/1205** depending on the desired output frequency. In some embodiments, a divider control input (e.g., N) may be determined from a look-up table based on a channel indicated by the application circuitry **1105/1205**.

Synthesizer circuitry **1306d** of the RF circuitry **1306** may include a divider, a delay-locked loop (DLL), a multiplexer and a phase accumulator. In some embodiments, the divider may be a dual modulus divider (DMD) and the phase accumulator may be a digital phase accumulator (DPA). In some embodiments, the DMD may be configured to divide the input signal by either N or N+1 (e.g., based on a carry out) to provide a fractional division ratio. In some example embodiments, the DLL may include a set of cascaded, tunable, delay elements, a phase detector, a charge pump and a D-type flip-flop. In these embodiments, the delay elements may be configured to break a VCO period up into Nd equal packets of phase, where Nd is the number of delay elements in the delay line. In this way, the DLL provides negative feedback to help ensure that the total delay through the delay line is one VCO cycle.

In some embodiments, synthesizer circuitry **1306d** may be configured to generate a carrier frequency as the output frequency, while in other embodiments, the output frequency may be a multiple of the carrier frequency (e.g., twice the carrier frequency, four times the carrier frequency) and used in conjunction with quadrature generator and divider circuitry to generate multiple signals at the carrier frequency with multiple different phases with respect to each other. In some embodiments, the output frequency may be a LO frequency (fLO). In some embodiments, the RF circuitry **1306** may include an IQ/polar converter.

FEM circuitry **1308** may include a receive signal path, which may include circuitry configured to operate on RF signals received from antenna array **1311**, amplify the received signals and provide the amplified versions of the received signals to the RF circuitry **1306** for further processing. FEM circuitry **1308** may also include a transmit signal path, which may include circuitry configured to amplify signals for transmission provided by the RF cir-

cuitry **1306** for transmission by one or more of antenna elements of antenna array **1311**. In various embodiments, the amplification through the transmit or receive signal paths may be done solely in the RF circuitry **1306**, solely in the FEM circuitry **1308**, or in both the RF circuitry **1306** and the FEM circuitry **1308**.

In some embodiments, the FEM circuitry **1308** may include a TX/RX switch to switch between transmit mode and receive mode operation. The FEM circuitry **1308** may include a receive signal path and a transmit signal path. The receive signal path of the FEM circuitry **1308** may include an LNA to amplify received RF signals and provide the amplified received RF signals as an output (e.g., to the RF circuitry **1306**). The transmit signal path of the FEM circuitry **1308** may include a power amplifier (PA) to amplify input RF signals (e.g., provided by RF circuitry **1306**), and one or more filters to generate RF signals for subsequent transmission by one or more antenna elements of the antenna array **1311**.

The antenna array **1311** comprises one or more antenna elements, each of which is configured convert electrical signals into radio waves to travel through the air and to convert received radio waves into electrical signals. For example, digital baseband signals provided by the baseband circuitry **1310** is converted into analog RF signals (e.g., modulated waveform) that will be amplified and transmitted via the antenna elements of the antenna array **1311** including one or more antenna elements (not shown). The antenna elements may be omnidirectional, direction, or a combination thereof. The antenna elements may be formed in a multitude of arrangements as are known and/or discussed herein. The antenna array **1311** may comprise microstrip antennas or printed antennas that are fabricated on the surface of one or more printed circuit boards. The antenna array **1311** may be formed in as a patch of metal foil (e.g., a patch antenna) in a variety of shapes, and may be coupled with the RF circuitry **1306** and/or FEM circuitry **1308** using metal transmission lines or the like.

Processors of the application circuitry **1105/1205** and processors of the baseband circuitry **1310** may be used to execute elements of one or more instances of a protocol stack. For example, processors of the baseband circuitry **1310**, alone or in combination, may be used execute Layer 3, Layer 2, or Layer 1 functionality, while processors of the application circuitry **1105/1205** may utilize data (e.g., packet data) received from these layers and further execute Layer 4 functionality (e.g., TCP and UDP layers). As referred to herein, Layer 3 may comprise a RRC layer, described in further detail below. As referred to herein, Layer 2 may comprise a MAC layer, an RLC layer, and a PDCP layer, described in further detail below. As referred to herein, Layer 1 may comprise a PHY layer of a UE/RAN node, described in further detail below.

FIG. **14** illustrates various protocol functions that may be implemented in a wireless communication device according to various embodiments. In particular, FIG. **14** includes an arrangement **1400** showing interconnections between various protocol layers/entities. The following description of FIG. **14** is provided for various protocol layers/entities that operate in conjunction with the 5G/NR system standards and LTE system standards, but some or all of the aspects of FIG. **14** may be applicable to other wireless communication network systems as well.

The protocol layers of arrangement **1400** may include one or more of PHY **1410**, MAC **1420**, RLC **1430**, PDCP **1440**, SDAP **1447**, RRC **1455**, and NAS layer **1457**, in addition to other higher layer functions not illustrated. The protocol

layers may include one or more service access points (e.g., items **1459**, **1456**, **1450**, **1449**, **1445**, **1435**, **1425**, and **1415** in FIG. **14**) that may provide communication between two or more protocol layers.

The PHY **1410** may transmit and receive physical layer signals **1405** that may be received from or transmitted to one or more other communication devices. The physical layer signals **1405** may comprise one or more physical channels, such as those discussed herein. The PHY **1410** may further perform link adaptation or adaptive modulation and coding (AMC), power control, cell search (e.g., for initial synchronization and handover purposes), and other measurements used by higher layers, such as the RRC **1455**. The PHY **1410** may still further perform error detection on the transport channels, forward error correction (FEC) coding/decoding of the transport channels, modulation/demodulation of physical channels, interleaving, rate matching, mapping onto physical channels, and MIMO antenna processing. In embodiments, an instance of PHY **1410** may process requests from and provide indications to an instance of MAC **1420** via one or more PHY-SAP **1415**. According to some embodiments, requests and indications communicated via PHY-SAP **1415** may comprise one or more transport channels.

Instance(s) of MAC **1420** may process requests from, and provide indications to, an instance of RLC **1430** via one or more MAC-SAPs **1425**. These requests and indications communicated via the MAC-SAP **1425** may comprise one or more logical channels. The MAC **1420** may perform mapping between the logical channels and transport channels, multiplexing of MAC SDUs from one or more logical channels onto TBs to be delivered to PHY **1410** via the transport channels, de-multiplexing MAC SDUs to one or more logical channels from TBs delivered from the PHY **1410** via transport channels, multiplexing MAC SDUs onto TBs, scheduling information reporting, error correction through HARQ, and logical channel prioritization.

Instance(s) of RLC **1430** may process requests from and provide indications to an instance of PDCP **1440** via one or more radio link control service access points (RLC-SAP) **1435**. These requests and indications communicated via RLC-SAP **1435** may comprise one or more RLC channels. The RLC **1430** may operate in a plurality of modes of operation, including: Transparent Mode (TM), Unacknowledged Mode (UM), and Acknowledged Mode (AM). The RLC **1430** may execute transfer of upper layer protocol data units (PDUs), error correction through automatic repeat request (ARQ) for AM data transfers, and concatenation, segmentation and reassembly of RLC SDUs for UM and AM data transfers. The RLC **1430** may also execute re-segmentation of RLC data PDUs for AM data transfers, reorder RLC data PDUs for UM and AM data transfers, detect duplicate data for UM and AM data transfers, discard RLC SDUs for UM and AM data transfers, detect protocol errors for AM data transfers, and perform RLC re-establishment.

Instance(s) of PDCP **1440** may process requests from and provide indications to instance(s) of RRC **1455** and/or instance(s) of SDAP **1447** via one or more packet data convergence protocol service access points (PDCP-SAP) **1445**. These requests and indications communicated via PDCP-SAP **1445** may comprise one or more radio bearers. The PDCP **1440** may execute header compression and decompression of IP data, maintain PDCP Sequence Numbers (SNs), perform in-sequence delivery of upper layer PDUs at re-establishment of lower layers, eliminate duplicates of lower layer SDUs at re-establishment of lower

layers for radio bearers mapped on RLC AM, cipher and decipher control plane data, perform integrity protection and integrity verification of control plane data, control timer-based discard of data, and perform security operations (e.g., ciphering, deciphering, integrity protection, integrity verification, etc.).

Instance(s) of SDAP **1447** may process requests from and provide indications to one or more higher layer protocol entities via one or more SDAP-SAP **1449**. These requests and indications communicated via SDAP-SAP **1449** may comprise one or more QoS flows. The SDAP **1447** may map QoS flows to DRBs, and vice versa, and may also mark QFIs in DL and UL packets. A single SDAP entity **1447** may be configured for an individual PDU session. In the UL direction, the NG-RAN **810** may control the mapping of QoS Flows to DRB(s) in two different ways, reflective mapping or explicit mapping. For reflective mapping, the SDAP **1447** of a UE **801** may monitor the QFIs of the DL packets for each DRB, and may apply the same mapping for packets flowing in the UL direction. For a DRB, the SDAP **1447** of the UE **801** may map the UL packets belonging to the QoS flows(s) corresponding to the QoS flow ID(s) and PDU session observed in the DL packets for that DRB. To enable reflective mapping, the NG-RAN **1010** may mark DL packets over the Uu interface with a QoS flow ID. The explicit mapping may involve the RRC **1455** configuring the SDAP **1447** with an explicit QoS flow to DRB mapping rule, which may be stored and followed by the SDAP **1447**. In embodiments, the SDAP **1447** may only be used in NR implementations and may not be used in LTE implementations.

The RRC **1455** may configure, via one or more management service access points (M-SAP), aspects of one or more protocol layers, which may include one or more instances of PHY **1410**, MAC **1420**, RLC **1430**, PDCP **1440** and SDAP **1447**. In embodiments, an instance of RRC **1455** may process requests from and provide indications to one or more NAS entities **1457** via one or more RRC-SAPs **1456**. The main services and functions of the RRC **1455** may include broadcast of system information (e.g., included in MIBs or SIBs related to the NAS), broadcast of system information related to the access stratum (AS), paging, establishment, maintenance and release of an RRC connection between the UE **801** and RAN **810** (e.g., RRC connection paging, RRC connection establishment, RRC connection modification, and RRC connection release), establishment, configuration, maintenance and release of point to point Radio Bearers, security functions including key management, inter-RAT mobility, and measurement configuration for UE measurement reporting. The MIBs and SIBs may comprise one or more IEs, which may each comprise individual data fields or data structures.

The NAS **1457** may form the highest stratum of the control plane between the UE **801** and the AMF **1021**. The NAS **1457** may support the mobility of the UEs **801** and the session management procedures to establish and maintain IP connectivity between the UE **801** and a P-GW in LTE systems.

According to various embodiments, one or more protocol entities of arrangement **1400** may be implemented in UEs **801**, RAN nodes **811**, AMF **1021** in NR implementations or MME **921** in LTE implementations, UPF **1002** in NR implementations or S-GW **922** and P-GW **923** in LTE implementations, or the like to be used for control plane or user plane communications protocol stack between the aforementioned devices. In such embodiments, one or more protocol entities that may be implemented in one or more of UE **801**, gNB **811**, AMF **1021**, etc. may communicate with

a respective peer protocol entity that may be implemented in or on another device using the services of respective lower layer protocol entities to perform such communication. In some embodiments, a gNB-CU of the gNB **811** may host the RRC **1455**, SDAP **1447**, and PDCP **1440** of the gNB that controls the operation of one or more gNB-DUs, and the gNB-DUs of the gNB **811** may each host the RLC **1430**, MAC **1420**, and PHY **1410** of the gNB **811**.

In a first example, a control plane protocol stack may comprise, in order from highest layer to lowest layer, NAS **1457**, RRC **1455**, PDCP **1440**, RLC **1430**, MAC **1420**, and PHY **1410**. In this example, upper layers **1460** may be built on top of the NAS **1457**, which includes an IP layer **1461**, an SCTP **1462**, and an application layer signaling protocol (AP) **1463**.

In NR implementations, the AP **1463** may be an NG application protocol layer (NGAP or NG-AP) **1463** for the NG interface **813** defined between the NG-RAN node **811** and the AMF **1021**, or the AP **1463** may be an Xn application protocol layer (XnAP or Xn-AP) **1463** for the Xn interface **812** that is defined between two or more RAN nodes **811**.

The NG-AP **1463** may support the functions of the NG interface **813** and may comprise Elementary Procedures (EPs). An NG-AP EP may be a unit of interaction between the NG-RAN node **811** and the AMF **1021**. The NG-AP **1463** services may comprise two groups: UE-associated services (e.g., services related to a UE **801**) and non-UE-associated services (e.g., services related to the whole NG interface instance between the NG-RAN node **811** and AMF **1021**). These services may include functions including, but not limited to: a paging function for the sending of paging requests to NG-RAN nodes **811** involved in a particular paging area; a UE context management function for allowing the AMF **1021** to establish, modify, and/or release a UE context in the AMF **1021** and the NG-RAN node **811**; a mobility function for UEs **801** in ECM-CONNECTED mode for intra-system HOs to support mobility within NG-RAN and inter-system HOs to support mobility from/to EPS systems; a NAS Signaling Transport function for transporting or rerouting NAS messages between UE **801** and AMF **1021**; a NAS node selection function for determining an association between the AMF **1021** and the UE **801**; NG interface management function(s) for setting up the NG interface and monitoring for errors over the NG interface; a warning message transmission function for providing means to transfer warning messages via NG interface or cancel ongoing broadcast of warning messages; a Configuration Transfer function for requesting and transferring of RAN configuration information (e.g., SON information, performance measurement (PM) data, etc.) between two RAN nodes **811** via CN **820**; and/or other like functions.

The XnAP **1463** may support the functions of the Xn interface **812** and may comprise XnAP basic mobility procedures and XnAP global procedures. The XnAP basic mobility procedures may comprise procedures used to handle UE mobility within the NG RAN **811** (or E-UTRAN **910**), such as handover preparation and cancellation procedures, SN Status Transfer procedures, UE context retrieval and UE context release procedures, RAN paging procedures, dual connectivity related procedures, and the like. The XnAP global procedures may comprise procedures that are not related to a specific UE **801**, such as Xn interface setup and reset procedures, NG-RAN update procedures, cell activation procedures, and the like.

In LTE implementations, the AP **1463** may be an S1 Application Protocol layer (S1-AP) **1463** for the S1 interface **813** defined between an E-UTRAN node **811** and an MME,

or the AP **1463** may be an X2 application protocol layer (X2AP or X2-AP) **1463** for the X2 interface **812** that is defined between two or more E-UTRAN nodes **811**.

The S1 Application Protocol layer (S1-AP) **1463** may support the functions of the S1 interface, and similar to the NG-AP discussed previously, the S1-AP may comprise S1-AP EPs. An S1-AP EP may be a unit of interaction between the E-UTRAN node **811** and an MME **921** within an LTE CN **820**. The S1-AP **1463** services may comprise two groups: UE-associated services and non UE-associated services. These services perform functions including, but not limited to: E-UTRAN Radio Access Bearer (E-RAB) management, UE capability indication, mobility, NAS signaling transport, RAN Information Management (RIM), and configuration transfer.

The X2AP **1463** may support the functions of the X2 interface **812** and may comprise X2AP basic mobility procedures and X2AP global procedures. The X2AP basic mobility procedures may comprise procedures used to handle UE mobility within the E-UTRAN **820**, such as handover preparation and cancellation procedures, SN Status Transfer procedures, UE context retrieval and UE context release procedures, RAN paging procedures, dual connectivity related procedures, and the like. The X2AP global procedures may comprise procedures that are not related to a specific UE **801**, such as X2 interface setup and reset procedures, load indication procedures, error indication procedures, cell activation procedures, and the like.

The SCTP layer (alternatively referred to as the SCTP/IP layer) **1462** may provide guaranteed delivery of application layer messages (e.g., NGAP or XnAP messages in NR implementations, or S1-AP or X2AP messages in LTE implementations). The SCTP **1462** may ensure reliable delivery of signaling messages between the RAN node **811** and the AMF **1021**/MME **921** based, in part, on the IP protocol, supported by the IP **1461**. The Internet Protocol layer (IP) **1461** may be used to perform packet addressing and routing functionality. In some implementations the IP layer **1461** may use point-to-point transmission to deliver and convey PDUs. In this regard, the RAN node **811** may comprise L2 and L1 layer communication links (e.g., wired or wireless) with the MME/AMF to exchange information.

In a second example, a user plane protocol stack may comprise, in order from highest layer to lowest layer, SDAP **1447**, PDCP **1440**, RLC **1430**, MAC **1420**, and PHY **1410**. The user plane protocol stack may be used for communication between the UE **801**, the RAN node **811**, and UPF **1002** in NR implementations or an S-GW **922** and P-GW **923** in LTE implementations. In this example, upper layers **1451** may be built on top of the SDAP **1447**, and may include a user datagram protocol (UDP) and IP security layer (UDP/IP) **1452**, a General Packet Radio Service (GPRS) Tunneling Protocol for the user plane layer (GTP-U) **1453**, and a User Plane PDU layer (UP PDU) **1463**.

The transport network layer **1454** (also referred to as a "transport layer") may be built on IP transport, and the GTP-U **1453** may be used on top of the UDP/IP layer **1452** (comprising a UDP layer and IP layer) to carry user plane PDUs (UP-PDUs). The IP layer (also referred to as the "Internet layer") may be used to perform packet addressing and routing functionality. The IP layer may assign IP addresses to user data packets in any of IPv4, IPv6, or PPP formats, for example.

The GTP-U **1453** may be used for carrying user data within the GPRS core network and between the radio access network and the core network. The user data transported can be packets in any of IPv4, IPv6, or PPP formats, for

example. The UDP/IP **1452** may provide checksums for data integrity, port numbers for addressing different functions at the source and destination, and encryption and authentication on the selected data flows. The RAN node **811** and the S-GW **922** may utilize an S1-U interface to exchange user plane data via a protocol stack comprising an L1 layer (e.g., PHY **1410**), an L2 layer (e.g., MAC **1420**, RLC **1430**, PDCP **1440**, and/or SDAP **1447**), the UDP/IP layer **1452**, and the GTP-U **1453**. The S-GW **922** and the P-GW **923** may utilize an S5/S8a interface to exchange user plane data via a protocol stack comprising an L1 layer, an L2 layer, the UDP/IP layer **1452**, and the GTP-U **1453**. As discussed previously, NAS protocols may support the mobility of the UE **801** and the session management procedures to establish and maintain IP connectivity between the UE **801** and the P-GW **923**.

Moreover, although not shown by FIG. **14**, an application layer may be present above the AP **1463** and/or the transport network layer **1454**. The application layer may be a layer in which a user of the UE **801**, RAN node **811**, or other network element interacts with software applications being executed, for example, by application circuitry **1105** or application circuitry **1205**, respectively. The application layer may also provide one or more interfaces for software applications to interact with communications systems of the UE **801** or RAN node **811**, such as the baseband circuitry **1310**. In some embodiments the IP layer and/or the application layer may provide the same or similar functionality as layers 5-7, or portions thereof, of the Open Systems Interconnection (OSI) model (e.g., OSI Layer 7—the application layer, OSI Layer 6—the presentation layer, and OSI Layer 5—the session layer).

FIG. **15** illustrates components of a core network in accordance with various embodiments. The components of the CN **920** may be implemented in one physical node or separate physical nodes including components to read and execute instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium). In embodiments, the components of CN **1020** may be implemented in a same or similar manner as discussed herein with regard to the components of CN **920**. In some embodiments, NFV is utilized to virtualize any or all of the above-described network node functions via executable instructions stored in one or more computer-readable storage mediums (described in further detail below). A logical instantiation of the CN **920** may be referred to as a network slice **1501**, and individual logical instantiations of the CN **920** may provide specific network capabilities and network characteristics. A logical instantiation of a portion of the CN **920** may be referred to as a network sub-slice **1502** (e.g., the network sub-slice **1502** is shown to include the P-GW **923** and the PCRF **926**).

As used herein, the terms “instantiate,” “instantiation,” and the like may refer to the creation of an instance, and an “instance” may refer to a concrete occurrence of an object, which may occur, for example, during execution of program code. A network instance may refer to information identifying a domain, which may be used for traffic detection and routing in case of different IP domains or overlapping IP addresses. A network slice instance may refer to a set of network functions (NFs) instances and the resources (e.g., compute, storage, and networking resources) required to deploy the network slice.

With respect to 5G systems (see, e.g., FIG. **10**), a network slice always comprises a RAN part and a CN part. The support of network slicing relies on the principle that traffic for different slices is handled by different PDU sessions. The

network can realize the different network slices by scheduling and also by providing different L1/L2 configurations. The UE **1001** provides assistance information for network slice selection in an appropriate RRC message, if it has been provided by NAS. While the network can support large number of slices, the UE need not support more than 8 slices simultaneously.

A network slice may include the CN **1020** control plane and user plane NFs, NG-RANs **1010** in a serving PLMN, and a N3IWF functions in the serving PLMN. Individual network slices may have different S-NSSAI and/or may have different SSTs. NSSAI includes one or more S-NSSAIs, and each network slice is uniquely identified by an S-NSSAI. Network slices may differ for supported features and network functions optimizations, and/or multiple network slice instances may deliver the same service/features but for different groups of UEs **1001** (e.g., enterprise users). For example, individual network slices may deliver different committed service(s) and/or may be dedicated to a particular customer or enterprise. In this example, each network slice may have different S-NSSAIs with the same SST but with different slice differentiators. Additionally, a single UE may be served with one or more network slice instances simultaneously via a 5G AN and associated with eight different S-NSSAIs. Moreover, an AMF **1021** instance serving an individual UE **1001** may belong to each of the network slice instances serving that UE.

Network Slicing in the NG-RAN **1010** involves RAN slice awareness. RAN slice awareness includes differentiated handling of traffic for different network slices, which have been pre-configured. Slice awareness in the NG-RAN **1010** is introduced at the PDU session level by indicating the S-NSSAI corresponding to a PDU session in all signaling that includes PDU session resource information. How the NG-RAN **1010** supports the slice enabling in terms of NG-RAN functions (e.g., the set of network functions that comprise each slice) is implementation dependent. The NG-RAN **1010** selects the RAN part of the network slice using assistance information provided by the UE **1001** or the 5GC **1020**, which unambiguously identifies one or more of the pre-configured network slices in the PLMN. The NG-RAN **1010** also supports resource management and policy enforcement between slices as per SLAs. A single NG-RAN node may support multiple slices, and the NG-RAN **1010** may also apply an appropriate RRM policy for the SLA in place to each supported slice. The NG-RAN **1010** may also support QoS differentiation within a slice.

The NG-RAN **1010** may also use the UE assistance information for the selection of an AMF **1021** during an initial attach, if available. The NG-RAN **1010** uses the assistance information for routing the initial NAS to an AMF **1021**. If the NG-RAN **1010** is unable to select an AMF **1021** using the assistance information, or the UE **1001** does not provide any such information, the NG-RAN **1010** sends the NAS signaling to a default AMF **1021**, which may be among a pool of AMFs **1021**. For subsequent accesses, the UE **1001** provides a temp ID, which is assigned to the UE **1001** by the 5GC **1020**, to enable the NG-RAN **1010** to route the NAS message to the appropriate AMF **1021** as long as the temp ID is valid. The NG-RAN **1010** is aware of, and can reach, the AMF **1021** that is associated with the temp ID. Otherwise, the method for initial attach applies.

The NG-RAN **1010** supports resource isolation between slices. NG-RAN **1010** resource isolation may be achieved by means of RRM policies and protection mechanisms that should avoid that shortage of shared resources if one slice breaks the service level agreement for another slice. In some

embodiments, it is possible to fully dedicate NG-RAN 1010 resources to a certain slice. How NG-RAN 1010 supports resource isolation is implementation dependent.

Some slices may be available only in part of the network. Awareness in the NG-RAN 1010 of the slices supported in the cells of its neighbors may be beneficial for inter-frequency mobility in connected mode. The slice availability may not change within the UE's registration area. The NG-RAN 1010 and the 5GC 1020 are responsible to handle a service request for a slice that may or may not be available in a given area. Admission or rejection of access to a slice may depend on factors such as support for the slice, availability of resources, support of the requested service by NG-RAN 1010.

The UE 1001 may be associated with multiple network slices simultaneously. In case the UE 1001 is associated with multiple slices simultaneously, only one signaling connection is maintained, and for intra-frequency cell reselection, the UE 1001 tries to camp on the best cell. For inter-frequency cell reselection, dedicated priorities can be used to control the frequency on which the UE 1001 camps. The 5GC 1020 is to validate that the UE 1001 has the rights to access a network slice. Prior to receiving an Initial Context Setup Request message, the NG-RAN 1010 may be allowed to apply some provisional/local policies, based on awareness of a particular slice that the UE 1001 is requesting to access. During the initial context setup, the NG-RAN 1010 is informed of the slice for which resources are being requested.

NFV architectures and infrastructures may be used to virtualize one or more NFs, alternatively performed by proprietary hardware, onto physical resources comprising a combination of industry-standard server hardware, storage hardware, or switches. In other words, NFV systems can be used to execute virtual or reconfigurable implementations of one or more EPC components/functions.

FIG. 16 is a block diagram illustrating components, according to some example embodiments, of a system 1600 to support NFV. The system 1600 is illustrated as including a VIM 1602, an NFVI 1604, an VNFM 1606, VNFs 1608, an EM 1610, an NFVO 1612, and a NM 1614.

The VIM 1602 manages the resources of the NFVI 1604. The NFVI 1604 can include physical or virtual resources and applications (including hypervisors) used to execute the system 1600. The VIM 1602 may manage the life cycle of virtual resources with the NFVI 1604 (e.g., creation, maintenance, and tear down of VMs associated with one or more physical resources), track VM instances, track performance, fault and security of VM instances and associated physical resources, and expose VM instances and associated physical resources to other management systems.

The VNFM 1606 may manage the VNFs 1608. The VNFs 1608 may be used to execute EPC components/functions. The VNFM 1606 may manage the life cycle of the VNFs 1608 and track performance, fault and security of the virtual aspects of VNFs 1608. The EM 1610 may track the performance, fault and security of the functional aspects of VNFs 1608. The tracking data from the VNFM 1606 and the EM 1610 may comprise, for example, PM data used by the VIM 1602 or the NFVI 1604. Both the VNFM 1606 and the EM 1610 can scale up/down the quantity of VNFs of the system 1600.

The NFVO 1612 may coordinate, authorize, release and engage resources of the NFVI 1604 in order to provide the requested service (e.g., to execute an EPC function, component, or slice). The NM 1614 may provide a package of end-user functions with the responsibility for the manage-

ment of a network, which may include network elements with VNFs, non-virtualized network functions, or both (management of the VNFs may occur via the EM 1610).

FIG. 17 is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. 17 shows a diagrammatic representation of hardware resources 1700 including one or more processors (or processor cores) 1710, one or more memory/storage devices 1720, and one or more communication resources 1730, each of which may be communicatively coupled via a bus 1740. For embodiments where node virtualization (e.g., NFV) is utilized, a hypervisor 1702 may be executed to provide an execution environment for one or more network slices/sub-slices to utilize the hardware resources 1700.

The processors 1710 may include, for example, a processor 1712 and a processor 1714. The processor(s) 1710 may be, for example, a central processing unit (CPU), a reduced instruction set computing (RISC) processor, a complex instruction set computing (CISC) processor, a graphics processing unit (GPU), a DSP such as a baseband processor, an ASIC, an FPGA, a radio-frequency integrated circuit (RFIC), another processor (including those discussed herein), or any suitable combination thereof.

The memory/storage devices 1720 may include main memory, disk storage, or any suitable combination thereof. The memory/storage devices 1720 may include, but are not limited to, any type of volatile or nonvolatile memory such as dynamic random access memory (DRAM), static random access memory (SRAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), Flash memory, solid-state storage, etc.

The communication resources 1730 may include interconnection or network interface components or other suitable devices to communicate with one or more peripheral devices 1704 or one or more databases 1706 via a network 1708. For example, the communication resources 1730 may include wired communication components (e.g., for coupling via USB), cellular communication components, NFC components, Bluetooth® (or Bluetooth® Low Energy) components, Wi-Fi® components, and other communication components.

Instructions 1750 may comprise software, a program, an application, an applet, an app, or other executable code for causing at least any of the processors 1710 to perform any one or more of the methodologies discussed herein. The instructions 1750 may reside, completely or partially, within at least one of the processors 1710 (e.g., within the processor's cache memory), the memory/storage devices 1720, or any suitable combination thereof. Furthermore, any portion of the instructions 1750 may be transferred to the hardware resources 1700 from any combination of the peripheral devices 1704 or the databases 1706. Accordingly, the memory of processors 1710, the memory/storage devices 1720, the peripheral devices 1704, and the databases 1706 are examples of computer-readable and machine-readable media.

In some embodiments, the electronic device(s), network (s), system(s), chip(s) or component(s), or portions or implementations thereof, of FIGS. 8-17, or some other figure herein, may be configured to perform one or more processes, techniques, or methods as described herein, or portions thereof. One such example method 1800 is depicted in FIG.

18. For example, at 1802, the method 1800 may include processing, or causing to process, a received a random positioning request message. At 1804, the method 1800 may include transmitting, or causing to transmit, a random positioning response message, to determine estimated position coordinates. FIG. 19 depicts an example method 1900 according to embodiments. At 1902, the method 1900 can include transmitting, or causing to transmit, a random positioning request message. At 1904, the method 1900 can include processing, or causing to process, a received random positioning response message, to determine estimated position coordinates.

For one or more embodiments, at least one of the components set forth in one or more of the preceding figures may be configured to perform one or more operations, techniques, processes, and/or methods as set forth in the example section below. For example, the baseband circuitry as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below. For another example, circuitry associated with a UE, base station, network element, etc. as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below in the example section.

EXAMPLES

Example 1 may include a method for advanced NR based RAT dependent positioning, comprising: A 2 step RACH procedure to trigger DL only (D-TDOA), UL only (U-TDOA) or DL&UL positioning (RTT) or any other positioning approach; and NR signaling to trigger DL only (D-TDOA), UL only (U-TDOA) or DL&UL positioning (RTT) or any other positioning approach comprising any of the following: RRC; MAC; LPP; NPP; Physical layer signaling.

Example 2 may include the method of example 1 or some other example herein, wherein the 2 step RACH procedure comprises two messages Msg-A and Msg-B.

Example 3 may include the method of example 2 or some other example herein, wherein Msg-A is sent by a UE to a gNB, and comprises: PRACH transmission (RACH Preamble); An associated PUSCH transmission with embedded MAC IE; and An associate UL SRS transmission.

Example 4 may include the method of example 3 or some other example herein, wherein Msg-A with associated PUSCH transmission with embedded MAC IE/RRC IE comprises:

A type of requested positioning procedure (e.g., UL only-UTDOA, DL&UL-RTT with serving cell, RTT with serving+neighboring cells, DL only-D-TDOA, etc.); A request for additional allocation of reference signals or their activation if preconfigured (e.g., Request for UL SRS/PRS resource allocation; Request for DL SRS/PRS resource allocation; Request for UL SRS/PRS resource allocation followed by DL CSI-RS/PRS; Request for DL CSI-RS/PRS resource allocation followed by UL SRS/PRS); A type of requested UL positioning reference resources (periodic/semi-persistent/aperiodic, etc); A type of requested DL positioning reference resources (periodic/semi-persistent/aperiodic, etc); Initial SLP Measurements (e.g., Initial timestamps t_1 of RACH preamble transmission; Initial timestamps t_1 of UL SRS/PRS transmission, if RACH transmission is also associated with UL SRS/PRS transmission; A priori measurement on DL PRS/CSI-RS positioning resource and associated resource IDs (e.g., RSRP, time

stamps t_2), if available; List of cell IDs detected by UE; List of DL PRS/CSI-RS Resource IDs; List of SSB indexes); A UE ID; Other positioning related parameters; Previous UE coordinate; UE-based location capability; Positioning service related info (e.g., QoS for positioning procedure, e.g., latency, accuracy levels); PUSCH message type for positioning; Depending on positioning procedure not all fields may be present. Given that PUSCH payload size can be predetermined the message may need to have a fixed size and padded to fixed size if payload is smaller. In order to properly interpret the message the message type information can be added; Msg B resource; The UE may indicate time interval or specific resources when it expects from gNB to receive MsgB or monitor PDCCH channel for MsgB reception; Wherein this may be useful to reduce UE power consumption and allocate some processing time for network/gNB; and Information for contention resolution if contention based 2-step RACH positioning procedure is employed

Example 5 may include the method of example 2 or some other example herein, wherein Msg-B is sent by a gNB to a UE in response of Msg-A and comprises: Estimated UE coordinate based on serving/neighboring cell measurements; Metric characterizing coordinate estimation quality or uncertainty; Time stamps t_{2i} of PRACH or UL reference signal reception and corresponding Resource IDs and QCL information if any; Time stamps t_{ai} of DL reference signal transmission (SSB/DL PRS/DL CSI-RS) and corresponding Resource IDs and QCL information if any; Request/trigger for additional positioning measurements and its type; Configuration or activation of additional set of DL/UL resource for transmission of positioning reference signals to conduct additional measurements of SLPs and their type; UL SRS/PRS resource set configuration for UE requested positioning procedure (e.g., Information on UL SRS/PRS transmission resources (resource ID), their type (periodic/semi-persistent/aperiodic); UL PRS/SRS sequence initialization information (sequence ID)); UL SRS/PRS measurement data (e.g., Information on timestamps of UL SRS/PRS signal reception as well as other SLP measured by each TRP/gNB; Information on geolocation coordinates of TRPs/gNBs that measured SLPs); Timestamp of MsgB transmission; Configuration of DL PRS/CSI-RS resources for UE requested positioning procedure (e.g., DL PRS transmission resources and the ID (Resource set for DL positioning); DL PRS sequence initialization parameters (sequence IDs)); DL PRS/CSI-RS measurement data (e.g., Information on timestamps of UL SRS/PRS signal reception as well as other SLP measured by each TRP/gNB; Information on geolocation coordinates of TRPs/gNBs that measured SLPs); Timestamp of MsgB transmission or any associated with MsgB DL PRS or CSI-RS signal; and The same contention resolution information that is initiated and carried in MsgA from UE.

Example 6 is an apparatus, comprising: means to transmit a random positioning request message; and means to process a received random positioning response message, to determine estimated position coordinates.

Example 7 includes the subject matter of example 6, or some other example herein, wherein the estimated position coordinates are determined based on a positioning reference signal and one or more measured signal location parameters (SLP).

Example 8 includes the subject matter of example 7, or some other example herein, wherein the SLPs are measured on a downlink, an uplink, or both.

Example 9 includes the subject matter of example 7 or 8, or some other example herein, wherein the positioning reference signal is a PRACH, SRS, or ranging signal.

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Example 10 includes the subject matter of any of examples 6-9, or some other example herein, wherein the apparatus is a user equipment, or a portion thereof.

Example 11 is an apparatus, comprising: means to process a received a random positioning request message; and means to transmit a random positioning response message, to determine estimated position coordinates.

Example 12 includes the subject matter of example 11, or some other example herein, wherein the estimated position coordinates are determined based on a positioning reference signal and one or more measured signal location parameters (SLP).

Example 13 includes the subject matter of example 12, or some other example herein, wherein the SLPs are measured on a downlink, an uplink, or both.

Example 14 includes the subject matter of example 12 or 13, or some other example herein, wherein the positioning reference signal is a PRACH, SRS, or ranging signal.

Example 15 includes the subject matter of any of examples 11-14, or some other example herein, wherein the apparatus is a base station, or a portion thereof.

Example 16 is an apparatus, configured to: transmit a random positioning request message; and process a received random positioning response message, to determine estimated position coordinates.

Example 17 includes the subject matter of example 16, or some other example herein, wherein the estimated position coordinates are determined based on a positioning reference signal and one or more measured signal location parameters (SLP).

Example 18 includes the subject matter of example 17, or some other example herein, wherein the SLPs are measured on a downlink, an uplink, or both.

Example 19 includes the subject matter of example 17 or 18, or some other example herein, wherein the positioning reference signal is a PRACH, SRS, or ranging signal.

Example 20 includes the subject matter of any of examples 16-19, or some other example herein, wherein the apparatus is a user equipment, or a portion thereof.

Example 21 is an apparatus, configured to: process a received a random positioning request message; and transmit a random positioning response message, to determine estimated position coordinates.

Example 22 includes the subject matter of example 21, or some other example herein, wherein the estimated position coordinates are determined based on a positioning reference signal and one or more measured signal location parameters (SLP).

Example 23 includes the subject matter of example 22, or some other example herein, wherein the SLPs are measured on a downlink, an uplink, or both.

Example 24 includes the subject matter of example 22 or 23, or some other example herein, wherein the positioning reference signal is a PRACH, SRS, or ranging signal.

Example 25 includes the subject matter of any of examples 21-24, or some other example herein, wherein the apparatus is a base station, or a portion thereof.

Example 26 is a method, comprising: transmitting, or causing to transmit, a random positioning request message; and processing, or causing to process, a received random positioning response message, to determine estimated position coordinates.

Example 27 includes the subject matter of example 26, or some other example herein, further comprising determining, or causing to determine, the estimated position coordinates based on a positioning reference signal and one or more measured signal location parameters (SLP).

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Example 28 includes the subject matter of example 27, or some other example herein, further comprising measuring, or causing to measure, the SLPs on a downlink, an uplink, or both.

Example 29 includes the subject matter of example 27 or 28, or some other example herein, wherein the positioning reference signal is a PRACH, SRS, or ranging signal.

Example 30 includes the subject matter of any of examples 6-9, or some other example herein, wherein the method is performed, in whole or in part, by a user equipment, or a portion thereof.

Example 31 is a method, comprising: processing, or causing to process, a received a random positioning request message; and transmitting, or causing to transmit, a random positioning response message, to determine estimated position coordinates.

Example 32 includes the subject matter of example 31, or some other example herein, further comprising determining, or causing to determine, the estimated position coordinates based on a positioning reference signal and one or more measured signal location parameters (SLP).

Example 33 includes the subject matter of example 32, or some other example herein, further comprising measuring, or causing to measure, the SLPs on a downlink, an uplink, or both.

Example 34 includes the subject matter of example 32 or 33, or some other example herein, wherein the positioning reference signal is a PRACH, SRS, or ranging signal.

Example 35 includes the subject matter of any of examples 31-34, or some other example herein, wherein the method is performed, in whole or in part, by a base station, or a portion thereof.

Example Z01 may include an apparatus comprising means to perform one or more elements of a method described in or related to any of examples 1-35, or any other method or process described herein.

Example Z02 may include one or more non-transitory computer-readable media comprising instructions to cause an electronic device, upon execution of the instructions by one or more processors of the electronic device, to perform one or more elements of a method described in or related to any of examples 1-35, or any other method or process described herein.

Example Z03 may include an apparatus comprising logic, modules, or circuitry to perform one or more elements of a method described in or related to any of examples 1-35, or any other method or process described herein.

Example Z04 may include a method, technique, or process as described in or related to any of examples 1-35, or portions or parts thereof.

Example Z05 may include an apparatus comprising: one or more processors and one or more computer-readable media comprising instructions that, when executed by the one or more processors, cause the one or more processors to perform the method, techniques, or process as described in or related to any of examples 1-35, or portions thereof.

Example Z06 may include a signal as described in or related to any of examples 1-35, or portions or parts thereof.

Example Z07 may include a signal in a wireless network as shown and described herein.

Example Z08 may include a method of communicating in a wireless network as shown and described herein.

Example Z09 may include a system for providing wireless communication as shown and described herein.

Example Z10 may include a device for providing wireless communication as shown and described herein.

Any of the above-described examples may be combined with any other example (or combination of examples), unless explicitly stated otherwise. The foregoing description of one or more embodiments provides illustration and description, but is not intended to be exhaustive or to limit the scope of embodiments to the precise form disclosed. Modifications and variations are possible in light of the above teachings or may be acquired from practice of various embodiments.

Abbreviations and Terminology

For the purposes of the present document, the following abbreviations may apply to the examples and embodiments discussed herein, but is not meant to be limiting.

3GPP Third Generation Partnership Project
 4G Fourth Generation
 5G Fifth Generation
 5GC 5G Core network
 ACK Acknowledgement
 AF Application Function
 AM Acknowledged Mode
 AMBR Aggregate Maximum Bit Rate
 AMF Access and Mobility Management Function
 AN Access Network
 ANR Automatic Neighbour Relation
 AP Application Protocol, Antenna Port, Access Point
 API Application Programming Interface
 APN Access Point Name
 ARP Allocation and Retention Priority
 ARQ Automatic Repeat Request
 AS Access Stratum
 ASN.1 Abstract Syntax Notation One
 AUSF Authentication Server Function
 AWGN Additive White Gaussian Noise
 BCH Broadcast Channel
 BER Bit Error Ratio
 BFD Beam Failure Detection
 BLER Block Error Rate
 BPSK Binary Phase Shift Keying
 BRAS Broadband Remote Access Server
 BSS Business Support System
 BS Base Station
 BSR Buffer Status Report
 BW Bandwidth
 BWP Bandwidth Part
 C-RNTI Cell Radio Network Temporary Identity
 CA Carrier Aggregation, Certification Authority
 CAPEX CAPital EXpenditure
 CBRA Contention Based Random Access
 CC Component Carrier, Country Code, Cryptographic Checksum
 CCA Clear Channel Assessment
 CCE Control Channel Element
 CCH Common Control Channel
 CE Coverage Enhancement
 CDM Content Delivery Network
 CDMA Code-Division Multiple Access
 CFRA Contention Free Random Access
 CG Cell Group
 CI Cell Identity
 CID Cell-ID (e.g., positioning method)
 CIM Common Information Model
 CIR Carrier to Interference Ratio
 CK Cipher Key
 CM Connection Management, Conditional Mandatory
 CMAS Commercial Mobile Alert Service

CMD Command
 CMS Cloud Management System
 CO Conditional Optional
 CoMP Coordinated Multi-Point
 CORESET Control Resource Set
 COTS Commercial Off-The-Shelf
 CP Control Plane, Cyclic Prefix, Connection Point
 CPD Connection Point Descriptor
 CPE Customer Premise Equipment
 CPICH Common Pilot Channel
 CQI Channel Quality Indicator
 CPU CSI processing unit, Central Processing Unit
 C/R Command/Response field bit
 CRAN Cloud Radio Access Network, Cloud RAN
 CRB Common Resource Block
 CRC Cyclic Redundancy Check
 CRI Channel-State Information Resource Indicator, CSI-RS Resource Indicator
 C-RNTI Cell RNTI
 CS Circuit Switched
 CSAR Cloud Service Archive
 CSI Channel-State Information
 CSI-IM CSI Interference Measurement
 CSI-RS CSI Reference Signal
 CSI-RSRP CSI reference signal received power
 CSI-RSRQ CSI reference signal received quality
 CSI-SINR CSI signal-to-noise and interference ratio
 CSMA Carrier Sense Multiple Access
 CSMA/CA CSMA with collision avoidance
 CSS Common Search Space, Cell-specific Search Space
 CTS Clear-to-Send
 CW Codeword
 CWS Contention Window Size
 D2D Device-to-Device
 DC Dual Connectivity, Direct Current
 DCI Downlink Control Information
 DF Deployment Flavour
 DL Downlink
 DMTF Distributed Management Task Force
 DPDK Data Plane Development Kit
 DM-RS, DMRS Demodulation Reference Signal
 DN Data network
 DRB Data Radio Bearer
 DRS Discovery Reference Signal
 DRX Discontinuous Reception
 DSL Domain Specific Language, Digital Subscriber Line
 DSLAM DSL Access Multiplexer
 DwPTS Downlink Pilot Time Slot
 E-LAN Ethernet Local Area Network
 E2E End-to-End
 ECCA extended clear channel assessment, extended CCA
 ECCE Enhanced Control Channel Element, Enhanced CCE
 ED Energy Detection
 EDGE Enhanced Data Rates for GSM Evolution (GSM Evolution)
 EGMF Exposure Governance Management Function
 EGPRS Enhanced GPRS
 EIR Equipment Identity Register
 eLAA enhanced Licensed Assisted Access, enhanced LAA
 EM Element Manager
 eMBB Enhanced Mobile Broadband
 EMS Element Management System
 eNB evolved NodeB, E-UTRAN Node B
 EN-DC E-UTRA-NR Dual Connectivity

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EPC Evolved Packet Core
 EPDCCH enhanced PDCCH, enhanced Physical Down-
 link Control Channel
 EPRE Energy per resource element
 EPS Evolved Packet System
 EREG enhanced REG, enhanced resource element groups
 ETSI European Telecommunications Standards Institute
 ETWS Earthquake and Tsunami Warning System
 eUICC embedded UICC, embedded Universal Integrated
 Circuit Card
 E-UTRA Evolved UTRA
 E-UTRAN Evolved UTRAN
 EV2X Enhanced V2X
 FLAP F1 Application Protocol
 F1-C F1 Control plane interface
 F1-U F1 User plane interface
 FACCH Fast Associated Control Channel
 FACCH/F Fast Associated Control Channel/Full rate
 FACCH/H Fast Associated Control Channel/Half rate
 FACH Forward Access Channel
 FAUSCH Fast Uplink Signalling Channel
 FB Functional Block
 FBI Feedback Information
 FCC Federal Communications Commission
 FCCH Frequency Correction CHannel
 FDD Frequency Division Duplex
 FDM Frequency Division Multiplex
 FDMA Frequency Division Multiple Access
 FE Front End
 FEC Forward Error Correction
 FFS For Further Study
 FFT Fast Fourier Transformation
 feLAA further enhanced Licensed Assisted Access, fur-
 ther enhanced LAA
 FN Frame Number
 FPGA Field-Programmable Gate Array
 FR Frequency Range
 G-RNTI GERAN Radio Network Temporary Identity
 GERAN GSM EDGE RAN, GSM EDGE Radio Access
 Network
 GGSN Gateway GPRS Support Node
 GLONASS GLObal'naya NAVigatsionnaya Sputnikovaya
 Sistema (Engl.: Global Navigation Satellite System)
 gNB Next Generation NodeB
 gNB-CU gNB-centralized unit, Next Generation NodeB
 centralized unit
 gNB-DU gNB-distributed unit, Next Generation NodeB
 distributed unit
 GNSS Global Navigation Satellite System
 GPRS General Packet Radio Service
 GSM Global System for Mobile Communications,
 Groupe Special
 Mobile
 GTP GPRS Tunneling Protocol
 GTP-U GPRS Tunnelling Protocol for User Plane
 GTS Go To Sleep Signal (related to WUS)
 GUMMEI Globally Unique MME Identifier
 GUTI Globally Unique Temporary UE Identity
 HARQ Hybrid ARQ, Hybrid Automatic Repeat Request
 HANDO, HO Handover
 HFN HyperFrame Number
 HHO Hard Handover
 HLR Home Location Register
 HN Home Network
 HO Handover
 HPLMN Home Public Land Mobile Network
 HSDPA High Speed Downlink Packet Access

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HSN Hopping Sequence Number
 HSPA High Speed Packet Access
 HSS Home Subscriber Server
 HSUPA High Speed Uplink Packet Access
 HTTP Hyper Text Transfer Protocol
 HTTPS Hyper Text Transfer Protocol Secure (https is
 http/1.1 over SSL, i.e., port 443)
 I-Block Information Block
 ICCID Integrated Circuit Card Identification
 ICIC Inter-Cell Interference Coordination
 ID Identity, identifier
 IDFT Inverse Discrete Fourier Transform
 IE Information element
 IBE In-Band Emission
 IEEE Institute of Electrical and Electronics Engineers
 IEI Information Element Identifier
 IEIDL Information Element Identifier Data Length
 IETF Internet Engineering Task Force
 IF Infrastructure
 IM Interference Measurement, Intermodulation, IP Mul-
 timedia
 IMC IMS Credentials
 IMEI International Mobile Equipment Identity
 IMGI International mobile group identity
 IMPI IP Multimedia Private Identity
 IMPU IP Multimedia PUBLIC identity
 IMS IP Multimedia Subsystem
 IMSI International Mobile Subscriber Identity
 IoT Internet of Things
 IP Internet Protocol
 Ipsec IP Security, Internet Protocol Security
 IP-CAN IP-Connectivity Access Network
 IP-M IP Multicast
 IPv4 Internet Protocol Version 4
 IPv6 Internet Protocol Version 6
 IR Infrared
 IS In Sync
 IRP Integration Reference Point
 ISDN Integrated Services Digital Network
 ISIM IM Services Identity Module
 ISO International Organisation for Standardisation
 ISP Internet Service Provider
 IWF Interworking-Function
 I-WLAN Interworking WLAN
 K Constraint length of the convolutional code, USIM
 Individual key
 kB Kilobyte (1000 bytes)
 kbps kilo-bits per second
 Kc Ciphering key
 Ki Individual subscriber authentication key
 KPI Key Performance Indicator
 KQI Key Quality Indicator
 KSI Key Set Identifier
 ksps kilo-symbols per second
 KVM Kernel Virtual Machine
 L1 Layer 1 (physical layer)
 L1-RSRP Layer 1 reference signal received power
 L2 Layer 2 (data link layer)
 L3 Layer 3 (network layer)
 LAA Licensed Assisted Access
 LAN Local Area Network
 LBT Listen Before Talk
 LCM LifeCycle Management
 LCR Low Chip Rate
 LCS Location Services
 LCID Logical Channel ID
 LI Layer Indicator

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LLC Logical Link Control, Low Layer Compatibility
 LPLMN Local PLMN
 LPP LTE Positioning Protocol
 LSB Least Significant Bit
 LTE Long Term Evolution
 LWA LTE-WLAN aggregation
 LWIP LTE/WLAN Radio Level Integration with IPsec Tunnel
 LTE Long Term Evolution
 M2M Machine-to-Machine
 MAC Medium Access Control (protocol layering context)
 MAC Message authentication code (security/encryption context)
 MAC-A MAC used for authentication and key agreement (TSG T WG3 context)
 MAC-I MAC used for data integrity of signalling messages (TSG T WG3 context)
 MANO Management and Orchestration
 MBMS Multimedia Broadcast and Multicast Service
 MBSFN Multimedia Broadcast multicast service Single Frequency Network
 MCC Mobile Country Code
 MCG Master Cell Group
 MCOT Maximum Channel Occupancy Time
 MCS Modulation and coding scheme
 MDAF Management Data Analytics Function
 MDAS Management Data Analytics Service
 MDT Minimization of Drive Tests
 ME Mobile Equipment
 MeNB master eNB
 MER Message Error Ratio
 MGL Measurement Gap Length
 MGRP Measurement Gap Repetition Period
 MIB Master Information Block, Management Information Base
 MIMO Multiple Input Multiple Output
 MLC Mobile Location Centre
 MM Mobility Management
 MME Mobility Management Entity
 MN Master Node
 MO Measurement Object, Mobile Originated
 MPBCH MTC Physical Broadcast CHannel
 MPDCCH MTC Physical Downlink Control CHannel
 MPDSCH MTC Physical Downlink Shared CHannel
 MPRACH MTC Physical Random Access CHannel
 MPUSCH MTC Physical Uplink Shared Channel
 MPLS MultiProtocol Label Switching
 MS Mobile Station
 MSB Most Significant Bit
 MSC Mobile Switching Centre
 MSI Minimum System Information, MCH Scheduling Information
 MSID Mobile Station Identifier
 MSIN Mobile Station Identification Number
 MSISDN Mobile Subscriber ISDN Number
 MT Mobile Terminated, Mobile Termination
 MTC Machine-Type Communications
 mMTC massive MTC, massive Machine-Type Communications
 MU-MIMO Multi User MIMO
 MWUS MTC wake-up signal, MTC WUS
 NACK Negative Acknowledgement
 NAI Network Access Identifier
 NAS Non-Access Stratum, Non-Access Stratum layer
 NCT Network Connectivity Topology
 NEC Network Capability Exposure

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NE-DC NR-E-UTRA Dual Connectivity
 NEF Network Exposure Function
 NF Network Function
 NFP Network Forwarding Path
 NFPD Network Forwarding Path Descriptor
 NFV Network Functions Virtualization
 NFVI NFV Infrastructure
 NFVO NFV Orchestrator
 NG Next Generation, Next Gen
 NGEN-DC NG-RAN E-UTRA-NR Dual Connectivity
 NM Network Manager
 NMS Network Management System
 N-PoP Network Point of Presence
 NMIB, N-MIB Narrowband MIB
 NPBCH Narrowband Physical Broadcast Channel
 NPDCCH Narrowband Physical Downlink Control Channel
 NPDSCH Narrowband Physical Downlink Shared Channel
 NPRACH Narrowband Physical Random Access Channel
 NPUSCH Narrowband Physical Uplink Shared CHannel
 NPSS Narrowband Primary Synchronization Signal
 NSSS Narrowband Secondary Synchronization Signal
 NR New Radio, Neighbour Relation
 NRF NF Repository Function
 NRS Narrowband Reference Signal
 NS Network Service
 NSA Non-Standalone operation mode
 NSD Network Service Descriptor
 NSR Network Service Record
 NSSAI 'Network Slice Selection Assistance Information
 S-NNSAI Single-NSSAI
 NSSF Network Slice Selection Function
 NW Network
 NWUS Narrowband wake-up signal, Narrowband WUS
 NZP Non-Zero Power
 O&M Operation and Maintenance
 ODU2 Optical channel Data Unit—type 2
 OFDM Orthogonal Frequency Division Multiplexing
 OFDMA Orthogonal Frequency Division Multiple Access
 OOB Out-of-band
 OOS Out of Sync
 OPEX OPERating EXpense
 OSI Other System Information
 OSS Operations Support System
 OTA over-the-air
 PAPR Peak-to-Average Power Ratio
 PAR Peak to Average Ratio
 PBCH Physical Broadcast Channel
 PC Power Control, Personal Computer
 PCC Primary Component Carrier, Primary CC
 PCell Primary Cell
 PCI Physical Cell ID, Physical Cell Identity
 PCEF Policy and Charging Enforcement Function
 PCF Policy Control Function
 PCRF Policy Control and Charging Rules Function
 PDCP Packet Data Convergence Protocol, Packet Data Convergence Protocol layer
 PDCCH Physical Downlink Control Channel
 PDCP Packet Data Convergence Protocol
 PDN Packet Data Network, Public Data Network
 PDSCH Physical Downlink Shared Channel
 PDU Protocol Data Unit
 PEI Permanent Equipment Identifiers
 PFD Packet Flow Description
 P-GW PDN Gateway

PHICH Physical hybrid-ARQ indicator channel
 PHY Physical layer
 PLMN Public Land Mobile Network
 PIN Personal Identification Number
 PM Performance Measurement
 PMI Precoding Matrix Indicator
 PNF Physical Network Function
 PNFD Physical Network Function Descriptor
 PNFR Physical Network Function Record
 POC PTT over Cellular
 PP, PTP Point-to-Point
 PPP Point-to-Point Protocol
 PRACH Physical RACH
 PRB Physical resource block
 PRG Physical resource block group
 ProSe Proximity Services, Proximity-Based Service
 PRS Positioning Reference Signal
 PRR Packet Reception Radio
 PS Packet Services
 PSBCH Physical Sidelink Broadcast Channel
 PSDCH Physical Sidelink Downlink Channel
 PSCCH Physical Sidelink Control Channel
 PSSCH Physical Sidelink Shared Channel
 PSCell Primary SCell
 PSS Primary Synchronization Signal
 PSTN Public Switched Telephone Network
 PT-RS Phase-tracking reference signal
 PTT Push-to-Talk
 PUCCH Physical Uplink Control Channel
 PUSCH Physical Uplink Shared Channel
 QAM Quadrature Amplitude Modulation
 QCI QoS class of identifier
 QCL Quasi co-location
 QFI QoS Flow ID, QoS Flow Identifier
 QoS Quality of Service
 QPSK Quadrature (Quaternary) Phase Shift Keying
 QZSS Quasi-Zenith Satellite System
 RA-RNTI Random Access RNTI
 RAB Radio Access Bearer, Random Access Burst
 RACH Random Access Channel
 RADIUS Remote Authentication Dial In User Service
 RAN Radio Access Network
 RAND RANDom number (used for authentication)
 RAR Random Access Response
 RAT Radio Access Technology
 RAU Routing Area Update
 RB Resource block, Radio Bearer
 RBG Resource block group
 REG Resource Element Group
 Rel Release
 REQ REQuest
 RF Radio Frequency
 RI Rank Indicator
 RIV Resource indicator value
 RL Radio Link
 RLC Radio Link Control, Radio Link Control layer
 RLC AM RLC Acknowledged Mode
 RLC UM RLC Unacknowledged Mode
 RLF Radio Link Failure
 RLM Radio Link Monitoring
 RLM-RS Reference Signal for RLM
 RM Registration Management
 RMC Reference Measurement Channel
 RMSI Remaining MSI, Remaining Minimum System Information
 RN Relay Node
 RNC Radio Network Controller

RNL Radio Network Layer
 RNTI Radio Network Temporary Identifier
 ROHC ROBust Header Compression
 RRC Radio Resource Control, Radio Resource Control layer
 RRM Radio Resource Management
 RS Reference Signal
 RSRP Reference Signal Received Power
 RSRQ Reference Signal Received Quality
 RSSI Received Signal Strength Indicator
 RSU Road Side Unit
 RSTD Reference Signal Time difference
 RTP Real Time Protocol
 RTS Ready-To-Send
 RTT Round Trip Time
 Rx Reception, Receiving, Receiver
 S1AP S1 Application Protocol
 S1-MME S1 for the control plane
 S1-U S1 for the user plane
 S-GW Serving Gateway
 S-RNTI SRNC Radio Network Temporary Identity
 S-TMSI SAE Temporary Mobile Station Identifier
 SA Standalone operation mode
 SAE System Architecture Evolution
 SAP Service Access Point
 SAPD Service Access Point Descriptor
 SAPI Service Access Point Identifier
 SCC Secondary Component Carrier, Secondary CC
 SCell Secondary Cell
 SC-FDMA Single Carrier Frequency Division Multiple Access
 SCG Secondary Cell Group
 SCM Security Context Management
 SCS Subcarrier Spacing
 SCTP Stream Control Transmission Protocol
 SDAP Service Data Adaptation Protocol, Service Data Adaptation Protocol layer
 SDL Supplementary Downlink
 SDNF Structured Data Storage Network Function
 SDP Service Discovery Protocol (Bluetooth related)
 SDSF Structured Data Storage Function
 SDU Service Data Unit
 SEAF Security Anchor Function
 SeNB secondary eNB
 SEPP Security Edge Protection Proxy
 SFI Slot format indication
 SFTD Space-Frequency Time Diversity, SFN and frame timing difference
 SFN System Frame Number
 SgNB Secondary gNB
 SGSN Serving GPRS Support Node
 S-GW Serving Gateway
 SI System Information
 SI-RNTI System Information RNTI
 SIB System Information Block
 SIM Subscriber Identity Module
 SIP Session Initiated Protocol
 SiP System in Package
 SL Sidelink
 SLA Service Level Agreement
 SM Session Management
 SMF Session Management Function
 SMS Short Message Service
 SMSF SMS Function
 SMTC SSB-based Measurement Timing Configuration
 SN Secondary Node, Sequence Number

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SoC System on Chip
 SON Self-Organizing Network
 SpCell Special Cell
 SP-CSI-RNTI Semi-Persistent CSI RNTI
 SPS Semi-Persistent Scheduling
 SQN Sequence number
 SR Scheduling Request
 SRB Signalling Radio Bearer
 SRS Sounding Reference Signal
 SS Synchronization Signal
 SSB Synchronization Signal Block, SS/PBCH Block
 SSBRI SS/PBCH Block Resource Indicator, Synchroni-
 zation Signal Block Resource Indicator
 SSC Session and Service Continuity
 SS-RSRP Synchronization Signal based Reference Signal
 Received Power
 SS-RSRQ Synchronization Signal based Reference Sig-
 nal Received Quality
 SS-SINR Synchronization Signal based Signal to Noise
 and Interference Ratio
 SSS Secondary Synchronization Signal
 SSSG Search Space Set Group
 SSSIF Search Space Set Indicator
 SST Slice/Service Types
 SU-MIMO Single User MIMO
 SUL Supplementary Uplink
 TA Timing Advance, Tracking Area
 TAC Tracking Area Code
 TAG Timing Advance Group
 TAU Tracking Area Update
 TB Transport Block
 TBS Transport Block Size
 TBD To Be Defined
 TCI Transmission Configuration Indicator
 TCP Transmission Communication Protocol
 TDD Time Division Duplex
 TDM Time Division Multiplexing
 TDMA Time Division Multiple Access
 TE Terminal Equipment
 TEID Tunnel End Point Identifier
 TFT Traffic Flow Template
 TMSI Temporary Mobile Subscriber Identity
 TNL Transport Network Layer
 TPC Transmit Power Control
 TPMI Transmitted Precoding Matrix Indicator
 TR Technical Report
 TRP, TRxP Transmission Reception Point
 TRS Tracking Reference Signal
 TRx Transceiver
 TS Technical Specifications, Technical Standard
 TTI Transmission Time Interval
 Tx Transmission, Transmitting, Transmitter
 U-RNTI UTRAN Radio Network Temporary Identity
 UART Universal Asynchronous Receiver and Transmitter
 UCI Uplink Control Information
 UE User Equipment
 UDM Unified Data Management
 UDP User Datagram Protocol
 UDSF Unstructured Data Storage Network Function
 UICC Universal Integrated Circuit Card
 UL Uplink
 UM Unacknowledged Mode
 UML Unified Modelling Language
 UMTS Universal Mobile Telecommunications System
 UP User Plane
 UPF User Plane Function
 URI Uniform Resource Identifier

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URL Uniform Resource Locator
 URLLC Ultra-Reliable and Low Latency
 USB Universal Serial Bus
 USIM Universal Subscriber Identity Module
 5 USS UE-specific search space
 UTRA UMTS Terrestrial Radio Access
 UTRAN Universal Terrestrial Radio Access Network
 UwPTS Uplink Pilot Time Slot
 V2I Vehicle-to-Infrastructure
 10 V2P Vehicle-to-Pedestrian
 V2V Vehicle-to-Vehicle
 V2X Vehicle-to-everything
 VIM Virtualized Infrastructure Manager
 VL Virtual Link,
 VLAN Virtual LAN, Virtual Local Area Network
 VM Virtual Machine
 VNF Virtualized Network Function
 VNFFG VNF Forwarding Graph
 VNFFGD VNF Forwarding Graph Descriptor
 20 VNFM VNF Manager
 VoIP Voice-over-IP, Voice-over-Internet Protocol
 VPLMN Visited Public Land Mobile Network
 VPN Virtual Private Network
 VRB Virtual Resource Block
 25 WiMAX Worldwide Interoperability for Microwave
 Access
 WLAN Wireless Local Area Network
 WMAN Wireless Metropolitan Area Network
 WPAN Wireless Personal Area Network
 30 X2-C X2-Control plane
 X2-U X2-User plane
 XML eXtensible Markup Language
 XRES EXpected user RESponse
 XOR eXclusive OR
 35 ZC Zadoff-Chu
 ZP Zero Power

For the purposes of the present document, the following terms and definitions are applicable to the examples and embodiments discussed herein, but is not meant to be limiting.

The term “circuitry” as used herein refers to, is part of, or includes hardware components such as an electronic circuit, a logic circuit, a processor (shared, dedicated, or group) and/or memory (shared, dedicated, or group), an Application Specific Integrated Circuit (ASIC), a field-programmable device (FPD) (e.g., a field-programmable gate array (FPGA), a programmable logic device (PLD), a complex PLD (CPLD), a high-capacity PLD (HCPLD), a structured ASIC, or a programmable SoC), digital signal processors (DSPs), etc., that are configured to provide the described functionality. In some embodiments, the circuitry may execute one or more software or firmware programs to provide at least some of the described functionality. The term “circuitry” may also refer to a combination of one or more hardware elements (or a combination of circuits used in an electrical or electronic system) with the program code used to carry out the functionality of that program code. In these embodiments, the combination of hardware elements and program code may be referred to as a particular type of circuitry.

The term “processor circuitry” as used herein refers to, is part of, or includes circuitry capable of sequentially and automatically carrying out a sequence of arithmetic or logical operations, or recording, storing, and/or transferring digital data. The term “processor circuitry” may refer to one or more application processors, one or more baseband processors, a physical central processing unit (CPU), a

single-core processor, a dual-core processor, a triple-core processor, a quad-core processor, and/or any other device capable of executing or otherwise operating computer-executable instructions, such as program code, software modules, and/or functional processes. The terms “application circuitry” and/or “baseband circuitry” may be considered synonymous to, and may be referred to as, “processor circuitry.”

The term “interface circuitry” as used herein refers to, is part of, or includes circuitry that enables the exchange of information between two or more components or devices. The term “interface circuitry” may refer to one or more hardware interfaces, for example, buses, I/O interfaces, peripheral component interfaces, network interface cards, and/or the like.

The term “user equipment” or “UE” as used herein refers to a device with radio communication capabilities and may describe a remote user of network resources in a communications network. The term “user equipment” or “UE” may be considered synonymous to, and may be referred to as, client, mobile, mobile device, mobile terminal, user terminal, mobile unit, mobile station, mobile user, subscriber, user, remote station, access agent, user agent, receiver, radio equipment, reconfigurable radio equipment, reconfigurable mobile device, etc. Furthermore, the term “user equipment” or “UE” may include any type of wireless/wired device or any computing device including a wireless communications interface.

The term “network element” as used herein refers to physical or virtualized equipment and/or infrastructure used to provide wired or wireless communication network services. The term “network element” may be considered synonymous to and/or referred to as a networked computer, networking hardware, network equipment, network node, router, switch, hub, bridge, radio network controller, RAN device, RAN node, gateway, server, virtualized VNF, NFVI, and/or the like.

The term “computer system” as used herein refers to any type interconnected electronic devices, computer devices, or components thereof. Additionally, the term “computer system” and/or “system” may refer to various components of a computer that are communicatively coupled with one another. Furthermore, the term “computer system” and/or “system” may refer to multiple computer devices and/or multiple computing systems that are communicatively coupled with one another and configured to share computing and/or networking resources.

The term “appliance,” “computer appliance,” or the like, as used herein refers to a computer device or computer system with program code (e.g., software or firmware) that is specifically designed to provide a specific computing resource. A “virtual appliance” is a virtual machine image to be implemented by a hypervisor-equipped device that virtualizes or emulates a computer appliance or otherwise is dedicated to provide a specific computing resource.

The term “resource” as used herein refers to a physical or virtual device, a physical or virtual component within a computing environment, and/or a physical or virtual component within a particular device, such as computer devices, mechanical devices, memory space, processor/CPU time, processor/CPU usage, processor and accelerator loads, hardware time or usage, electrical power, input/output operations, ports or network sockets, channel/link allocation, throughput, memory usage, storage, network, database and applications, workload units, and/or the like. A “hardware resource” may refer to compute, storage, and/or network resources provided by physical hardware element(s). A

“virtualized resource” may refer to compute, storage, and/or network resources provided by virtualization infrastructure to an application, device, system, etc. The term “network resource” or “communication resource” may refer to resources that are accessible by computer devices/systems via a communications network. The term “system resources” may refer to any kind of shared entities to provide services, and may include computing and/or network resources. System resources may be considered as a set of coherent functions, network data objects or services, accessible through a server where such system resources reside on a single host or multiple hosts and are clearly identifiable.

The term “channel” as used herein refers to any transmission medium, either tangible or intangible, which is used to communicate data or a data stream. The term “channel” may be synonymous with and/or equivalent to “communications channel,” “data communications channel,” “transmission channel,” “data transmission channel,” “access channel,” “data access channel,” “link,” “data link,” “carrier,” “radiofrequency carrier,” and/or any other like term denoting a pathway or medium through which data is communicated. Additionally, the term “link” as used herein refers to a connection between two devices through a RAT for the purpose of transmitting and receiving information.

The terms “instantiate,” “instantiation,” and the like as used herein refers to the creation of an instance. An “instance” also refers to a concrete occurrence of an object, which may occur, for example, during execution of program code.

The terms “coupled,” “communicatively coupled,” along with derivatives thereof are used herein. The term “coupled” may mean two or more elements are in direct physical or electrical contact with one another, may mean that two or more elements indirectly contact each other but still cooperate or interact with each other, and/or may mean that one or more other elements are coupled or connected between the elements that are said to be coupled with each other. The term “directly coupled” may mean that two or more elements are in direct contact with one another. The term “communicatively coupled” may mean that two or more elements may be in contact with one another by a means of communication including through a wire or other interconnect connection, through a wireless communication channel or ink, and/or the like.

The term “information element” refers to a structural element containing one or more fields. The term “field” refers to individual contents of an information element, or a data element that contains content.

The term “SMTC” refers to an SSB-based measurement timing configuration configured by SSB-MeasurementTimingConfiguration.

The term “SSB” refers to an SS/PBCH block.

The term “a ‘Primary Cell’” refers to the MCG cell, operating on the primary frequency, in which the UE either performs the initial connection establishment procedure or initiates the connection re-establishment procedure.

The term “Primary SCG Cell” refers to the SCG cell in which the UE performs random access when performing the Reconfiguration with Sync procedure for DC operation.

The term “Secondary Cell” refers to a cell providing additional radio resources on top of a Special Cell for a UE configured with CA.

The term “Secondary Cell Group” refers to the subset of serving cells comprising the PSCell and zero or more secondary cells for a UE configured with DC.

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The term “Serving Cell” refers to the primary cell for a UE in RRC_CONNECTED not configured with CA/DC there is only one serving cell comprising of the primary cell.

The term “serving cell” or “serving cells” refers to the set of cells comprising the Special Cell(s) and all secondary cells for a UE in RRC_CONNECTED configured with CA/

The term “Special Cell” refers to the PCell of the MCG or the PSCell of the SCG for DC operation; otherwise, the term “Special Cell” refers to the Pcell.

What is claimed is:

1. A base station in a wireless network, comprising:
one or more processors configured to:
receive a positioning request from a user equipment (UE), the positioning request implemented in a random access channel (RACH) message A (MsgA);
provide an uplink (UL) positioning reference signal (PRS) resource configuration to the UE;
receive UL PRS data from the UE based at least in part on the UL PRS resource configuration; and
provide a positioning response to the UE, the positioning response implemented in a RACH message B (MsgB),
wherein the UL PRS data is received from the UE prior to the providing of the positioning response.
2. The base station of claim 1, wherein the RACH MsgA comprises a physical random access channel (PRACH) preamble.
3. The base station of claim 1, wherein the RACH MsgA comprises at least one of a requested type of positioning procedure, a request for additional allocation of reference signals, one or more signal location parameter measurements, a UE identification, and a requested type of positioning reference resources.
4. The base station of claim 1, wherein a requested type of positioning procedure comprises a downlink (DL) only positioning technique, a UL only positioning technique, or a DL and UL positioning technique.
5. The base station of claim 1, wherein the positioning response comprises an estimated coordinate of the UE, a metric characterizing coordinate estimation quality, time-stamps indicative of a time of reception of one or more reference signals, a request for additional positioning measurements, a configuration or activation of additional resources for transmission of positioning reference signals, or location information associated with downlink (DL) positioning reference resources.
6. The base station of claim 1, wherein the RACH MsgB comprises at least one of an estimated location of the UE, a metric characterizing a quality or uncertainty of the estimated location, a timestamp indicating a time of reception of the RACH MsgA by the base station, a timestamp indicating a time of transmission of the RACH MsgB by the base station, a request for additional positioning measurements, or location information associated with downlink (DL) positioning reference resources.
7. The base station of claim 6, wherein the estimated location of the UE is determined by a location function entity, and provided to the base station.
8. The base station of claim 1, wherein the UL PRS resource configuration is implemented in a second RACH MsgB or through upper layer signaling.
9. A method for determining an estimated location of a user equipment (UE) in a wireless network, the method comprising:

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receiving, by a base station in the wireless network, a positioning request from a UE, the positioning request implemented in a random access channel (RACH) message A (MsgA);

providing, by the base station, an uplink (UL) positioning reference signal (PRS) resource configuration to the UE;

receiving, by the base station, UL PRS data from the UE based at least in part on the UL PRS resource configuration; and

providing, by the base station, a positioning response to the UE, the positioning response implemented in a RACH message B (MsgB),

wherein the UL PRS data is received from the UE prior to the providing of the positioning response.

10. The method of claim 9, wherein the RACH MsgA comprises a physical random access channel (PRACH) preamble.

11. The method of claim 9, wherein the RACH MsgA comprises at least one of a requested type of positioning procedure, a request for additional allocation of reference signals, one or more signal location parameter measurements, a UE identification, and a requested type of positioning reference resources.

12. The method of claim 9, wherein a requested type of positioning procedure comprises a downlink (DL) only positioning technique, a UL only positioning technique, or a DL and UL positioning technique.

13. The method of claim 9, wherein the positioning response comprises an estimated coordinate of the UE, a metric characterizing coordinate estimation quality, time-stamps indicative of a time of reception of one or more reference signals, a request for additional positioning measurements, a configuration or activation of additional resources for transmission of positioning reference signals, or location information associated with downlink (DL) positioning reference resources.

14. The method of claim 9, wherein the UL PRS resource configuration is implemented in a second RACH MsgB or through upper layer signaling.

15. One or more non-transitory computer-readable media (CRM) comprising instructions to, upon execution of the instructions by one or more processors of a base station in a wireless network, cause the base station to:

receive a positioning request from a user equipment (UE), the positioning request implemented in a random access channel (RACH) message A (MsgA);

provide an uplink (UL) positioning reference signal (PRS) resource configuration to the UE;

receive UL PRS data from the UE based at least in part on the UL PRS resource configuration; and

provide a positioning response to the UE, the positioning response implemented in a RACH message B (MsgB), wherein the UL PRS data is received from the UE prior to the providing of the positioning response.

16. The one or more non-transitory CRM of claim 15, wherein the RACH MsgB comprises at least one of an estimated location of the UE, a metric characterizing a quality or uncertainty of the estimated location, a timestamp indicating a time of reception of the RACH MsgA by the base station, a timestamp indicating a time of transmission of the RACH MsgB by the base station, a request for additional positioning measurements, or location information associated with downlink (DL) positioning reference resources.

17. The one or more non-transitory CRM of claim 15, wherein the UL PRS resource configuration is implemented in a second RACH MsgB or through upper layer signaling.

18. The one or more non-transitory CRM of claim 15, wherein the positioning response comprises an estimated coordinate of the UE, a metric characterizing coordinate estimation quality, timestamps indicative of a time of reception of one or more reference signals, a request for additional positioning measurements, a configuration or activation of additional resources for transmission of positioning reference signals, or location information associated with downlink (DL) positioning reference resources.

19. The one or more non-transitory CRM of claim 15, wherein the RACH MsgA comprises at least one of a requested type of positioning procedure, a request for additional allocation of reference signals, one or more signal location parameter measurements, a UE identification, and a requested type of positioning reference resources.

20. The one or more non-transitory CRM of claim 15, wherein the RACH MsgA comprises a physical random access channel (PRACH) preamble.

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